

Commutative Algebra

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1 Introduction.

1.1 Multiplicative sets and Prime ideals

Definition 1.1. A subset $S \subset A$ is called multiplicatively closed iff

1. $1 \in S$
2. if $s, t \in S$, then $st \in S$.

Proposition 1.2. Let S be multiplicatively closed, and let a be an ideal with $a \cap S = \emptyset$. Consider all ideals b s.t

$$\begin{cases} a \subseteq b \\ b \cap S = \emptyset \end{cases}$$

Equivalently $a \subset b \subset A - S$. Then the poset of such ideals satisfies the conditions of Zorn's lemma, and any maximal such element is a prime ideal of A .

Proof. The poset is nonempty because it contains a , every totally ordered subset $\{b_i : i \in I\}$, has an upper bound $\bigcup_{i \in I} b_i$. Therefore by Zorn's lemma there exists some maximal element p of this set.

Suppose that $x, y \in A$ with $xy \in p$ but $x, y \notin p$: Since $x, y \notin p$ then the ideal sums

$$p + \langle x \rangle, p + \langle y \rangle$$

Strictly contain p , so they must contain each an element of S , therefore $\exists s \in S$ of the form

$$s = p_1 + xa_1 \quad p_1 \in p, a_1 \in R$$

and similarly $\exists t \in s$ with

$$t = p_2 + ya_2$$

but then $st \in S$, therefore

$$p_1p_2 + p_1ya_2 + p_2xa_1 + xya_1a_2 \in P \implies st \in p$$

contradicting that $p \cap S = \emptyset$.

Therefore p is prime. □

Example 1.3. • A a domain and $S = A - \{0\}$.

- In $\mathbb{Z}$: $S = \{1, 6, 6^2, \dots\}$, or more generally in any A if $x \in A$ is not nilpotent, then

$$S = \{1, x, x^2, \dots\}, \quad S \cap \{0\} = \emptyset$$

There exists a prime ideal with $p \cap S = \emptyset \implies x \notin p$ (obviously $0 \in p$).

Therefore

$$\{\text{all nilpotent } x\} = \text{nilrad}(0) = \bigcap_{p \text{ prime ideal}} p$$

- $p \subset A$ prime then $A - p$ is multiplicatively closed.

Theorem 1.4. If $a \subset A$ is an ideal, then

$$\text{rad } a := \{x \in A \mid \exists N \geq 1 \text{ with } x^N \in a\}$$

satisfies

$$\text{rad } a = \bigcap_{\text{all } p, p \supseteq a} p$$

Proof. • Slick proof: apply the conclusion above to the quotient ring A/a .

- Pedestrian proof: Show both inclusions directly : If $x^N \in a$ and $a \subset p$, then $x \in p$ (for all primes $p \supseteq a$), so $x \in \bigcap_{p \supseteq a} p$.

If $\forall N, x^N \notin a$ then take $S = \{1, x, x^2, \dots\}$ we have $a \cap S = \emptyset$; so $\exists p$ ($p \supseteq a$ and $p \cap S = \emptyset$) which is prime with $p \supseteq a$ and $p \cap S = \emptyset$ so $\exists p \supseteq a$ with $x \notin p$ so

$$x \notin \bigcap_{p \supseteq a} p$$

□

1.2 The Jacobson radical of a commutative ring

Definition 1.5.

$$\text{Jrad } A = \bigcap_{\text{all maximal ideals } m \subset A} m \supseteq \text{nilrad } A$$

Proposition 1.6. $x \in \text{Jrad } A \iff \forall y \in A, 1 - xy$ is a unit in A .

Proof. $\implies$ Let $x \in \text{Jrad } A$, then $\forall m$ maximal ideal, $x \in m$. If $y \in A$ satisfies $1 - xy$ not a unit, then

$$\langle 1 - xy \rangle \subsetneq A$$

so $\exists m$ maximal with $1 - xy \in m$ but $x \in m$. This would force $1 \in m$ since

$$1 - xy + xy = 1$$

$\Leftarrow$ Suppose that $\forall y \in A, 1 - xy$ is a unit in A . We want to show that if M is maximal, is $x \in m$? Work in A/m , field: $\bar{x}$ satisfies that $\forall \bar{y} \in A/m$,

$$\bar{1} - \bar{x}\bar{y} \text{ is a unit in } A/m$$

but in a field, this can only happen if $\bar{x} = \bar{0}$. Therefore $x \in m$, for all m .

□

1.3 Extension and contraction of ideals

Setup. $A \xrightarrow{f} B$ a homomorphism of rings (so B is an A -algebra) and

$$\mathfrak{a} \subseteq A \text{ ideal} \rightarrow \mathfrak{a}^e = \mathfrak{a}B = \langle f(a) : a \in A \rangle$$

$$\mathfrak{b} \subseteq B \rightarrow \mathfrak{b}^c = f^{-1}(\mathfrak{b}) = \ker \left(A \xrightarrow{f} B \xrightarrow{\nu} B/\mathfrak{b} \right)$$

Basic observations

$$\mathfrak{a}^{ec} \supseteq \mathfrak{a}, \quad \mathfrak{b}^{ce} \subseteq \mathfrak{b}$$

but notice that

$$\mathfrak{a} = \mathfrak{a}^e, \quad \mathfrak{b}^{cec} = \mathfrak{b}^c$$

important fact. if $q \subseteq B$ is prime then so is q^c therefore

$$A/q^c \xrightarrow{\text{subdomain}} B/q$$

Example 1.7. $\mathbb{Z} \rightarrow \mathbb{Q}, (6\mathbb{Z})^e = \mathbb{Q}, (6\mathbb{Z})^{ec} = \mathbb{Z}$.

Example 1.8. $\mathbb{Z} \rightarrow \mathbb{Z}/10\mathbb{Z} (4\mathbb{Z})^e = \frac{4\mathbb{Z}+10\mathbb{Z}}{10\mathbb{Z}} = 2\mathbb{Z}/10\mathbb{Z}$ so

$$(4\mathbb{Z})^{ec} = 2\mathbb{Z}$$

Example 1.9. $A \rightarrow A[x]$ take $\mathfrak{a} \subseteq A$ then $\mathfrak{a}^e = \mathfrak{a}[x]$ which is the set of all polynomials with coefficients in $\mathfrak{a}$. And $\{0_A\} = \langle x+1 \rangle^c; \langle x+1 \rangle^{ce} = \{0_{A[x]}\}$.

1.4 Modules.

View Math 341 Notes, I won't be taking notes here.

Let A be given and $a_1, \dots, a_r$ ideals in A , we say that these ideals are relatively prime iff $\forall i, j$ with $i \neq j$,

$$a_i + a_j = A \iff \exists x \in a_i, y \in a_j \text{ s.t } x + y = 1$$

Proposition 1.10. In the above situation we have that

$$a_1 \cdots a_r = a_1 \cap a_2 \cap \cdots \cap a_r$$

Proof. Induction on r :

$$a_1 a_2 \subseteq a_1 \cap a_2 \quad \text{easy}$$

Conversely if $z \in a_1 \cap a_2$ we have that $x \in a_1, y \in a_2$ s.t $x + y = 1$ then

$$z = z \cdot 1 = zx + zy \in a_1 a_2$$

for the inductive step, we have already $b = a_1 \cap \cdots \cap a_{r-1} = a_1 \cdots a_{r-1}$ need to observe that a_r is still relatively prime to b .

Indeed, we know that

$$a_1 + a_r = A, \dots, a_{r-1} + a_r = A$$

therefore we chose elements in $x_1 \in a_1, \dots, x_{r-1} \in a_{r-1}$ and $y_1, \dots, y_{r-1} \in a_r$ with $x_i + y_i = 1$. We can take

$$\begin{aligned} 1 &= 1^{r-1} \\ &= (x_1 + y_1)(x_2 + y_2) \cdots (x_{r-1} + y_{r-1}) \\ &= x_1 \cdots x_{r-1} + (\text{terms involving some } y_i) \\ &\in a_1 \cdots a_{r-1} + a_r \end{aligned}$$

□

Theorem 1.11 (Chinese Remainder Theorem). *Let $a_1 \cdots a_r$ be pairwise relatively prime, consider the homomorphism of A modules*

$$A \xrightarrow{f} A/a_1 \oplus \cdots \oplus A/a_r$$

Hence

$$A/a_1 \oplus \cdots \oplus A/a_r \cong A/\ker = A/a_1 \cdots a_r$$

1.5 Local Rings

Definition 1.12. A ring A is said to be *local* iff it has exactly one maximal ideal (called m).

Example 1.13. • $A = K$ a field, then $m = 0$

- Fix p prime $A \{a/b \in \mathbb{Q} \mid a, b \in \mathbb{Z}, b \notin p\mathbb{Z}\} \supset m = \{pc/b \mid c, b \in \mathbb{Z}, p \nmid b\} = pA$. Check that the only ideals are

$$A \supsetneq pA \supsetneq p^2A \supsetneq \cdots$$

•

$$\begin{aligned} A &= k[[x_1, x_2]] \\ &= \{\text{formal power series}\} \\ &= \{a_0 + b_1x_1 + b_2x_2 + c_{11}x_1 + \cdots\} \end{aligned}$$

Where $a_i, b_j, c_{k,l}, \dots \in K$ with

$$m = \langle x_1, x_2 \rangle = \{\text{formal power series with } a_0 = 0\}$$

This is the only maximal ideal because m is the set of non-units, think about it. Same for the example above.

Proposition 1.14. A is local $\iff$ the set of non-units of A is an ideal (which turns out to be the unique maximal ideal).

Proof. We denote by V the set of non-units of A

$\implies$ Let A be local, let m be its unique maximal ideal we clearly have that $m \subseteq V$. Now take any $v \in V$, then by Zorn's lemma we can find some maximal ideal $I \ni v$. But since m is the unique maximal ideal then we have that $m \ni v$.

$\impliedby$ Trivial.

□

1.6 Algebras

Definition 1.15. An A -algebra is a ring R (not necc. commutative) with a ring homomorphism $f: A \rightarrow R$ s.t

$$\text{im } f \subseteq \text{center of } R$$

We will generally stick to the case where R is commutative.

Example 1.16. All rings are $\mathbb{Z}$ -algebras, $M_n(A)$, $A[x_1, \dots, x_r]$. Moreover $\mathbb{C}$, $\mathbb{H}$ and matrices in $\mathbb{C}$ are all $\mathbb{R}$ algebras

Definition 1.17. A *finitely generated A -algebra* means that $\exists a \subseteq A$ an ideal s.t

$$R \cong A[x_1, \dots, x_n]/a$$

Example 1.18. A $\mathbb{Z}$ -algebra $\mathbb{Z}/10\mathbb{Z}[y_1, y_2]$ where $y_1^2 - 1 = 0$ and $y_2^3 - 2y_1y_2 - 3 = 0$ which is isomorphic to

$$\mathbb{Z}[x_1, x_2]/\langle 10, x_1^2 - 1, x_2^3 - 2x_1x_2 - 3 \rangle$$

Definition 1.19. a, b ideals of A then the *colon ideal* $(a : b)$ is defined to be

$$(a : b) = \{x \in A : xb \subseteq a\} = \bigcap_{y \in b} \text{Ann}(y + a \in A/a)$$

Example 1.20. In $\mathbb{Z}$,

$$(4\mathbb{Z} : 10\mathbb{Z}) = 2\mathbb{Z}, \quad (10\mathbb{Z} : 4\mathbb{Z}) = 5\mathbb{Z}$$

Theorem 1.21. Let $p_1, \dots, p_r$ be prime ideals of A and suppose that a satisfies

$$a \subseteq p_1 \cup \dots \cup p_n$$

Then $\exists j$ s.t $a \subseteq p_j$.

(Note that the proof below also works if all the p_i 's are prime except for 1 or 2 p_i 's).

Proof. By induction on n :

- $n = 1$: trivial.
- $n = 2$: Suppose $a \subseteq p_1 \cup p_2$ but $a \not\subseteq p_1$ and $a \not\subseteq p_2$ so $\exists x_1, x_2 \in a$ s.t $x_1 \notin p_1, x_2 \notin p_2$. Therefore we must have $x_1 \in p_2$ and $x_2 \in p_1$, then

$$a \ni z := x_1 + x_2 \notin p_1 \cup p_2$$

(If $z \in p_1$ then $x_1 = z - x_2 \in p_1$).

- Now suppose $n \geq 3$ and we know the result for $n - 1$ primes. Suppose that

$$a \subseteq p_1 \cup \cdots \cup p_n$$

if $a \subseteq p_1 \cup \cdots \cup p_{n-1}$ (after potentially reordering), then we are okay. So suppose that $a \not\subseteq p_1 \cup \cdots \cup p_{n-1}$ for any reordering. so $\exists x_j \in a$ with $x_j \in p_j$ for every j .

Consider $z = x_1 + x_2x_3 \cdots x_n$, of course $z \in a$ but $z \notin p_j$ for any j (contradiction)!

Now let $x_1 \in p_1$ and $x_2, \dots, x_n \notin p_1$ and p_1 is prime therefore

$$x_2 \cdots x_n \notin p_1$$

hence $z = x_1 + (x_2 \cdots x_n) \notin p_1$. Now for p_2 , we have that

$$x_1 \notin p_2, x_2 \in p_2 \implies x_1 + x_2x_3 \cdots x_n \notin p_2$$

Similarly for the other p 's.

□

2 Modules

Lemma 2.1 (Nakayama's lemma). *Let A be a local ring, with maximal ideal m ; and Let M be a finitely generated A -module. Suppose that*

$$mM = M$$

(where mM is the set of all finite sums $m_1y_1 + \cdots + m_ry_r$ with $m_i \in m$ and $y_i \in M$.)

Then $M = 0$.

Cheap proof. Suppose $M = \langle x_1, \dots, x_r \rangle$, we can remove redundant x_i 's, therefore we can assume that

$$\forall j, x_j \in \langle x_1, \dots, \hat{x}_j, \dots, x_r \rangle$$

If $M \neq 0$ we are left with this set of generators with $r \geq 1$. But then $x_1 \in M = mM$ therefore

$$x_1 = \sum_{i=1}^r m_i x_i \quad m_i \in m$$

Therefore $(1 - m_1)x_1 = m_2x_2 + \cdots + m_rx_r$ but $1 - m_1 \notin m$ must be a unit in A ! Therefore

$$x_1 \in \langle x_2, \dots, x_r \rangle \quad \text{impossible}$$

□

Corollary 2.2. *Fix (A, m) a local ring. Suppose that $M \supseteq N$, M finitely generated and $M = N + mM$ (Equivalently in M/mM the images of N generate everything). Then $M = N$.*

Proof. Apply Nakayama to M/N ; note that

$$m \cdot (M/N) = M/N, \quad M/N \text{ is finitely generated}$$

□

Theorem 2.3 (General Statement of Nakayama's Lemma). *Let A be any (commutative!) ring and let M be a finitely generated A -module, let $a \subseteq A$ be an ideal (usually $a = \text{Jrad } A$) and suppose*

$$aM = M$$

then $\exists b \in 1 + a$ s.t $bM = 0$.

In the case where a is the Jrad A in which case $b \in 1 + a$ is a unit, so $M = 0$.

Proof. Let $M = \langle x_1, \dots, x_r \rangle$, since $M = aM$, we see that $\exists a_{ij} \in a$ s.t

$$\begin{aligned} x_1 &= a_{11}x_1 + \dots + a_{1r}x_r \in aM \\ x_2 &= a_{21}x_1 + \dots + a_{2r}x_r \in aM \\ &\vdots \\ x_r &= a_{r1}x_1 + \dots + a_{rr}x_r \in aM \end{aligned}$$

Let $P = (a_{ij})_{ij}$ be the matrix of a_{ij} 's, rewrite the system above as

$$\begin{aligned} (1 - a_{11})x_1 - a_{12}x_2 + \dots - a_{1r}x_r &= 0 \\ a_{21}x_1 + (1 - a_{22})x_2 + \dots - a_{2r}x_r &= 0 \\ &\vdots \end{aligned}$$

Symbolically we get that

$$(I - P) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_r \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Multiplying on the left by the matrix $Q = (I - P)^{adj} \in M_r(A)$; it has the property that

$$Q(I - P) = \text{diag}(b)$$

where $b = \det(I - P) \in 1 + a$; we deduce

$$bx_1 = 0, \quad bx_2 = 0, \dots, \quad bx_r = 0$$

so b annihilates every generator of M , and therefore annihilates all of M .

□

Example 2.4. We can use this so that if (A, m) then

$$m \supseteq m^2 \supseteq m^3 \supseteq \dots$$

if $m = m^{n+1}$ then $m^n = 0$. Indeed m_i/m_{i+1} is an A/m -module.

2.1 Hom functors and exactness

Quick reminder. M, N are A Modules, therefore $\text{Hom}_A(M, N)$ is an A -module (since A is commutative). Some facts:

- We have a natural isomorphism $\text{Hom}(A, N) \cong N$ (as A -modules).
- Have an natural isomorphism

$$\text{Hom}\left(\bigoplus_{\alpha \in I} M_{\alpha}, N\right) \cong \prod_{\alpha \in I} \text{Hom}(M_{\alpha}, N)$$

$$(\dots x_{\alpha} \dots)_{\text{almost all zeroes}} \xrightarrow{\varphi} \sum_{\alpha} \varphi_{\alpha}(x_{\alpha})$$

But also

$$\text{Hom}(M, \prod_{\beta} N_{\beta}) \cong \prod_{\beta} \text{Hom}(M, N_{\beta})$$

$$x \in M \rightarrow (\dots \psi_{\beta}(x) \dots) \in \prod_{\beta} N_{\beta}$$

- $0 \rightarrow N' \xrightarrow{f} N \xrightarrow{g} N''$ exact implies that

$$0 \rightarrow \text{Hom}(M, N') \xrightarrow{f_*} \text{Hom}(M, N) \xrightarrow{g_*} \text{Hom}(M, N'')$$

(even if g is surjective, then g_* need not be surjective).

Similarly if

$$M' \xrightarrow{u} M \xrightarrow{w} M'' \rightarrow 0 \quad \text{exact}$$

then

$$0 \rightarrow \text{Hom}(M'', N) \xrightarrow{w^*} \text{Hom}(M, N) \xrightarrow{u^*} \text{Hom}(M', N)$$

(even if U is injective then U^* need not be surjective).

2.2 Presentations of Modules

Let A be a (commutative) ring and M an A -module.

Finite presentation. Say M is generated by $x_1, \dots, x_n$ with relations $r_1, \dots, r_k$, where

$$r_j : a_{j1}x_1 + \dots + a_{jn}x_n = 0 \quad (1 \leq j \leq k).$$

Let A^n have basis $e_1, \dots, e_n$ and A^k have basis $f_1, \dots, f_k$. Define $\eta : A^n \rightarrow M$ by $\eta(e_i) = x_i$ and define $\Phi : A^k \rightarrow A^n$ by

$$\Phi(f_j) = a_{j1}e_1 + \dots + a_{jn}e_n.$$

Then $\eta \circ \Phi = 0$, and $\text{im}(\Phi) = \ker(\eta)$ (the relations generate all relations), hence we have an exact sequence

$$A^k \xrightarrow{\Phi} A^n \xrightarrow{\eta} M \rightarrow 0, \quad \text{so } M \cong \text{coker}(\Phi) = A^n / \text{im}(\Phi).$$

If we write Φ as a matrix, it is the $n \times k$ matrix (a_{ji}) with the convention that the j -th column is the coordinate vector of $\Phi(f_j)$ in the basis (e_i) .

Infinite presentations. Nothing changes: if M is generated by a family $(x_i)_{i \in I}$ with relations indexed by J , one has

$$A^{(J)} \xrightarrow{\Phi} A^{(I)} \xrightarrow{\eta} M \rightarrow 0,$$

where $A^{(I)} = \bigoplus_{i \in I} A$ and $A^{(J)} = \bigoplus_{j \in J} A$ are free modules on those bases.

Computing $\text{Hom}_A(M, N)$ from a presentation

Given an exact sequence

$$A^k \xrightarrow{\Phi} A^n \xrightarrow{\eta} M \rightarrow 0,$$

apply the contravariant left-exact functor $\text{Hom}_A(-, N)$ to obtain the exact sequence

$$0 \rightarrow \text{Hom}_A(M, N) \xrightarrow{\eta^*} \text{Hom}_A(A^n, N) \xrightarrow{\Phi^*} \text{Hom}_A(A^k, N),$$

where $\eta^*(\psi) = \psi \circ \eta$ and $\Phi^*(\varphi) = \varphi \circ \Phi$.

Identifications.

- If $n, k < \infty$ then $\text{Hom}_A(A^n, N) \cong N^n$ and $\text{Hom}_A(A^k, N) \cong N^k$ via

$$\text{Hom}_A(A^n, N) \ni \psi \longleftrightarrow (\psi(e_1), \dots, \psi(e_n)) \in N^n.$$

- More generally, for a free module $A^{(I)} = \bigoplus_{i \in I} A$, one has

$$\text{Hom}_A\left(\bigoplus_{i \in I} A, N\right) \cong \prod_{i \in I} \text{Hom}_A(A, N) \cong \prod_{i \in I} N.$$

(important: *direct sum* on the left becomes a *direct product* on the right.)

Under the finite identifications, the map $\Phi^* : N^n \rightarrow N^k$ is given explicitly by

$$(y_1, \dots, y_n) \longmapsto (z_1, \dots, z_k), \quad z_j = a_{j1}y_1 + \dots + a_{jn}y_n.$$

Therefore

$$\text{Hom}_A(M, N) \cong \ker(N^n \xrightarrow{\Phi^*} N^k).$$

Equivalently: an A -linear map $M \rightarrow N$ is determined by the images of the generators x_i , and the only constraints are that the relations r_j hold after applying the map.

2.3 Tensor Product of Modules (review)

Assume A is commutative (so that left/right issues disappear).

Bilinear maps. For A -modules M, N, P , let $\text{Bil}_A(M \times N, P)$ be the set of maps $\beta : M \times N \rightarrow P$ which are A -linear in each variable:

$$\beta(x_1 + x_2, y) = \beta(x_1, y) + \beta(x_2, y), \quad \beta(ax, y) = a\beta(x, y),$$

and similarly in the second variable.

There is a natural bijection

$$\text{Bil}_A(M \times N, P) \cong_{\text{nat}} \text{Hom}_A(M, \text{Hom}_A(N, P)),$$

sending β to the map $x \mapsto (y \mapsto \beta(x, y))$.

Universal property of the tensor product. The tensor product $M \otimes_A N$ is an A -module equipped with a bilinear map

$$\tau : M \times N \rightarrow M \otimes_A N, \quad (x, y) \mapsto x \otimes y,$$

such that for every A -module P , composition with τ induces a natural isomorphism

$$\text{Hom}_A(M \otimes_A N, P) \cong_{\text{nat}} \text{Bil}_A(M \times N, P).$$

Concrete construction. One can construct $M \otimes_A N$ as the quotient of the free A -module F on symbols (x, y) , by the submodule generated by the bilinearity relations:

$$\begin{aligned} (x_1 + x_2, y) - (x_1, y) - (x_2, y), & \quad (ax, y) - a(x, y), \\ (x, y_1 + y_2) - (x, y_1) - (x, y_2), & \quad (x, ay) - a(x, y). \end{aligned}$$

The class of (x, y) is denoted $x \otimes y$.

Standard isomorphisms. There are natural isomorphisms

$$A \otimes_A N \cong N, \quad \left(\bigoplus_{\alpha} M_{\alpha} \right) \otimes_A N \cong \bigoplus_{\alpha} (M_{\alpha} \otimes_A N), \quad M \otimes_A N \cong N \otimes_A M,$$

and associativity up to canonical isomorphism:

$$(M \otimes_A N) \otimes_A Z \cong M \otimes_A (N \otimes_A Z).$$

Functoriality of $\otimes$

Given A -linear maps $f : M_1 \rightarrow M_2$ and $g : N_1 \rightarrow N_2$, there is a unique A -linear map

$$f \otimes g : M_1 \otimes_A N_1 \rightarrow M_2 \otimes_A N_2$$

satisfying on simple tensors

$$(f \otimes g)(x \otimes y) = f(x) \otimes g(y).$$

Hence for a general tensor $t = \sum_i a_i(x_i \otimes y_i)$,

$$(f \otimes g)(t) = \sum_i a_i(f(x_i) \otimes g(y_i)).$$

Extension of scalars

Let $f : A \rightarrow B$ be a ring homomorphism and let M be an A -module. Then $B \otimes_A M$ is naturally a B -module via multiplication on the left factor:

$$b \cdot (b' \otimes m) := (bb') \otimes m.$$

(Equivalently: the action comes from $\mu_b \otimes 1_M$, where $\mu_b : B \rightarrow B$ is multiplication by b .)

Example 2.5 (Polynomials). If $B = A[x]$, then as an A -module,

$$A[x] \cong \bigoplus_{d \geq 0} A \cdot x^d,$$

so

$$A[x] \otimes_A M \cong \bigoplus_{d \geq 0} (A \cdot x^d) \otimes_A M \cong \bigoplus_{d \geq 0} M.$$

Under this identification, $x^d \otimes m$ corresponds to $m x^d$, and one gets

$$A[x] \otimes_A M \cong M[x]$$

(the module of polynomials with coefficients in M).

If $\mathfrak{a} \subseteq A$ is an ideal (viewed as an A -module), then

$$A[x] \otimes_A \mathfrak{a} \cong \mathfrak{a}[x].$$

Inside $A[x]$, this is the extended ideal $\mathfrak{a}^e := \mathfrak{a} \cdot A[x]$.

Tensoring algebras. If A is commutative and B, C are (commutative) A -algebras, then $B \otimes_A C$ is an A -algebra with multiplication determined by

$$(b \otimes c) \cdot (b' \otimes c') := (bb') \otimes (cc').$$

Example 2.6. There is a natural isomorphism of A -algebras

$$A[x] \otimes_A B \cong B[x].$$

2.4 Exactness properties of $- \otimes_A N$

Right exactness. The functor $- \otimes_A N$ is *right exact*: if

$$M' \xrightarrow{f} M \xrightarrow{g} M'' \rightarrow 0$$

is exact, then

$$M' \otimes_A N \xrightarrow{f \otimes 1} M \otimes_A N \xrightarrow{g \otimes 1} M'' \otimes_A N \rightarrow 0$$

is exact.

Warning: it need not be left exact (injectivity can fail).

Example: quotients / extended ideals. From $0 \rightarrow \mathfrak{a} \rightarrow A \rightarrow A/\mathfrak{a} \rightarrow 0$ we get

$$\mathfrak{a} \otimes_A B \rightarrow A \otimes_A B \rightarrow (A/\mathfrak{a}) \otimes_A B \rightarrow 0.$$

Using $A \otimes_A B \cong B$, the last map identifies $(A/\mathfrak{a}) \otimes_A B$ with

$$B/\mathfrak{a}^e, \quad \mathfrak{a}^e = \mathfrak{a} \cdot B.$$

Computing $M \otimes_A N$ from a presentation

If M has a presentation

$$A^k \xrightarrow{\Phi} A^n \rightarrow M \rightarrow 0,$$

then tensoring with N gives an exact sequence

$$N^k \cong A^k \otimes_A N \xrightarrow{\Phi \otimes 1} A^n \otimes_A N \cong N^n \rightarrow M \otimes_A N \rightarrow 0.$$

Hence

$$M \otimes_A N \cong \text{coker}(\Phi \otimes 1),$$

and under the identifications $A^k \otimes N \cong N^k$, $A^n \otimes N \cong N^n$, the map $\Phi \otimes 1 : N^k \rightarrow N^n$ is given by the *same matrix* (a_{ji}) :

$$(u_1, \dots, u_k) \mapsto \left(\sum_{j=1}^k a_{j1} u_j, \dots, \sum_{j=1}^k a_{jn} u_j \right).$$

Equivalently: $M \otimes_A N$ is generated by $x_i \otimes y$ ($i = 1, \dots, n$, $y \in N$), with relations coming from:

- bilinearity in y (so $x_i \otimes (y + y') = x_i \otimes y + x_i \otimes y'$ and $x_i \otimes (ay) = a(x_i \otimes y)$),
- and the original relations of M : if $\sum_i c_i x_i = 0$ in M , then for every $y \in N$,

$$\sum_i c_i (x_i \otimes y) = 0 \quad \text{in } M \otimes_A N.$$

Conceptual proof of right exactness (via adjunction)

Conceptual proof of right exactness (via adjunction)

Recall the adjunction: for any A -modules X, N, P there is a natural isomorphism

$$\mathrm{Hom}_A(X \otimes_A N, P) \cong \mathrm{Hom}_A(X, \mathrm{Hom}_A(N, P)),$$

equivalently

$$\mathrm{Hom}_A(X \otimes_A N, P) \cong \mathrm{Bil}_A(X \times N, P).$$

Lemma 2.7. *Let $Q \xrightarrow{\lambda} R \xrightarrow{\mu} S \rightarrow 0$ be a sequence of A -modules. Assume that for every A -module P the sequence*

$$0 \rightarrow \mathrm{Hom}_A(S, P) \xrightarrow{\mu^*} \mathrm{Hom}_A(R, P) \xrightarrow{\lambda^*} \mathrm{Hom}_A(Q, P)$$

is exact. Then $Q \xrightarrow{\lambda} R \xrightarrow{\mu} S \rightarrow 0$ is exact.

Sketch. Surjectivity of μ follows by taking $P = S/\mathrm{im}(\mu)$ and considering the natural quotient map. To show $\ker(\mu) \subseteq \mathrm{im}(\lambda)$, take $P = \mathrm{coker}(\lambda) = R/\mathrm{im}(\lambda)$ and let $\varphi : R \rightarrow \mathrm{coker}(\lambda)$ be the quotient map. Then $\varphi \circ \lambda = 0$, so $\varphi \in \ker(\lambda^*) = \mathrm{im}(\mu^*)$, hence $\varphi = \psi \circ \mu$ for some $\psi : S \rightarrow \mathrm{coker}(\lambda)$. If $r \in \ker(\mu)$ then $\varphi(r) = \psi(\mu(r)) = 0$, so $r \in \ker(\varphi) = \mathrm{im}(\lambda)$. $\square$

Now suppose $M' \xrightarrow{f} M \xrightarrow{g} M'' \rightarrow 0$ is exact. Fix an A -module P and apply the left exact functor $\mathrm{Hom}_A(-, \mathrm{Hom}_A(N, P))$ to obtain an exact sequence

$$0 \rightarrow \mathrm{Hom}_A(M'', \mathrm{Hom}_A(N, P)) \rightarrow \mathrm{Hom}_A(M, \mathrm{Hom}_A(N, P)) \rightarrow \mathrm{Hom}_A(M', \mathrm{Hom}_A(N, P)).$$

Using the adjunction isomorphisms, this becomes

$$0 \rightarrow \mathrm{Hom}_A(M'' \otimes_A N, P) \rightarrow \mathrm{Hom}_A(M \otimes_A N, P) \rightarrow \mathrm{Hom}_A(M' \otimes_A N, P).$$

Since this holds for all P , the lemma implies that

$$M' \otimes_A N \xrightarrow{f \otimes 1} M \otimes_A N \xrightarrow{g \otimes 1} M'' \otimes_A N \rightarrow 0$$

is exact, i.e. $- \otimes_A N$ is right exact.

3 Localization of rings and modules

Intuition for example. Let A be a ring (pretend $A = k[x_1, \dots, x_n]/\mathfrak{a}$ = polynomial functions on $V(\mathfrak{a}) \subseteq k^n$). So $f \in A$ is basically $f : V(\mathfrak{a}) \rightarrow k$, let $V(f) = V(\langle f \rangle + \mathfrak{a})$ be the set where $f = 0$. We can look at

$$V(\mathfrak{a}) - V(f).$$

What are polynomial functions on this? Intuitively they look like fractions

$$a/f^n, \quad a \in A, n \geq 0,$$

i.e. we allow denominators which are powers of f (so we “ignore” what happens on $V(f)$).

We try to define

$$A_f = \left\{ \frac{a}{f^n} : a \in A, n \geq 0 \right\}.$$

Question: when is $\frac{a}{f^n} = \frac{b}{f^m}$? Note we can rewrite as

$$\frac{af^{m+k}}{f^{n+m+k}} = \frac{bf^{n+k}}{f^{n+m+k}}.$$

Therefore $\frac{a}{f^n} = \frac{b}{f^m}$ iff $\exists k$ s.t

$$f^k(af^m) = f^k(bf^n),$$

i.e.

$$f^k(af^m - bf^n) = 0.$$

Remark 3.1. In our hw look at 2.1, this describes exactly $A[x]/\langle xf - 1 \rangle$.

Definition 3.2. Let A be a ring, let $S \subseteq A$ be a multiplicatively closed set, define a relation $\sim$ on $A \times S$ as follows:

$$(a, s) \sim (b, t) \iff \exists u \in S \text{ s.t } u(ta - sb) = 0.$$

Example 3.3. if $f \in A$, the set

$$\{1, f, f^2, \dots\}$$

is multiplicatively closed.

If $\mathfrak{p} \subset A$ is prime, then $A - \mathfrak{p}$ is multiplicatively closed.

Proposition 3.4. $\sim$ is an equivalence relation.

Proof. Exercise. □

Definition 3.5. We define $S^{-1}A$ (also written as A_S) to be the set of equivalence classes of $\sim$. So an element of $S^{-1}A$ is a symbol

$$\frac{a}{s}.$$

Note that

$$\frac{a}{s} = \frac{b}{t} \iff \exists u \in S \text{ s.t } u(ta - sb) = 0.$$

Proposition 3.6. The operations

$$\begin{aligned} \frac{a}{s} + \frac{b}{t} &= \frac{ta + sb}{st} \\ \frac{a}{s} \cdot \frac{b}{t} &= \frac{ab}{st} \end{aligned}$$

are well defined, and they turn $S^{-1}A$ into a ring with

$$0 = \frac{0}{1}, 1 = \frac{1}{1}.$$

Notation. If $S = \{1, f, f^2, \dots\}$ then we write A_f instead of $S^{-1}A$. If $S = A - \mathfrak{p}$ we write $A_{\mathfrak{p}}$.

Example 3.7. In $\mathbb{Z}$ we write

$$\mathbb{Z}_f = \left\{ \frac{a}{f^n} \in \mathbb{Q} \mid a \in \mathbb{Z}, n \geq 0 \right\}.$$

And

$$\mathbb{Z}_{\langle p \rangle} = \left\{ \frac{a}{b} \in \mathbb{Q} \mid a, b \in \mathbb{Z}, p \nmid b \right\}.$$

Remark 3.8. There is a ring homomorphism $\varphi: A \rightarrow S^{-1}A$ by $\varphi(a) = a/1$. But note that φ is not necessarily injective.

For example take $(\mathbb{Z}/6\mathbb{Z})$ localized at $S = \{1, 3\}$. Then

$$S^{-1}(\mathbb{Z}/6\mathbb{Z}) = \{0/1, 1/1\} \cong \mathbb{Z}/2\mathbb{Z}.$$

This is because 3 becomes a unit, so the $\mathbb{Z}/3\mathbb{Z}$ -part disappears. Concretely:

$$2/1 = \frac{3 \cdot 2}{3} = 0,$$

hence $\{0, 2, 4\} \subseteq \ker \varphi$. Also

$$\frac{1}{3} = \frac{3 \cdot 1}{3 \cdot 3} = \frac{3}{3} = \frac{1}{1}.$$

Using CRT: $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. Inverting 3 makes $(1, 0)$ invertible, which forces the second factor to be 0, leaving $\mathbb{Z}/2\mathbb{Z}$.

Example 3.9. Let $A = \mathbb{C}[x, y]/\langle xy \rangle$, these are functions on the union of the two coordinate axes in $\mathbb{C}^2$. Localize at x :

$$A_x \cong \mathbb{C}[x, x^{-1}].$$

Therefore

$$\frac{y}{1} = \frac{xy}{x} = 0,$$

so $y \in \ker \varphi$ (intuitively: on U_x we are away from the y -axis, so y vanishes there).

We also have the UMP. Let $\iota: A \rightarrow S^{-1}A$ be $\iota(a) = a/1$. Then for any ring B and any ring map $g: A \rightarrow B$ such that $g(S) \subseteq B^\times$, there exists a unique ring map $\tilde{g}: S^{-1}A \rightarrow B$ with $\tilde{g} \circ \iota = g$. Concretely

$$\tilde{g}(a/s) = g(a) g(s)^{-1}.$$

The diagram is:

$$\begin{array}{ccc} A & \xrightarrow{\iota} & S^{-1}A \\ & \searrow g & \downarrow \tilde{g} \\ & & B \end{array}$$

Remark 3.10. $0 \in S \iff S^{-1}A$ is the zero ring.

Soon we will also localize modules; we will get a module over $S^{-1}A$. Define

$$S^{-1}M = \left\{ \frac{x}{s} \mid x \in M, s \in S \right\}$$

with

$$\frac{x}{s} = \frac{y}{t} \iff \exists u \in S \text{ s.t. } u(tx - sy) = 0.$$

3.1 Ideals in the Localization

We will especially look at extensions and contractions of ideals for the homomorphism

$$\varphi: A \rightarrow S^{-1}A.$$

Proposition 3.11. *If $\mathfrak{b} \subseteq S^{-1}A$ is an ideal, then*

$$\mathfrak{b}^{ce} = \mathfrak{b}.$$

Consequently all ideals of a localization are extensions of something.

Proof. $\mathfrak{b} \supseteq \mathfrak{b}^{ce}$ (always).

Conversely, let $\frac{a}{s} \in \mathfrak{b}$. Then $s(\frac{a}{s}) = \frac{a}{1}$ so $a \in \mathfrak{b}^c$, hence $\frac{a}{1} \in \mathfrak{b}^{ce}$ and so

$$\frac{1}{s} \cdot \frac{a}{1} = \frac{a}{s} \in \mathfrak{b}^{ce}.$$

□

Proposition 3.12. *Let $\mathfrak{a} \subseteq A$ be an ideal. Then*

$$\mathfrak{a}^e = \left\{ \frac{a}{s} \mid a \in \mathfrak{a}, s \in S \right\}$$

(i.e. $S^{-1}\mathfrak{a}$ inside $S^{-1}A$), and

$$\mathfrak{a}^{ec} = \{x \in A \mid \exists s \in S \text{ s.t. } sx \in \mathfrak{a}\}.$$

Example 3.13. Think of $A = \mathbb{Z}$ and $S = \{1, 6, 6^2, \dots\}$ therefore

$$S^{-1}A = \mathbb{Z}[1/6].$$

Then $(99\mathbb{Z})^{ec} = 11\mathbb{Z} = (11\mathbb{Z})^{ec}$. Indeed $x \in (99\mathbb{Z})^{ec}$ iff $\exists n$ such that $6^n x \in 99\mathbb{Z}$. For $n \geq 2$, 6^n contributes a factor 3^n , so the 3^2 part is automatic, and the condition becomes exactly $11 \mid x$.

Notice that elements of $\mathfrak{a}^e$ are of the form

$$\varphi(a_1) \cdot \left(\frac{b_1}{s_1}\right) + \dots + \varphi(a_k) \cdot \left(\frac{b_k}{s_k}\right)$$

with $a_i \in \mathfrak{a}$, $b_i \in A$, $s_i \in S$. Take a common denominator

$$s = s_1 \cdots s_k,$$

so $\frac{b_i}{s_i} = \frac{c_i}{s}$ with $c_i = b_i \prod_{j \neq i} s_j$, hence wlog we have

$$\frac{a_1}{1} \cdot \frac{c_1}{s} + \cdots + \frac{a_k}{1} \cdot \frac{c_k}{s}.$$

Corollary 3.14. *If $\mathfrak{a} = \langle x_1, \dots, x_r \rangle \subset A$ then*

$$\mathfrak{a}^e = \langle \frac{x_1}{1}, \dots, \frac{x_r}{1} \rangle.$$

Corollary 3.15. *If A is Noetherian, then so is $S^{-1}A$ (and every ideal of $S^{-1}A$) by the above corollary.*

Prime ideals in $S^{-1}A$

Theorem 3.16. 1. *If $\mathfrak{q} \subset S^{-1}A$ is prime, then $\mathfrak{p} := \mathfrak{q}^c \subset A$ is prime and*

$$\mathfrak{p} \cap S = \emptyset.$$

2. *If $\mathfrak{p} \subset A$ is prime and*

$$\mathfrak{p} \cap S = \emptyset,$$

then $\mathfrak{p}^e \subset S^{-1}A$ is also prime.

3. *Extension and contraction are mutually inverse bijections between the sets*

$$\{\mathfrak{p} \subset A \text{ prime} \mid \mathfrak{p} \cap S = \emptyset\} \leftrightarrow \{\mathfrak{q} \subset S^{-1}A \text{ prime}\}.$$

Remark 3.17. This clarifies: if $\mathfrak{a} \subset A$ and $\mathfrak{a} \cap S = \emptyset$, then $\mathfrak{a}^e \subsetneq S^{-1}A$. Hence $\mathfrak{a}^e$ is contained in some maximal ideal $\mathfrak{m} \subset S^{-1}A$. Contracting gives a prime $\mathfrak{p} = \mathfrak{m}^c \subset A$ with $\mathfrak{a} \subseteq \mathfrak{p}$ and $\mathfrak{p} \cap S = \emptyset$.

Proof. 1. We are given $\mathfrak{q}$ is prime, then $\mathfrak{q}^c$ is prime (contraction preserves primality). Also if $\exists s \in S \cap \mathfrak{q}^c$ then $s/1 \in \mathfrak{q}$, but $s/1$ is a unit in $S^{-1}A$ (inverse $1/s$), contradiction since $\mathfrak{q} \subsetneq S^{-1}A$.

2. Suppose we have two elements of $S^{-1}A$ whose product is in $\mathfrak{p}^e$. Call these elements a/s and b/t with $a, b \in A$ and $s, t \in S$. Then

$$\frac{ab}{st} \in \mathfrak{p}^e.$$

So $\exists u \in S$ such that $uab \in \mathfrak{p}$ (clearing denominators). Since $u \in S$ and $\mathfrak{p} \cap S = \emptyset$, we have $u \notin \mathfrak{p}$; by primality,

$$a \in \mathfrak{p} \text{ or } b \in \mathfrak{p}.$$

Hence a/s or b/t lies in $\mathfrak{p}^e$.

3. For any ideal $\mathfrak{q} \subset S^{-1}A$, we already proved $\mathfrak{q}^{ce} = \mathfrak{q}$. Now let $\mathfrak{p} \subset A$ be prime with $\mathfrak{p} \cap S = \emptyset$. Clearly $\mathfrak{p}^{ec} \supseteq \mathfrak{p}$. Take $a \in \mathfrak{p}^{ec}$, so $a/1 \in \mathfrak{p}^e$. Then $\exists s \in S$ such that $sa \in \mathfrak{p}$. But $s \notin \mathfrak{p}$, so $a \in \mathfrak{p}$ by primality. Thus $\mathfrak{p}^{ec} = \mathfrak{p}$.

□

Proposition 3.18. *There is a natural isomorphism of $S^{-1}A$ -modules*

$$S^{-1}M \cong_{\text{nat}} (S^{-1}A) \otimes_A M, \quad \frac{ax}{s} \mapsto \frac{a}{s} \otimes x.$$

Proof. Define a map

$$\Phi : (S^{-1}A) \otimes_A M \longrightarrow S^{-1}M, \quad \Phi\left(\frac{a}{s} \otimes x\right) = \frac{ax}{s}.$$

First observe that this map exists and is an A -module homomorphism (in fact, it is $S^{-1}A$ -linear once it is well-defined). Indeed, it is induced by the bilinear map

$$\beta : S^{-1}A \times M \rightarrow S^{-1}M, \quad \beta\left(\frac{a}{s}, x\right) = \frac{ax}{s}.$$

We check that β is well-defined. If

$$\frac{a}{s} = \frac{a'}{s'},$$

then by definition there exists $u \in S$ such that

$$u(s'a - sa') = 0.$$

Multiplying by $x \in M$ gives

$$u(s'ax - sa'x) = 0,$$

hence in $S^{-1}M$ we have

$$\frac{ax}{s} = \frac{a'x}{s'}.$$

Therefore β is well-defined, so it induces the A -linear map Φ .

Surjectivity is trivial: every element of $S^{-1}M$ has the form x/s , and

$$\Phi\left(\frac{1}{s} \otimes x\right) = \frac{x}{s}.$$

Reduction to simple tensors. A general element of $(S^{-1}A) \otimes_A M$ is a finite sum

$$t = \frac{a_1}{s_1} \otimes x_1 + \cdots + \frac{a_n}{s_n} \otimes x_n.$$

Passing to a common denominator $s = s_1 \cdots s_n$, we may rewrite

$$t = \frac{b_1}{s} \otimes x_1 + \cdots + \frac{b_n}{s} \otimes x_n = \frac{1}{s} \otimes \left(\sum_i b_i x_i\right).$$

Thus every tensor may be represented in the form $\frac{1}{s} \otimes y$.

Alternate approach (constructing the inverse). Define

$$\Psi : S^{-1}M \rightarrow (S^{-1}A) \otimes_A M, \quad \Psi\left(\frac{x}{s}\right) = \frac{1}{s} \otimes x.$$

We check that Ψ is well-defined. Suppose

$$\frac{x}{s} = \frac{x'}{s'}.$$

Then there exists $u \in S$ such that

$$u(s'x - sx') = 0,$$

i.e.

$$us'x = usx'.$$

Then in the tensor product,

$$\frac{1}{s} \otimes x = \frac{u}{us} \otimes x = \frac{1}{us} \otimes ux = \frac{1}{us'} \otimes ux' = \frac{1}{s'} \otimes x'.$$

Hence Ψ is well-defined.

Finally,

$$(\Phi \circ \Psi)\left(\frac{x}{s}\right) = \Phi\left(\frac{1}{s} \otimes x\right) = \frac{x}{s}$$

and

$$(\Psi \circ \Phi)\left(\frac{a}{s} \otimes x\right) = \Psi\left(\frac{ax}{s}\right) = \frac{1}{s} \otimes ax = \frac{a}{s} \otimes x.$$

Therefore Φ is an isomorphism, natural in M . □

Remark 3.19. Why is it called *localization*? Recall that there is an order-preserving bijection between primes:

$$\{\mathfrak{p} \subseteq A \mid \mathfrak{p} \cap S = \emptyset\} \longleftrightarrow \{\mathfrak{p} \subseteq S^{-1}A\}, \quad \mathfrak{p} \mapsto S^{-1}\mathfrak{p}.$$

In particular, if $S = A - \mathfrak{p}$, then

$$A_{\mathfrak{p}} = S^{-1}A$$

is a *local ring*. Indeed, its unique maximal ideal is

$$\mathfrak{m}_{\mathfrak{p}} = \left\{ \frac{a}{s} \in A_{\mathfrak{p}} \mid a \in \mathfrak{p}, s \notin \mathfrak{p} \right\}.$$

Moreover,

$$A_{\mathfrak{p}} - \mathfrak{m}_{\mathfrak{p}} = \left\{ \frac{u}{s} \mid u \notin \mathfrak{p}, s \notin \mathfrak{p} \right\}$$

is exactly the set of units of $A_{\mathfrak{p}}$.

A quick example: let $A = k[x, y, z]$ and $\mathfrak{p} = \langle x, y \rangle$. Then

$$V(\mathfrak{p}) = \text{“the } z\text{-axis”}.$$

Also $A/\mathfrak{p} \simeq k[z]$, via

$$f(x, y, z) + \mathfrak{p} \longleftrightarrow f(0, 0, z).$$

Thus

$$A_{\mathfrak{p}} = \left\{ \frac{f(x, y, z)}{s(x, y, z)} \mid s \notin \mathfrak{p} \right\}.$$

The maximal ideal consists of fractions whose numerator vanishes on the z -axis:

$$\mathfrak{m}_{\mathfrak{p}} = \left\{ \frac{f(x, y, z)}{s(x, y, z)} \mid f \in \mathfrak{p} \right\}.$$

Moreover one checks that

$$\frac{A_{\mathfrak{p}}}{\mathfrak{m}_{\mathfrak{p}}} \cong \text{Frac}(A/\mathfrak{p}) \cong k(z),$$

the field of rational functions in z .

Therefore we can define the *localization functor*

$$M \longmapsto S^{-1}M,$$

which is secretly the functor

$$M \longmapsto (S^{-1}A) \otimes_A M.$$

Given a homomorphism $M \xrightarrow{f} N$, we define

$$S^{-1}f : S^{-1}M \rightarrow S^{-1}N, \quad (S^{-1}f)\left(\frac{x}{s}\right) = \frac{f(x)}{s}.$$

Theorem 3.20. *The localization functor is exact. Equivalently, $S^{-1}A$ is a flat A -algebra.*

Proof. Let

$$M \xrightarrow{f} N \xrightarrow{g} P$$

be exact at N , i.e. $\text{im}(f) = \ker(g)$.

We claim that

$$S^{-1}M \xrightarrow{S^{-1}f} S^{-1}N \xrightarrow{S^{-1}g} S^{-1}P$$

is exact at $S^{-1}N$.

Trivially $\text{im}(S^{-1}f) \subseteq \ker(S^{-1}g)$ since $gf = 0$.

Now let $y/s \in \ker(S^{-1}g)$. Then

$$(S^{-1}g)\left(\frac{y}{s}\right) = \frac{g(y)}{s} = 0$$

in $S^{-1}P$, hence there exists $u \in S$ such that

$$ug(y) = 0, \quad \text{i.e.} \quad g(uy) = 0.$$

Thus $uy \in \ker(g) = \text{im}(f)$, so $uy = f(x)$ for some $x \in M$. Therefore

$$\frac{y}{s} = \frac{uy}{us} = \frac{f(x)}{us} = (S^{-1}f)\left(\frac{x}{us}\right),$$

proving $\ker(S^{-1}g) \subseteq \text{im}(S^{-1}f)$. □

immediate consequences of exactness.

Let $M \xrightarrow{f} N$ be any homomorphism. Then localization preserves kernels, cokernels, and images:

$$S^{-1}(\ker f) = \ker(S^{-1}f), \quad S^{-1}(\text{coker } f) = \text{coker}(S^{-1}f), \quad S^{-1}(\text{im } f) = \text{im}(S^{-1}f).$$

If $N \subseteq M$, then

$$S^{-1}(M/N) \cong (S^{-1}M)/(S^{-1}N),$$

since localizing the short exact sequence

$$0 \rightarrow N \rightarrow M \rightarrow M/N \rightarrow 0$$

remains exact.

Similarly,

$$S^{-1}(M_1 \oplus M_2) \cong S^{-1}M_1 \oplus S^{-1}M_2.$$

The direct sum maps fit into the standard diagram:

$$\begin{array}{ccccc} M_1 & \xrightarrow{i_1} & M_1 \oplus M_2 & \xleftarrow{i_2} & M_2 \\ & \searrow p_1 & & \swarrow p_2 & \\ & & & & \end{array}$$

with relations

$$p_1 i_1 = 1, \quad p_2 i_2 = 1, \quad p_1 i_2 = 0, \quad p_2 i_1 = 0, \quad i_1 p_1 + i_2 p_2 = 1_{M_1 \oplus M_2}.$$

More interestingly, if $N_1, N_2 \subseteq M$, then

$$S^{-1}(N_1 \cap N_2) = S^{-1}N_1 \cap S^{-1}N_2, \quad S^{-1}(N_1 + N_2) = S^{-1}N_1 + S^{-1}N_2.$$

Indeed we use the exact sequence

$$0 \rightarrow N_1 \cap N_2 \rightarrow N_1 \oplus N_2 \xrightarrow{(y_1, y_2) \mapsto y_1 - y_2} N_1 + N_2 \rightarrow 0,$$

where $z \mapsto (z, z)$ is the inclusion.

Proposition 3.21. *Let M, N be A -modules. Then $\text{Hom}_A(M, N)$ is an A -module, and there is a natural map*

$$S^{-1} \text{Hom}_A(M, N) \xrightarrow{\varphi} \text{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N),$$

given by

$$\frac{f}{s} \longmapsto \left(\frac{x}{t} \mapsto \frac{f(x)}{st} \right).$$

Moreover:

- If M is finitely generated, then φ is injective.
- If M is finitely presented, then φ is bijective.

(Sketch: finite generation controls kernels, finite presentation controls surjectivity.)

Local vs global properties

We know: for $x \in M$, when is $x/1 = 0$ in $S^{-1}M$? Precisely when

$$\exists s \in S \text{ such that } sx = 0, \quad \text{i.e.} \quad \text{Ann}(x) \cap S \neq \emptyset.$$

Fix $x \in M$. Which primes $\mathfrak{p}$ satisfy $x/1 \neq 0$ in $M_{\mathfrak{p}}$? These are exactly

$$\{\mathfrak{p} \mid \text{Ann}(x) \subseteq \mathfrak{p}\}.$$

Example 3.22. Let $A = \mathbb{Z}$ and $M = \mathbb{Z}/20\mathbb{Z}$. Then

$$\text{Ann}([1]) = 20\mathbb{Z} \subseteq 2\mathbb{Z}, 5\mathbb{Z}, \quad \text{Ann}([4]) = 5\mathbb{Z}.$$

Hence

$$M_{(2)} \cong \mathbb{Z}/4\mathbb{Z}, \quad M_{(5)} \cong \mathbb{Z}/5\mathbb{Z}.$$

Example 3.23. Let $A = \mathbb{C}[x, y]$ and maximal ideals

$$\mathfrak{m}_{a,b} = \langle x - a, y - b \rangle.$$

If

$$M = A/\langle x^2 + y^2 - 1 \rangle,$$

then $[1]$ lives at $\mathfrak{m}_{a,b}$ iff

$$x^2 + y^2 - 1 \in \mathfrak{m}_{a,b} \iff a^2 + b^2 = 1.$$

Example 3.24. Consider $[x] \in k[x, y]/\langle x, y \rangle$ as a $k[x, y]$ -module element. Then

$$\text{Ann}([x]) = \langle y \rangle,$$

so $[x]$ lives exactly along the x -axis.

Local-global principles

Proposition 3.25. Let M be an A -module. Then the following are equivalent:

- $M = 0$.
- For all primes $\mathfrak{p}$, $M_{\mathfrak{p}} = 0$.
- For all maximal ideals $\mathfrak{m}$, $M_{\mathfrak{m}} = 0$.

Equivalently, for $x \in M$, the following are equivalent:

- $x = 0$.
- For all primes $\mathfrak{p}$, $x/1 = 0$ in $M_{\mathfrak{p}}$.
- For all maximal ideals $\mathfrak{m}$, $x/1 = 0$ in $M_{\mathfrak{m}}$.

Proof. (1) $\implies$ (2) $\implies$ (3) is trivial.

For (3) $\implies$ (1): assume $M_{\mathfrak{m}} = 0$ for all maximal ideals. Let $x \in M$ with $x \neq 0$. Then $\text{Ann}(x)$ is a proper ideal, hence contained in some maximal ideal $\mathfrak{m}$:

$$\text{Ann}(x) \subseteq \mathfrak{m}.$$

But $x/1 = 0$ in $M_{\mathfrak{m}}$ means there exists $s \notin \mathfrak{m}$ such that $sx = 0$. Thus $s \in \text{Ann}(x) \subseteq \mathfrak{m}$, contradiction. Therefore $x = 0$, so $M = 0$. $\square$

Consequences.

Proposition 3.26. *A homomorphism $f : M \rightarrow N$ is injective (resp. surjective, resp. bijective) iff for all primes $\mathfrak{p}$ (equivalently, all maximal $\mathfrak{m}$),*

$$f_{\mathfrak{p}} : M_{\mathfrak{p}} \rightarrow N_{\mathfrak{p}}$$

is injective (resp. surjective, resp. bijective).

Proof. Apply localization to the exact sequences involving $\ker f$ and $\text{coker } f$. $\square$

Proposition 3.27. *If $N \subseteq M$, then*

$$N = M \iff N_{\mathfrak{p}} = M_{\mathfrak{p}} \forall \mathfrak{p} \iff N_{\mathfrak{m}} = M_{\mathfrak{m}} \forall \mathfrak{m}.$$

Proof. Apply the previous proposition to M/N . $\square$

Proposition 3.28. *If $x \in M$ satisfies $x/1 = 0$ in $M_{\mathfrak{m}}$, then the image of x in*

$$M/\mathfrak{m}M \cong (A/\mathfrak{m}) \otimes_A M$$

is zero.

Proof. If $x/1 = 0$ in $M_{\mathfrak{m}}$, then $\exists s \notin \mathfrak{m}$ with $sx = 0$. In $M/\mathfrak{m}M$, we have $s[x] = 0$, but also $\mathfrak{m}[x] = 0$. Since $s \notin \mathfrak{m}$, the ideal $\mathfrak{m} + \langle s \rangle = A$, hence $[x] = 0$. $\square$

Theorem 3.29. *Suppose M is a finitely generated A -module. Then*

$$M = 0 \iff \forall \mathfrak{m}, M/\mathfrak{m}M = 0.$$

Proof. Assume $M/\mathfrak{m}M = 0$ for all maximal ideals $\mathfrak{m}$.

Localizing at $\mathfrak{m}$, we get

$$M_{\mathfrak{m}}/\mathfrak{m}_{\mathfrak{m}}M_{\mathfrak{m}} \cong (M/\mathfrak{m}M)_{\mathfrak{m}} = 0.$$

Since $M_{\mathfrak{m}}$ is finitely generated over the local ring $A_{\mathfrak{m}}$, Nakayama's lemma implies

$$M_{\mathfrak{m}} = 0 \quad \forall \mathfrak{m}.$$

By the local-global principle, $M = 0$. $\square$

Remark 3.30. Finite generation is essential. For example, if $A = \mathbb{Z}$ and $M = \mathbb{Q}/\mathbb{Z}$, then multiplication by n is surjective for all n , hence

$$(\mathbb{Q}/\mathbb{Z})/p(\mathbb{Q}/\mathbb{Z}) = 0 \quad \forall p,$$

but $\mathbb{Q}/\mathbb{Z} \neq 0$.

3.2 The spectrum of a ring.

Definition 3.31. Let A be a ring. The *spectrum* of A is the set of all prime ideals:

$$\text{Spec}(A) = \{\mathfrak{p} \subseteq A \mid \mathfrak{p} \text{ is a prime ideal}\}.$$

The *maximal spectrum* is the subset of maximal ideals:

$$\text{maxSpec}(A) = \{\mathfrak{m} \subseteq A \mid \mathfrak{m} \text{ is maximal}\}.$$

Example 3.32. Let $A = \mathbb{C}[x, y]$. Then $\text{Spec}(A)$ consists of:

1. Maximal ideals

$$\mathfrak{m}_{a,b} = \langle x - a, y - b \rangle,$$

which correspond to points $(a, b) \in \mathbb{C}^2$.

2. Prime ideals of height 1, generated by irreducible polynomials:

$$\langle f(x, y) \rangle, \quad f \text{ irreducible},$$

which correspond to irreducible affine curves.

3. The zero ideal (0) , which corresponds to the generic point of A^2 .

Let $X = \text{Spec}(A)$. We now define a topology on X , called the *Zariski topology*.

Closed sets

Definition 3.33. If $\mathfrak{a} \subseteq A$ is an ideal, define its *vanishing set*:

$$V(\mathfrak{a}) = \{\mathfrak{p} \in X \mid \mathfrak{a} \subseteq \mathfrak{p}\}.$$

More generally, for any subset $\Sigma \subseteq A$, define

$$V(\Sigma) = \{\mathfrak{p} \in \text{Spec}(A) \mid \Sigma \subseteq \mathfrak{p}\}.$$

Equivalently,

$$V(\Sigma) = V(\langle \Sigma \rangle).$$

Basic open sets

Definition 3.34. For $f \in A$, define the *distinguished open set*

$$U_f = \{\mathfrak{p} \in X \mid f \notin \mathfrak{p}\}.$$

Then

$$U_f = X \setminus V(\langle f \rangle).$$

Intuitively, U_f is the set of primes where f does *not* vanish.

Proposition 3.35. The sets U_f form a basis of open sets, and

$$U_f \cap U_g = U_{fg}.$$

Moreover, for any $\Sigma \subseteq A$,

$$X \setminus V(\Sigma) = \bigcup_{f \in \Sigma} U_f.$$

The closed sets form a topology

Proposition 3.36. The sets $V(\mathfrak{a})$ satisfy the axioms for closed sets:

1.

$$V(0) = X, \quad V(1) = \emptyset.$$

2. For ideals $\mathfrak{a}_i$,

$$\bigcap_{i \in I} V(\mathfrak{a}_i) = V\left(\sum_{i \in I} \mathfrak{a}_i\right).$$

3. For ideals $\mathfrak{a}, \mathfrak{b}$,

$$V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}).$$

Useful special cases

•

$$V(\mathfrak{a}) \cap V(\mathfrak{b}) = V(\mathfrak{a} + \mathfrak{b}).$$

•

$$V(f_1 \cdots f_n) = V(f_1) \cup \cdots \cup V(f_n).$$

• If $\mathfrak{a} = \langle f_1, \dots, f_k \rangle$, then

$$V(\mathfrak{a}) = X \setminus (U_{f_1} \cup \cdots \cup U_{f_k}).$$

Restriction via localization

Proposition 3.37. There are natural homeomorphisms:

$$\mathrm{Spec}(A/\mathfrak{a}) \cong V(\mathfrak{a}), \quad \mathrm{Spec}(A_f) \cong U_f.$$

Radicals and closed sets**Remark 3.38.** For any ideal $\mathfrak{a}$,

$$V(\mathfrak{a}) = V(\text{rad } \mathfrak{a}).$$

Thus,

$$\mathfrak{a} \subseteq \mathfrak{b} \implies V(\mathfrak{a}) \supseteq V(\mathfrak{b}).$$

Proposition 3.39.

$$V(\mathfrak{a}) \supseteq V(\mathfrak{b}) \iff \text{rad } \mathfrak{a} \subseteq \text{rad } \mathfrak{b}.$$

Corollary 3.40.

$$V(\mathfrak{a}) = V(\mathfrak{b}) \iff \text{rad } \mathfrak{a} = \text{rad } \mathfrak{b}.$$

Thus closed sets in $\text{Spec}(A)$ correspond bijectively to radical ideals.

The Zariski topology is not Hausdorff**Example 3.41.** For $A = \mathbb{C}[x]$,

$$\text{Spec}(A) = \{(0)\} \cup \{\langle x - a \rangle \mid a \in \mathbb{C}\}.$$

Every nonempty open set contains (0) , so the space is not Hausdorff.

Example 3.42. For $A = \mathbb{Z}$,

$$\text{Spec}(\mathbb{Z}) = \{(0)\} \cup \{(p) \mid p \text{ prime}\}.$$

The basic opens are

$$U_n = \text{Spec}(\mathbb{Z}) \setminus \{(p) \mid p \mid n\}.$$

So U_n removes only finitely many primes. Every nonempty open set contains (0) .

Quasi-compactness**Proposition 3.43.** $\text{Spec}(A)$ is quasi-compact.*Proof.* Suppose

$$\text{Spec}(A) = \bigcup_{i \in I} U_{f_i}.$$

Then

$$\emptyset = \bigcap_{i \in I} V(f_i) = V(\langle f_i \mid i \in I \rangle).$$

Thus

$$1 \in \langle f_i \mid i \in I \rangle,$$

so finitely many $f_{i_1}, \dots, f_{i_k}$ generate the unit ideal, hence

$$\text{Spec}(A) = U_{f_{i_1}} \cup \dots \cup U_{f_{i_k}}.$$

□

Corollary 3.44. *If A is a Noetherian ring, then all subsets of $\operatorname{Spec} A$ are quasi-compact.*

3.3 Local vanishing and quasi-compactness

Proposition 3.45. *Let A be a ring, M an A -module, $x \in M$, and $\mathfrak{p} \in \operatorname{Spec} A$.*

1. *If $x/1 = 0$ in $M_{\mathfrak{p}}$, then there exists $f \notin \mathfrak{p}$ such that $x/1 = 0$ in M_f .*
2. *If $\operatorname{Spec}(A) = U_{f_1} \cup \cdots \cup U_{f_n}$ and $x/1 = 0$ in each M_{f_i} , then $x = 0$ in M . In particular, the same conclusion holds for any open cover (because $\operatorname{Spec} A$ is quasi-compact, so there is a finite subcover).*

Proof. 1. $x/1 = 0$ in $M_{\mathfrak{p}}$ means: there exists $s \in A \setminus \mathfrak{p}$ such that $sx = 0$ in M . Take $f = s$. Then in M_f we have

$$x/1 = (fx)/f = 0/f = 0,$$

so $x/1 = 0$ in M_f .

2. Since $\operatorname{Spec} A = \bigcup_{i=1}^n U_{f_i}$, the ideal $(f_1, \dots, f_n) = A$, so there exist $r_1, \dots, r_n \in A$ with

$$1 = r_1 f_1 + \cdots + r_n f_n.$$

For each i , the condition $x/1 = 0$ in M_{f_i} is equivalent to $\exists N_i \geq 0$ such that $f_i^{N_i} x = 0$ in M . Let $N = \max_i N_i$, so $f_i^N x = 0$ for all i .

Now expand

$$x = 1^{nN} x = (r_1 f_1 + \cdots + r_n f_n)^{nN} x.$$

Every monomial in the expansion is divisible by f_i^N for some i (pigeonhole principle: nN factors distributed among n indices), hence each monomial kills x . Therefore the whole sum is 0, so $x = 0$.

□

Remark 3.46. Fix $x \in M$. Consider the set

$$Z(x) = \{\mathfrak{p} \in \operatorname{Spec} A \mid x/1 = 0 \text{ in } M_{\mathfrak{p}}\}.$$

By Proposition 3.45(1), $Z(x)$ is open.

Moreover one checks:

$$Z(x) = \bigcup_{s \in \operatorname{Ann}(x)} U_s = U_{\operatorname{Ann}(x)},$$

and hence its complement is closed:

$$\{\mathfrak{p} \mid x/1 \neq 0\} = \operatorname{Spec} A \setminus Z(x) = V(\operatorname{Ann}(x)).$$

(So the “nonvanishing locus” of x is $V(\operatorname{Ann}(x))$, and the vanishing locus is $U_{\operatorname{Ann}(x)}$.)

Proposition 3.47. *Suppose M is finitely generated and $M_{\mathfrak{p}} = 0$ for some $\mathfrak{p} \in \text{Spec } A$. Then there exists $g \notin \mathfrak{p}$ such that $M_g = 0$ (equivalently, U_g is an open neighborhood of $\mathfrak{p}$ on which M localizes to zero).*

Proof. Choose generators $M = \langle x_1, \dots, x_n \rangle$. The assumption $M_{\mathfrak{p}} = 0$ means $x_i/1 = 0$ in $M_{\mathfrak{p}}$ for each i . By Proposition 3.45(1), for each i there exists $f_i \notin \mathfrak{p}$ such that $x_i/1 = 0$ in M_{f_i} . Set $g = f_1 \cdots f_n \notin \mathfrak{p}$. Then $M_g = (M_{f_i})_g$, so each $x_i/1$ maps to 0 in M_g . Since the images of the x_i generate M_g , we get $M_g = 0$. $\square$

Local freeness spreads out (Noetherian / finitely presented case).

Theorem 3.48. *Assume either:*

- *A is Noetherian and M is finitely generated, or*
- *M is finitely presented (no hypothesis on A).*

If for some $\mathfrak{p} \in \text{Spec } A$ the localized module $M_{\mathfrak{p}}$ is free of rank n , i.e. $M_{\mathfrak{p}} \cong A_{\mathfrak{p}}^n$, then there exists $f \notin \mathfrak{p}$ such that

$$M_f \cong A_f^n.$$

Proof. Pick elements $x_1, \dots, x_n \in M$ such that their images form a basis of $M_{\mathfrak{p}}$ over $A_{\mathfrak{p}}$.

Define $\varphi : A^n \rightarrow M$ by $\varphi(e_i) = x_i$. After localizing at $\mathfrak{p}$, the map

$$\varphi_{\mathfrak{p}} : (A^n)_{\mathfrak{p}} \rightarrow M_{\mathfrak{p}}$$

is an isomorphism by construction (the $x_i/1$ are a basis).

Let $K = \ker(\varphi)$ and $C = \text{coker}(\varphi)$. We have an exact sequence

$$0 \rightarrow K \rightarrow A^n \xrightarrow{\varphi} M \rightarrow C \rightarrow 0.$$

Localizing at $\mathfrak{p}$ preserves exactness, so

$$0 \rightarrow K_{\mathfrak{p}} \rightarrow (A^n)_{\mathfrak{p}} \xrightarrow{\varphi_{\mathfrak{p}}} M_{\mathfrak{p}} \rightarrow C_{\mathfrak{p}} \rightarrow 0.$$

Since $\varphi_{\mathfrak{p}}$ is an isomorphism, we get $K_{\mathfrak{p}} = 0$ and $C_{\mathfrak{p}} = 0$.

Now use Proposition 3.47 twice:

- C is finitely generated (e.g. quotient of M), so $C_{\mathfrak{p}} = 0$ implies $\exists h \notin \mathfrak{p}$ with $C_h = 0$.
- K is finitely generated in the two cases above: - if A is Noetherian, $K \subseteq A^n$ is a submodule of a finitely generated module, hence f.g.; - if M is finitely presented, then by definition K is f.g. Thus $K_{\mathfrak{p}} = 0$ implies $\exists g \notin \mathfrak{p}$ with $K_g = 0$.

Let $f = gh \notin \mathfrak{p}$. Then $K_f = 0$ and $C_f = 0$, and localizing the exact sequence at f gives

$$0 \rightarrow 0 \rightarrow A_f^n \xrightarrow{\varphi_f} M_f \rightarrow 0 \rightarrow 0,$$

so φ_f is an isomorphism. Hence $M_f \cong A_f^n$. $\square$

Remark 3.49 (Culture). Over a Noetherian ring, for finitely generated modules one has the chain of implications

$$\text{projective} \implies \text{locally free} \implies \text{flat},$$

and in fact for finitely generated modules over Noetherian rings,

$$M \text{ is projective} \iff M \text{ is locally free} \iff M \text{ is flat.}$$

(Precise hypotheses matter; e.g. “flat $\Rightarrow$ projective” needs finite presentation.)

3.4 Irreducible subsets of Spec A

Definition 3.50. Let X be a topological space. A nonempty subset $Y \subseteq X$ is *irreducible* if whenever $Y \subseteq Z_1 \cup Z_2$ with Z_1, Z_2 closed in X , then $Y \subseteq Z_1$ or $Y \subseteq Z_2$. Equivalently, Y (with the subspace topology) cannot be written as the union of two proper closed subsets.

Remark 3.51. Y is irreducible iff $\overline{Y}$ is irreducible. Also, if X is Hausdorff, then X is irreducible iff X is a single point (because distinct points admit disjoint open neighborhoods).

Proposition 3.52. Let $\mathfrak{a} \subseteq A$ be an ideal. The closed set $V(\mathfrak{a}) \subseteq \text{Spec } A$ is irreducible iff $\sqrt{\mathfrak{a}}$ is a prime ideal. In that case,

$$V(\mathfrak{a}) = V(\sqrt{\mathfrak{a}}) = V(\mathfrak{p}) \text{ for } \mathfrak{p} = \sqrt{\mathfrak{a}}.$$

Thus the irreducible closed subsets of Spec A are exactly the sets $V(\mathfrak{p})$ for prime $\mathfrak{p}$.

Proof. ($\Rightarrow$) Assume $V(\mathfrak{a})$ is irreducible. Set $\mathfrak{a}' = \sqrt{\mathfrak{a}}$, so $V(\mathfrak{a}) = V(\mathfrak{a}')$. We show $\mathfrak{a}'$ is prime.

Suppose $fg \in \mathfrak{a}'$. Then

$$V(\mathfrak{a}') \subseteq V(\mathfrak{a}' + (f)) \cup V(\mathfrak{a}' + (g)),$$

because a prime containing $\mathfrak{a}'$ must contain f or g if it contains fg . By irreducibility, $V(\mathfrak{a}') \subseteq V(\mathfrak{a}' + (f))$ or $V(\mathfrak{a}') \subseteq V(\mathfrak{a}' + (g))$. Say the first holds. Then every prime $\mathfrak{p} \supseteq \mathfrak{a}'$ contains f . Intersecting over all primes containing $\mathfrak{a}'$ gives

$$f \in \bigcap_{\mathfrak{p} \supseteq \mathfrak{a}'} \mathfrak{p} = \sqrt{\mathfrak{a}'} = \mathfrak{a}',$$

since $\mathfrak{a}'$ is radical. Hence $f \in \mathfrak{a}'$, proving primeness.

($\Leftarrow$) If $\mathfrak{p}$ is prime, then $V(\mathfrak{p})$ is irreducible: if $V(\mathfrak{p}) \subseteq V(\mathfrak{b}_1) \cup V(\mathfrak{b}_2) = V(\mathfrak{b}_1\mathfrak{b}_2)$, then $\mathfrak{b}_1\mathfrak{b}_2 \subseteq \mathfrak{p}$, so by primeness $\mathfrak{b}_1 \subseteq \mathfrak{p}$ or $\mathfrak{b}_2 \subseteq \mathfrak{p}$, hence $V(\mathfrak{p}) \subseteq V(\mathfrak{b}_1)$ or $V(\mathfrak{p}) \subseteq V(\mathfrak{b}_2)$. $\square$

Remark 3.53 (Generic points). For any prime $\mathfrak{p}$, one has

$$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}).$$

So irreducible closed subsets of Spec A have a natural “generic point”.

Remark 3.54 (Components and minimal primes (Noetherian case)). In a Noetherian topological space, every closed set is a finite union of irreducible closed subsets. In particular, if A is Noetherian then Spec A has finitely many irreducible components, and they are exactly

$$V(\mathfrak{p}_1) \cup \cdots \cup V(\mathfrak{p}_r),$$

where $\mathfrak{p}_i$ range over the *minimal primes* of A . Equivalently: every prime ideal contains some minimal prime.

Remark 3.55. If $\mathfrak{a} \subsetneq A$, then the canonical map induces a homeomorphism

$$V(\mathfrak{a}) \cong \text{Spec}(A/\mathfrak{a}),$$

and the irreducible components of $V(\mathfrak{a})$ correspond to primes minimal over $\mathfrak{a}$.

Goal (corrected): If A is Noetherian, then for every ideal $\mathfrak{a} \subseteq A$ there exist only finitely many *minimal primes over $\mathfrak{a}$* , say $\mathfrak{p}_1, \dots, \mathfrak{p}_r$, and

$$V(\mathfrak{a}) = V(\mathfrak{p}_1) \cup \dots \cup V(\mathfrak{p}_r).$$

In particular (taking $\mathfrak{a} = (0)$), A has only finitely many *minimal* prime ideals.

This is equivalent to the case $\mathfrak{a} = (0)$ because

$$V(\mathfrak{a}) = \text{Spec}(A/\mathfrak{a})$$

and minimal primes over $\mathfrak{a}$ correspond to minimal primes of $A/\mathfrak{a}$.

More generally, for an A -module M (assume f.g. to simplify), the support is

$$\text{Supp } M = \{\mathfrak{p} \in \text{Spec } A \mid M_{\mathfrak{p}} \neq 0\}.$$

If M is finitely generated, then

$$\text{Supp } M = V(\text{Ann } M).$$

(For non-f.g. modules this can fail.)

Example 3.56 (Support over $\mathbb{Z}$). Over $A = \mathbb{Z}$ we have $\text{Spec } \mathbb{Z} = \{(0)\} \cup \{(p) \mid p \text{ prime}\}$.

- $M = \mathbb{Z} \oplus \mathbb{Z}/45 \oplus \mathbb{Z}/15$ has $\text{Supp } M = \text{Spec } \mathbb{Z}$, since $(\mathbb{Z})_{(p)} \neq 0$ for every prime (p) and also at (0) .
- $M = \mathbb{Z}/45 \oplus \mathbb{Z}/15$ has

$$\text{Supp } M = \{(3), (5)\} = V(15),$$

because localizing at (p) kills a torsion module unless p divides the torsion.

- $M = \bigoplus_{n \geq 2} \mathbb{Z}/n\mathbb{Z}$ has

$$\text{Supp } M = \{(p) \mid p \text{ prime}\} = \text{Spec } \mathbb{Z} \setminus \{(0)\},$$

since for each prime p the summand $\mathbb{Z}/p\mathbb{Z}$ survives after localizing at (p) , while localization at (0) gives $\mathbb{Q} \otimes_{\mathbb{Z}} M = 0$ (torsion).

- $M = \bigoplus_{k \geq 1} \mathbb{Z}/2^k\mathbb{Z}$ has $\text{Supp } M = \{(2)\}$.

A special case: if $M = A/\mathfrak{a}$, then

$$\text{Supp}(A/\mathfrak{a}) = V(\mathfrak{a}).$$

Algebraic proof of finiteness of minimal primes over $\mathfrak{a}$. We prove: for every $\mathfrak{a} \subseteq A$ there are finitely many minimal primes over $\mathfrak{a}$.

Assume not. Consider the set of “bad” ideals

$$\mathcal{B} = \{\mathfrak{a} \subseteq A \mid \mathfrak{a} \text{ has infinitely many minimal primes above it}\}.$$

Since A is Noetherian, $\mathcal{B}$ has a maximal element (w.r.t. inclusion); call it $\mathfrak{b}$.

Claim: $\mathfrak{b}$ is not prime. Indeed, if $\mathfrak{b}$ were prime then it would be the unique minimal prime over itself, contradicting that $\mathfrak{b} \in \mathcal{B}$.

Thus $\mathfrak{b}$ is not prime, so there exist $f, g \notin \mathfrak{b}$ with $fg \in \mathfrak{b}$.

Now let $\mathfrak{p}$ be *minimal* over $\mathfrak{b}$. Then $fg \in \mathfrak{p}$, hence $f \in \mathfrak{p}$ or $g \in \mathfrak{p}$. Therefore

$$\mathfrak{p} \supseteq \mathfrak{b} + (f) \quad \text{or} \quad \mathfrak{p} \supseteq \mathfrak{b} + (g).$$

Since $f, g \notin \mathfrak{b}$, both $\mathfrak{b} + (f)$ and $\mathfrak{b} + (g)$ strictly contain $\mathfrak{b}$, so by maximality of $\mathfrak{b}$ they are not bad: each has only finitely many minimal primes over it.

Let $\{q_1, \dots, q_s\}$ be the minimal primes over $\mathfrak{b} + (f)$ and $\{q'_1, \dots, q'_t\}$ the minimal primes over $\mathfrak{b} + (g)$.

Key point: if $\mathfrak{p}$ is minimal over $\mathfrak{b}$ and $\mathfrak{p} \supseteq \mathfrak{b} + (f)$, then $\mathfrak{p}$ is minimal over $\mathfrak{b} + (f)$ as well: if $\mathfrak{q} \subsetneq \mathfrak{p}$ were prime with $\mathfrak{b} + (f) \subseteq \mathfrak{q}$, then $\mathfrak{b} \subseteq \mathfrak{q} \subsetneq \mathfrak{p}$, contradicting minimality of $\mathfrak{p}$ over $\mathfrak{b}$. Hence $\mathfrak{p}$ is one of the q_i 's; similarly in the g -case.

So the set of minimal primes over $\mathfrak{b}$ is contained in the finite set

$$\{q_1, \dots, q_s, q'_1, \dots, q'_t\},$$

contradiction. □

Geometric proof. Let $X = \text{Spec } A$. Since A is Noetherian, the ideals of A satisfy ACC, hence the closed sets $V(\mathfrak{a})$ satisfy DCC (equivalently: X is a Noetherian topological space).

General fact: if X is a topological space satisfying DCC on closed subsets, then every closed subset $Y \subseteq X$ is a finite union of irreducible closed subsets.

Proof. Suppose not. Among the closed subsets which are not a finite union of irreducibles, choose a minimal one Y . Then Y is not irreducible, so $Y = Z_1 \cup Z_2$ with Z_i proper closed subsets of Y . Set $W_i = Y \cap Z_i \subsetneq Y$, so $Y = W_1 \cup W_2$ with W_i closed and proper. By minimality of Y , each W_i is a finite union of irreducibles, hence so is Y — contradiction. □ □

Therefore, for A Noetherian and any ideal $\mathfrak{a}$,

$$V(\mathfrak{a}) = V(\mathfrak{p}_1) \cup \dots \cup V(\mathfrak{p}_r)$$

where $\mathfrak{p}_i$ are the minimal primes over $\mathfrak{a}$. Equivalently,

$$\sqrt{\mathfrak{a}} = \mathfrak{p}_1 \cap \dots \cap \mathfrak{p}_r.$$

But what can we say about $\mathfrak{a}$ itself? From the minimal primes you recover only $\sqrt{\mathfrak{a}}$ (i.e. the

reduced closed set). In general $\mathfrak{a}$ need not equal $\sqrt{\mathfrak{a}}$, and the extra information is precisely the “nilpotent/thickening” data. In a Noetherian ring this extra structure is governed by *primary decomposition*: one can write

$$\mathfrak{a} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$$

with each $\mathfrak{q}_i$ $\mathfrak{p}_i$ -primary and $\sqrt{\mathfrak{q}_i} = \mathfrak{p}_i$. The primes $\mathfrak{p}_i$ are (exactly) the associated primes of $A/\mathfrak{a}$; the minimal primes over $\mathfrak{a}$ correspond to the *minimal* elements among these associated primes (the “isolated components”), while the rest are *embedded primes*. So knowing only the minimal primes is not enough to recover $\mathfrak{a}$.

Also recall:

- Every element lying in a minimal prime of A is a zero divisor.
- The set of zero divisors of A is a union of (not necessarily minimal) primes; in the Noetherian case, it is the union of the associated primes of A .

Example 3.57. Let $R = k[x, y]/(x^2, xy)$ and write $\bar{x}, \bar{y}$ for the residue classes. Note that $(x^2, xy) = x(x, y)$, hence

$$\sqrt{(x^2, xy)} = (x), \quad V(x^2, xy) = V(x).$$

So there is only one minimal prime over (x^2, xy) , namely $(\bar{x})$. However there is also an *embedded* prime $(\bar{x}, \bar{y})$ showing up in $\text{Ass}(R)$ (equivalently in $\text{Ass}(R/(0))$).

Annihilators:

$$\text{Ann}_R(\bar{x}) = (\bar{x}, \bar{y})$$

since $f(\bar{x}, \bar{y})\bar{x} = 0$ iff $fx \in (x^2, xy) = x(x, y)$ iff $f \in (x, y)$. Also

$$\text{Ann}_R(\bar{y}) = (\bar{x}),$$

since $f\bar{y} = 0$ iff $fy \in (x^2, xy)$, which forces $f \in (x)$. The nilradical of R is $(\bar{x})$ (since $\bar{x}^2 = 0$ and $(\bar{x})$ is the intersection of the minimal primes).

3.5 Associated primes

Definition 3.58. Let M be an A -module. A prime $\mathfrak{p}$ is an *associated prime* of M if $\exists 0 \neq x \in M$ with

$$\text{Ann}_A(x) = \mathfrak{p}.$$

Equivalently, there exists an injective A -linear map

$$A/\mathfrak{p} \hookrightarrow M,$$

sending $1 \mapsto x$.

Example 3.59. For $M = \mathbb{Z} \oplus \mathbb{Z}/45 \oplus \mathbb{Z}/15$,

$$\text{Ass } M = \{(0), (3), (5)\}.$$

Indeed (0) comes from the $\mathbb{Z}$ summand, and $(3), (5)$ come from elements of exact order divisible by 3 or 5.

Example 3.60 (Important). If $M = A/\mathfrak{p}$ with $\mathfrak{p}$ prime, then $\text{Ass } M = \{\mathfrak{p}\}$, since every nonzero element has annihilator $\mathfrak{p}$. Also $\text{Ass}(0) = \emptyset$.

Proposition 3.61. If A is Noetherian and $M \neq 0$, then $\text{Ass } M \neq \emptyset$.

Proof. Consider the set of annihilators $\{\text{Ann}(x) \mid 0 \neq x \in M\}$. Since A is Noetherian, there is a maximal element $\text{Ann}(x)$ among them.

We show $\text{Ann}(x)$ is prime. Suppose $fg \in \text{Ann}(x)$ and $g \notin \text{Ann}(x)$. Then $gx \neq 0$ and $f(gx) = 0$, hence $f \in \text{Ann}(gx)$. But $\text{Ann}(x) \subseteq \text{Ann}(gx)$, and by maximality we must have $\text{Ann}(gx) = \text{Ann}(x)$, so $f \in \text{Ann}(x)$. Thus $\text{Ann}(x)$ is prime. $\square$

Remark 3.62. The same argument shows: if $0 \neq z \in M$, then $\text{Ann}(z)$ is contained in some associated prime of M .

Back to general theory. Let M be an A -module.

Proposition 3.63. 1. $\text{Ass } M \subseteq \text{Supp } M$.

2. If $N \subseteq M$, then $\text{Ass } N \subseteq \text{Ass } M$.

3. If $0 \rightarrow N \rightarrow M \rightarrow M/N \rightarrow 0$ is exact, then

$$\text{Ass } M \subseteq \text{Ass } N \cup \text{Ass}(M/N).$$

Proof. 1. If $\mathfrak{p} \in \text{Ass } M$, then $A/\mathfrak{p} \hookrightarrow M$. Localizing at $\mathfrak{p}$ yields an injection

$$(A/\mathfrak{p})_{\mathfrak{p}} \hookrightarrow M_{\mathfrak{p}}.$$

But $(A/\mathfrak{p})_{\mathfrak{p}} \neq 0$, hence $M_{\mathfrak{p}} \neq 0$, so $\mathfrak{p} \in \text{Supp } M$.

2. If $\mathfrak{p} = \text{Ann}(x)$ for some $0 \neq x \in N$, then also $x \in M$ has the same annihilator, so $\mathfrak{p} \in \text{Ass } M$.

3. Let $\mathfrak{p} = \text{Ann}(x) \in \text{Ass } M$. Consider the cyclic submodule $Ax \cong A/\mathfrak{p}$ and its intersection with N .

Case 1: $Ax \cap N = 0$. Then $Ax \rightarrow M \rightarrow M/N$ is injective, so $A/\mathfrak{p} \hookrightarrow M/N$ and $\mathfrak{p} \in \text{Ass}(M/N)$.

Case 2: $Ax \cap N \neq 0$. Choose $0 \neq y \in Ax \cap N$. Since $Ax \cong A/\mathfrak{p}$, any nonzero element of Ax has annihilator $\mathfrak{p}$, hence $\text{Ann}(y) = \mathfrak{p}$ and $\mathfrak{p} \in \text{Ass } N$. $\square$

Associated primes (Noetherian case)

Throughout, let A be Noetherian and M a finitely generated A -module.

1. $\text{Ass}_A(M)$ is finite.

2. If $\mathfrak{P}$ is minimal over $\text{Ann}_A(M)$, then $\mathfrak{P} \in \text{Ass}_A(M)$.

Recall:

$$\text{Ass}(M) \subseteq \text{Supp}(M) = V(\text{Ann } M).$$

If $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ are the primes minimal over $\text{Ann } M$, then

$$V(\text{Ann } M) = \bigcup_{i=1}^r V(\mathfrak{p}_i), \quad \sqrt{\text{Ann } M} = \bigcap_{i=1}^r \mathfrak{p}_i.$$

Also, for every $\mathfrak{p} \in \text{Ass}(M)$ we have $\text{Ann } M \subseteq \mathfrak{p}$, hence $\sqrt{\text{Ann } M} \subseteq \mathfrak{p}$. Therefore

$$\sqrt{\text{Ann } M} \subseteq \bigcap_{\mathfrak{p} \in \text{Ass}(M)} \mathfrak{p} \subseteq \bigcap_{i=1}^r \mathfrak{p}_i = \sqrt{\text{Ann } M},$$

so

$$\boxed{\sqrt{\text{Ann } M} = \bigcap_{\mathfrak{p} \in \text{Ass}(M)} \mathfrak{p}}.$$

Prime filtration and finiteness of $\text{Ass}(M)$

Proposition 3.64 (Prime filtration with associated primes). *There exists a finite chain of submodules*

$$0 = M_0 \subsetneq M_1 \subsetneq \dots \subsetneq M_n = M$$

such that for each i ,

$$M_i/M_{i-1} \cong A/\mathfrak{p}_i \quad \text{for some } \mathfrak{p}_i \in \text{Ass}(M/M_{i-1}).$$

In particular each $\mathfrak{p}_i$ is a prime ideal (and an associated prime of the corresponding quotient).

Proof. Assume $M_{i-1} \neq M$. Since M/M_{i-1} is a nonzero finitely generated module over a Noetherian ring, $\text{Ass}(M/M_{i-1}) \neq \emptyset$. Choose $\mathfrak{p}_i \in \text{Ass}(M/M_{i-1})$, so $\mathfrak{p}_i = \text{Ann}(\overline{x}_i)$ for some $\overline{x}_i \in M/M_{i-1}$. Let $M_i := M_{i-1} + A\overline{x}_i$. Then $M_i/M_{i-1} = A\overline{x}_i \cong A/\mathfrak{p}_i$. This strictly increases the chain. Since M is Noetherian, the process terminates. $\square$

Corollary 3.65 (Finiteness of $\text{Ass}(M)$). *$\text{Ass}(M)$ is finite.*

Proof. Using the short exact sequences

$$0 \rightarrow M_{i-1} \rightarrow M_i \rightarrow M_i/M_{i-1} \rightarrow 0,$$

we have the standard containment

$$\text{Ass}(M_i) \subseteq \text{Ass}(M_{i-1}) \cup \text{Ass}(M_i/M_{i-1}) = \text{Ass}(M_{i-1}) \cup \{\mathfrak{p}_i\}.$$

Inducting gives $\text{Ass}(M) \subseteq \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$, hence finite. $\square$

Behavior under localization

Let $S \subset A$ be multiplicatively closed. Then A_S is Noetherian and M_S is a finitely generated A_S -module.

Proposition 3.66 (Associated primes and localization).

$$\text{Ass}_{A_S}(M_S) = \{ \mathfrak{p}A_S \mid \mathfrak{p} \in \text{Ass}_A(M), \mathfrak{p} \cap S = \emptyset \}.$$

Equivalently, via contraction from A_S to A ,

$$\text{Ass}_{A_S}(M_S) \longleftrightarrow \{ \mathfrak{p} \in \text{Ass}_A(M) \mid \mathfrak{p} \cap S = \emptyset \}.$$

Proof. ($\supseteq$) If $\mathfrak{p} = \text{Ann}_A(x)$ with $\mathfrak{p} \cap S = \emptyset$, then in M_S ,

$$\text{Ann}_{A_S}(x/1) = \mathfrak{p}A_S.$$

Indeed, $a/s \in \text{Ann}_{A_S}(x/1)$ iff $(a/s)(x/1) = 0$ in M_S , i.e. $\exists t \in S$ with $tax = 0$ in M , so $ta \in \text{Ann}_A(x) = \mathfrak{p}$. Since $\mathfrak{p}$ is prime and disjoint from S , this forces $a \in \mathfrak{p}$.

($\subseteq$) Let $\mathfrak{q} \in \text{Ass}_{A_S}(M_S)$. Then $\mathfrak{q} = \text{Ann}_{A_S}(x/s)$ for some $x/s \in M_S$, and $\mathfrak{q} = \mathfrak{p}A_S$ where $\mathfrak{p} = \mathfrak{q}^c \subset A$ is prime with $\mathfrak{p} \cap S = \emptyset$. Set $N := \{m \in M \mid m/1 = 0 \text{ in } M_S\}$, so $M_S \cong (M/N)_S$ and x/s comes from some $\bar{x} \in M/N$. Then $\mathfrak{p}$ is the contraction of $\text{Ann}_{A_S}(\bar{x}/1)$, and a standard clearing-denominators argument produces $\bar{y} \in M/N$ with $\text{Ann}_A(\bar{y}) = \mathfrak{p}$; hence $\mathfrak{p} \in \text{Ass}_A(M/N) \subseteq \text{Ass}_A(M)$. Therefore $\mathfrak{q} = \mathfrak{p}A_S$ with $\mathfrak{p} \in \text{Ass}_A(M)$ and $\mathfrak{p} \cap S = \emptyset$. $\square$

Special case: localization at a prime. For $\mathfrak{P} \in \text{Spec } A$, taking $S = A \setminus \mathfrak{P}$ yields

$$\text{Ass}_{A_{\mathfrak{P}}}(M_{\mathfrak{P}}) = \{ \mathfrak{p}A_{\mathfrak{P}} \mid \mathfrak{p} \in \text{Ass}_A(M), \mathfrak{p} \subseteq \mathfrak{P} \}.$$

Equivalently (contracting back to A),

$$\text{Ass}_{A_{\mathfrak{P}}}(M_{\mathfrak{P}}) \leftrightarrow \{ \mathfrak{p} \in \text{Ass}_A(M) \mid \mathfrak{p} \subseteq \mathfrak{P} \}.$$

Minimal primes over $\text{Ann}(M)$ are associated

Proposition 3.67. If $\mathfrak{P}$ is minimal over $\text{Ann}(M)$, then $\mathfrak{P} \in \text{Ass}(M)$.

Proof. Localize at $\mathfrak{P}$. Since $\text{Ann}(M) \subseteq \mathfrak{P}$, we have $M_{\mathfrak{P}} \neq 0$. Moreover,

$$\text{Supp}_{A_{\mathfrak{P}}}(M_{\mathfrak{P}}) = \{ \mathfrak{q}A_{\mathfrak{P}} \mid \mathfrak{q} \in \text{Supp}_A(M), \mathfrak{q} \subseteq \mathfrak{P} \}.$$

But $\text{Supp}_A(M) = V(\text{Ann } M)$, and minimality of $\mathfrak{P}$ over $\text{Ann } M$ implies the only prime in $V(\text{Ann } M)$ contained in $\mathfrak{P}$ is $\mathfrak{P}$ itself. Hence

$$\text{Supp}_{A_{\mathfrak{P}}}(M_{\mathfrak{P}}) = \{ \mathfrak{P}A_{\mathfrak{P}} \}.$$

Since $\text{Ass}(M_{\mathfrak{P}}) \subseteq \text{Supp}(M_{\mathfrak{P}})$ and $\text{Ass}(M_{\mathfrak{P}}) \neq \emptyset$ (for a nonzero finitely generated module over a

Noetherian ring), we get

$$\text{Ass}_{A_{\mathfrak{P}}}(M_{\mathfrak{P}}) = \{\mathfrak{P}A_{\mathfrak{P}}\}.$$

By the localization formula above, this forces $\mathfrak{P} \in \text{Ass}_A(M)$. $\square$

Zero divisors and prime avoidance

Proposition 3.68. *Let $M \neq 0$ be finitely generated over Noetherian A , and let $\mathfrak{a} \subset A$ be an ideal. Then either:*

1. *there exists $a \in \mathfrak{a}$ such that $\mu_a : M \rightarrow M$ is injective; or*
2. *there exists $0 \neq x \in M$ such that $\mathfrak{a} \subseteq \text{Ann}(x)$.*

Proof. In the Noetherian, finitely generated setting, the set of zero divisors on M equals

$$\text{ZD}(M) = \bigcup_{\mathfrak{p} \in \text{Ass}(M)} \mathfrak{p}.$$

If no $a \in \mathfrak{a}$ acts injectively, then every $a \in \mathfrak{a}$ is a zero divisor on M , so $\mathfrak{a} \subseteq \bigcup_{\mathfrak{p} \in \text{Ass}(M)} \mathfrak{p}$. Since $\text{Ass}(M)$ is finite, prime avoidance gives $\mathfrak{a} \subseteq \mathfrak{p}$ for some $\mathfrak{p} \in \text{Ass}(M)$. Write $\mathfrak{p} = \text{Ann}(x)$ for some $0 \neq x \in M$, giving (2). $\square$

Remark 3.69. Combining this with the earlier discussion yields again

$$\sqrt{\text{Ann } M} = \bigcap_{\mathfrak{p} \in \text{Ass}(M)} \mathfrak{p}.$$

3.6 Primary decomposition

For simplicity we consider A Noetherian and M finitely generated.

Motivation (ideals). Classically, for an ideal $\mathfrak{a} \subset A$ (Noetherian),

$$\mathfrak{a} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r$$

with each $\mathfrak{q}_i$ primary. If $\mathfrak{q}$ is primary, then $\sqrt{\mathfrak{q}}$ is prime, and one has

$$\text{Ass}(A/\mathfrak{a}) = \{\sqrt{\mathfrak{q}_1}, \dots, \sqrt{\mathfrak{q}_r}\}$$

(after choosing a suitable minimal decomposition).

Definition 3.70. Let $N \subsetneq M$. We say N is $\mathfrak{p}$ -primary (as a submodule of M) if

$$\text{Ass}(M/N) = \{\mathfrak{p}\}.$$

Proposition 3.71. *If N is $\mathfrak{p}$ -primary, then*

$$\sqrt{\text{Ann}(M/N)} = \mathfrak{p}.$$

Equivalently, $a \in \mathfrak{p}$ iff multiplication by a on M/N is nilpotent.

Proof. In general, $\sqrt{\text{Ann}(M/N)}$ is the intersection of the primes minimal over $\text{Ann}(M/N)$, and those minimal primes are contained in $\text{Ass}(M/N)$. If $\text{Ass}(M/N) = \{\mathfrak{p}\}$, then the only minimal prime is $\mathfrak{p}$, so $\sqrt{\text{Ann}(M/N)} = \mathfrak{p}$. The nilpotence statement is the definition of radical: $a \in \sqrt{\text{Ann}(M/N)}$ iff $a^n \in \text{Ann}(M/N)$ for some n , i.e. $\mu_a^n = 0$ on M/N . $\square$

Proposition 3.72 (Useful characterization). *For $N \subsetneq M$, the following are equivalent:*

1. N is $\mathfrak{p}$ -primary.
2. Every zero divisor on M/N lies in $\sqrt{\text{Ann}(M/N)} = \{\mathfrak{p}\}$ (equivalently, every zero divisor acts nilpotently on M/N).

Proof. (1) $\Rightarrow$ (2) If $\text{Ass}(M/N) = \{\mathfrak{p}\}$, then the set of zero divisors on M/N is $\bigcup_{\mathfrak{q} \in \text{Ass}(M/N)} \mathfrak{q} = \mathfrak{p} = \sqrt{\text{Ann}(M/N)}$.

(2) $\Rightarrow$ (1) Write $\text{Ass}(M/N) = \{\mathfrak{q}_1, \dots, \mathfrak{q}_t\}$ (finite). Then the set of zero divisors on M/N is $\bigcup_{j=1}^t \mathfrak{q}_j$, so by assumption

$$\bigcup_{j=1}^t \mathfrak{q}_j \subseteq \sqrt{\text{Ann}(M/N)}.$$

But $\sqrt{\text{Ann}(M/N)} = \bigcap_{\mathfrak{r} \text{ minimal over } \text{Ann}(M/N)} \mathfrak{r}$ and every minimal prime over $\text{Ann}(M/N)$ lies in $\text{Ass}(M/N)$. If there were two distinct minimal primes among the $\mathfrak{q}_j$'s, their union cannot be contained in their intersection (prime avoidance / basic set-theoretic obstruction), contradicting the inclusion above. Hence there is a unique minimal prime, call it $\mathfrak{p}$, and then all associated primes must equal $\mathfrak{p}$, so $\text{Ass}(M/N) = \{\mathfrak{p}\}$. $\square$

Goal. We want to show that every $M' \subset M$ is an intersection of finitely many primary submodules.

$$M' = N_1 \cap \dots \cap N_r$$

Where each N_i is $\mathfrak{p}_i$ -primary, then $\mathfrak{p}_i$ are distinct, and

$$\text{Ass}(M/M') = \{\mathfrak{p}_1, \dots, \mathfrak{p}_r\}$$

Note that it is enough to show for $M' = 0$, work with M/M' in general.

Remark 3.73. For $\mathfrak{p}_i$ minimal, $N_i = \ker(M \rightarrow (M/M')_{\mathfrak{p}_i})$ is unique.

The proof is in stages: **M' is an intersection of some finite collection of primary submodules.** (This is analogous to if A is a field, and M is a finite dimensional vector space then M' is the $\cap$ of finitely many hyperplanes).

Definition 3.74. Let $N \subset M$ we say that N is an "old fashioned irreducible submodule $M \subset N$ " (ofi), iff

$$N \subsetneq M, \quad N \subsetneq N_1, N_2 \implies N \subsetneq N_1 \cap N_2$$

Proposition 3.75. $N \subset M$ is ofi $\implies N$ is primary.

Proof. Suppose that $N \subsetneq M$ so $M/N \neq \emptyset$ so $\text{Ass}(M/N) \neq \emptyset$. Suppose that we had two distinct associated primes $\mathfrak{p}_1, \mathfrak{p}_2 \in \text{Ass}(M/N)$. Then $\exists [x_1], [x_2] \in M/N$ s.t

$$\mathfrak{p}_1 = \text{Ann}(x_1), \quad \mathfrak{p}_2 = \text{Ann}(x_2)$$

in particular, $[x_1], [x_2] \neq 0$ so

$$N + \langle x_1 \rangle, N + \langle x_2 \rangle \supsetneq N$$

But Note that

$$N + \langle x_1 \rangle \cap N + \langle x_2 \rangle = N \quad \text{contradiction}$$

This is because in M/N we have that

$$\langle [x_1] \rangle \cap \langle [x_2] \rangle = 0$$

but every nonzero element in $\langle x_1 \rangle, \langle x_2 \rangle$ is annihilated by $\mathfrak{p}_1, \mathfrak{p}_2$ respectively. So their intersection must be $\{[0]\}$. $\square$

Proposition 3.76. *Given $M \supsetneq N$ then M' is an intersection of finitely many ofi submodules. (So M' is an intersection of finitely many primary submodules).*

Proof. Suppose that there is a proper submodule that isn't a finite intersection of ofi submodules, then there is a largest such

$$M' \subsetneq M \quad \text{not a finite intersection of ofi, therefore not ofi}$$

So $\exists N_1, N_2$ with $M' \subsetneq N_1, N_2 \subset M$ with

$$N_1 \cap N_2 = M'$$

by maximality of N_1, N_2 is an intersection of ofi's, we are done. $\square$

At this stage, $M' =$ a finite intersection of primary submodules; but the relevant primes might not be distinct and might include more than the associated primes of M/M' . However, it doesn't hurt to pretend $M' = 0$.

Proposition 3.77. *Suppose that $N_1, N_2 \subset M$ are $\mathfrak{p}$ -primary, then $N_1 \cap N_2$ is $\mathfrak{p}$ -primary.*

Proof.

$$M/(N_1 \cap N_2) \hookrightarrow M/N_1 \oplus M/N_2$$

$$x \mapsto (x + N_1, x + N_2)$$

Therefore

$$\text{Ass } M/(N_1 \cap N_2) \subseteq \text{Ass } M/N_1 \oplus M/N_2 = \text{Ass}(M/N_1) \cup \text{Ass}(M/N_2) = \{\mathfrak{p}\} \cup \{\mathfrak{p}\}$$

Where the last equality is from

$$0 \rightarrow M/N_1 \rightarrow M/N_1 \oplus M/N_2 \rightarrow M/N_2 \rightarrow 0$$

and the other direction

$$M/N_1 \hookrightarrow M/N_1 \oplus M/N_2$$

□

Therefore at this step,

$$M' = N_1 \cap \cdots \cap N_r \quad N_i \text{ is } \mathfrak{p}_i \text{ primary}$$

and all $\mathfrak{p}_i$ are distinct.

Remark 3.78. $\text{Ass}(M/M') \subseteq \{\mathfrak{p}_1, \dots, \mathfrak{p}_r\}$ because

$$M/M' \hookrightarrow M/N_1 \oplus \cdots \oplus M/N_r$$

uses $M' = N_1 \cap \cdots \cap N_r$.

Now, if any N_i is not need, throw it out. i.e

$$M' = N_1 \cap \cdots \cap N_{i-1} \cap N_{i+1} \cap \cdots \cap N_r$$

so at the end of this finite process, we end up with

$$M' = N'_1 \cap \cdots \cap N'_s$$

an irredundant (minimal) intersection of primary submodules, with distinct $\mathfrak{p}'_i$ s. (Note that we have that $\text{Ass } M/M' \subseteq \{\mathfrak{p}_i\}$ by the same argument as above). In particular

$$M' \subsetneq (N'_1 \cap \cdots \cap N'_{i-1} \cap N_{i+1} \cap \cdots \cap N_s)$$

Now let us show that these $\mathfrak{p}'_i$ s are all associated to M/M' .

For notational convenience, let us show that $\mathfrak{p}'_1 \in \text{Ass } M/M'$, so

$$M' \subsetneq N'_2 \cap \cdots \cap N'_s = Z$$

at this point, we pass to M/M' and $M' = 0$ for ease of notation. Therefore

$$0 = N'_1 \cap Z, \quad N'_1 \text{ is } \mathfrak{p}'_1 \text{ primary and } Z \neq 0$$

Therefore

$$Z \rightarrow M/N'_1 \quad \text{is an injection}$$

But since $Z \neq 0$ we have that

$$\emptyset \neq \text{Ass } Z \subset \text{Ass } M/N'_1 = \{\mathfrak{p}_1\}$$

We conclude that

$$\text{Ass } Z = \{\mathfrak{p}'_1\}$$

But $\text{Ass } Z \subseteq \text{Ass } M$ so $\mathfrak{p}'_1 \in \text{Ass } M$ as desired.

Remark 3.79. Note that $\exists A$'s with $\mathfrak{p}$ primes such that $\mathfrak{p}^2$ is not itself $\mathfrak{p}$ primary.

The correct thing to do is the symbolic n 'th power

$$\mathfrak{p}^{(n)} = \{a \in A \mid \exists s \notin \mathfrak{p} \text{ with } sa \in \mathfrak{p}^n\} = (\mathfrak{p}^{(n)})^{ec}$$

For

$$A \rightarrow A_{\mathfrak{p}}$$

4 Integral extensions.

Reminder. If k, L are fields, with $k \subset L$ and element $x \in L$ is algebraic over K iff $\exists$ nonzero polynomial $c_n x^n + \cdots + c_1 x + c_0 = 0$ with $c_0, \dots, c_n \in K$ (wlog $c_n = 0$, even $c_n = 1$).

Definition 4.1. let A be a subring of B (in particular $1_A = 1_B$) (Not necc domains). An element $b \in B$ is called integral over A , if $\exists$ monic polynomial in $f(x) \in A[x]$

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$$

s.t

$$f(b) = 0$$

Remark 4.2. if $A \xrightarrow{\varphi} B$ we say that φ is integral (or an integral extension), iff every element of B is integral over $\text{Im } \varphi \subseteq B$, note that φ need not be injective.

Theorem 4.3. Let $A \subset B$ and let $b \in B$. The following are equivalent:

1. b is integral over A .
2. The sub- A -algebra $A[b] \subset B$ is a finitely generated A -module. Recall that

$$A[b] = \text{Im}(A[x] \rightarrow B, g(x) \mapsto g(b)),$$

i.e. the smallest subring of B containing A and b .

3. There exists an intermediate subring

$$A[b] \subseteq C \subseteq B$$

such that C is a finitely generated A -module.

4. There exists an $A[b]$ -module M which is finitely generated as an A -module and satisfies

$$\text{Ann}_{A[b]}(M) = 0,$$

i.e. M is a faithful $A[b]$ -module.

5. Whenever $A \subset B \subset R$, there exists a finitely generated A -module $M \subset R$ such that

$$bM \subseteq M,$$

i.e. M is an $A[b]$ -submodule of R , and M is faithful as an $A[b]$ -module (in practice this holds for instance if $1 \in M$ or if R is a domain).

Pr(bf). $\Rightarrow$ (2) Suppose b satisfies a monic polynomial

$$b^n + a_{n-1}b^{n-1} + \cdots + a_0 = 0 \quad a_i \in A$$

Then every power b^m with $m \geq n$ can be written as an A -linear combination of $1, b, \dots, b^{n-1}$. Equivalently, if $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$, then for every $g(x) \in A[x]$ we can perform Euclidean division:

$$g(x) = f(x)q(x) + r(x), \quad \deg r < n.$$

Evaluating at b gives

$$g(b) = r(b),$$

so $A[b]$ is generated as an A -module by

$$1, b, \dots, b^{n-1}.$$

(2) $\Rightarrow$ (3) Immediate by taking $C = A[b]$.

(3) $\Rightarrow$ (4) Take $M = C$. Then M is finitely generated as an A -module by assumption. Faithfulness holds since $1 \in C$: if $c \in A[b]$ annihilates C , then in particular $c \cdot 1 = 0$, hence $c = 0$.

(4) $\Rightarrow$ (1) (Determinant trick.) Let $M = \langle m_1, \dots, m_k \rangle_A$. Since M is an $A[b]$ -module, multiplication by b sends generators to A -linear combinations:

$$\begin{aligned} bm_1 &= a_{11}m_1 + \cdots + a_{1k}m_k, \\ &\vdots \\ bm_k &= a_{k1}m_1 + \cdots + a_{kk}m_k, \end{aligned}$$

with $a_{ij} \in A$. Writing this in matrix form,

$$(bI - Q) \begin{pmatrix} m_1 \\ \vdots \\ m_k \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix},$$

where $Q = (a_{ij}) \in M_k(A)$. Multiply by the classical adjoint of $(bI - Q)$ to obtain

$$\det(bI - Q) \begin{pmatrix} m_1 \\ \vdots \\ m_k \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

The element $\det(bI - Q)$ is a monic polynomial in b with coefficients in A . It annihilates

all generators of M , hence annihilates M . By faithfulness,

$$\det(bI - Q) = 0,$$

so b satisfies a monic polynomial over A and is integral over A . □

Example 4.4. Let $A = \mathbb{Z} \subset \mathbb{Q}(\sqrt{5})$, and consider the element

$$\alpha = \frac{1 + \sqrt{5}}{2}.$$

Note that if $a, b \in \mathbb{Z}$, then $a + b\sqrt{5}$ is integral over $\mathbb{Z}$ (indeed $\sqrt{5}$ satisfies the monic polynomial $x^2 - 5 = 0$).

Let

$$M = 2\mathbb{Z} + (1 + \sqrt{5})\mathbb{Z} \subset \mathbb{Q}(\sqrt{5}),$$

so M is finitely generated as a $\mathbb{Z}$ -module. We claim that

$$\alpha M \subseteq M.$$

Indeed,

$$\frac{1 + \sqrt{5}}{2} \cdot 2 = 1 + \sqrt{5}, \quad \frac{1 + \sqrt{5}}{2} \cdot (1 + \sqrt{5}) = 3 + \sqrt{5} = 2 + (1 + \sqrt{5}),$$

hence $\alpha M \subseteq M$. Therefore $\frac{1+\sqrt{5}}{2}$ is integral over $\mathbb{Z}$.

Easy fact. $\mathbb{Z}[\frac{1+\sqrt{5}}{2}] = \left\{ \frac{a+b\sqrt{5}}{2} \mid a, b \in \mathbb{Z} \text{ } a \equiv b \pmod{2} \right\}$ is the set of all elements of $\mathbb{Q}(\sqrt{5})$ which are integral over $\mathbb{Z}$. (it's the integral closure over $\mathbb{Z}$).

Reminder. We know that $A \subset B$, $b \in B$ integral over A (say with $b^n + a_{n-1}b^{n-1} + \dots + a_0$), then $A[b]$ is generated over A by $1, b, b^2, \dots, b^{n-1}$. (Note that $\ker A[x] \rightarrow B, x \rightarrow b$ contains monic polynomial but it need not consist of multiples of a single monic polynomial).

Slight generalization.

Proposition 4.5. Say $A \subseteq B$ and $b_1, \dots, b_r \in B$ are all integral over A then $A \subset A[b_1, \dots, b_r] \subset B$ and $A[b_1, \dots, b_r]$ is f.g as an A -module.

Remark 4.6. Note that it is enough to have the weaker assumption b_1 integral over A , b_2 integral over $A[b_1]$, b_3 is integral over $A[b_1, b_2]$ etc..

Proof. Indeed

$$A[b_1] = A + Ab_1 + \dots + Ab_1^{n-1}$$

and

$$A[b_1, b_2] = (A[b_1])[b_2] = (A[b_1]) + (A[b_1])b_2 + \dots + (A[b_1])b_2^{m-1}$$

finitely generated over A by $\{b_i b_j\}_{\substack{0 \leq j \leq n-1 \\ 0 \leq j \leq m-1}}$ this patten continues. □

Proposition 4.7. *If $b_1, \dots, b_r$ are all integral over A then the map*

$$A \rightarrow A[b_1, \dots, b_r] \text{ is integral!}$$

i.e we get that $\forall \gamma \in A[b_1, \dots, b_r]$ are also integral over A .

Proof. $\gamma(A[b_1, \dots, b_r]) \subseteq A[b_1, \dots, b_r]$. □

Interpretation. Let $A \xrightarrow{f} B$ be any ring homomorphism, then B becomes a finite A -module iff

1. f is of finite type, B is a finitely generated A -algebra ($= A[b_1, \dots, b_r]$).
2. f is integral.

In this case we say that f is a "finite ring homomorphism".

Definition 4.8. If $A \subseteq B$ a subring, then the set C of elements of B that are integral over A is the integral closure of A in B ; hence

$$A \subseteq C \subseteq B$$

If $A \xrightarrow{f} B$ and f not injective, then C is the integral closure of $f(A)$ in B .

Theorem 4.9. C is a subring of B .

Proof. If $c_1, c_2 \in C$ then $A[c_1, c_2]$ is a finitely generated A -module, so all elements are integral over A , hence $c_1 + c_2, c_1 - c_2, c_1 c_2 \in C$. □

Definition 4.10. If A is a domain, with field of fractions K , we say that A is integral closed iff the integral closure of A in K is A .

Proposition 4.11. *Every UFD is integrally closed.*

Proof. (Pretend that $A = \mathbb{Z}$), suppose $\alpha = \frac{a}{b} \in \mathbb{Q}$ is integral over $\mathbb{Z}$. We can assume that this fraction is in lowest terms, i.e every reducible p with $p \mid b$ satisfies $p \mid a$. Our goal is to show that $\forall p$ we have $p \mid b$. We have an equation

$$\alpha^n + c_{n-1}\alpha^{n-1} + \dots + c_1\alpha + c_0 = 0 \quad c_i \in \mathbb{Z}$$

so

$$(a/b)^n + c_{n-1}(a/b)^{n-1} + c_1(a/b) + c_0 = 0$$

multiplying by b^n we get

$$a^n + c_{n-1}a^{n-1}b + \dots + c_1ab^{n-1} + c_0b^n = 0$$

if $p \mid b$ then p divides all the later terms. So $p \mid a^n$, therefore (by UFD) we have $p \mid a$ contradiction. □

Proposition 4.12. *if $A \rightarrow B$ and $B \rightarrow C$ are integral homomorphisms, then $A \xrightarrow{gf} C$ is also integral.*

Proof. (it doesn't hurt pretend $A \subset B \subset C$) let $c \in C$ we want to show that c is integral over A . We know that c is integral over B therefore c satisfies some equation

$$c^n + b_{n-1}c^{n-1} + \cdots + b_1c + b_0 = 0$$

let $B_0 = A[b_0, \dots, b_{n-1}]$, then B_0 is a finitely generated A -algebra. On the other hand, the b_i 's are integral over A . so B_0 is a finitely generated A -module, and $B_0[C]$ is a finitely generated B_0 module. Then $B_0[C] = A[b_0, \dots, b_{n-1}, c]$ is a finitely generated A module. Therefore C is integral over A . $\square$

Proposition 4.13. If $A \xrightarrow{f} B$ is integral and $\mathfrak{b} \subseteq B$ is an ideal and $\mathfrak{a} \subseteq A$ is an ideal with $\mathfrak{a} \subseteq \mathfrak{b}^c$ then f induces a ring homomorphism.

$$A/\mathfrak{a} \xrightarrow{\bar{f}} B/\mathfrak{b}$$

then $\bar{f}$ is also integral.

Proof. Trivial, take $\bar{b} \in B/\mathfrak{b}$ and take some representative b . Find an equation for b with coefficients in A and reduce mod $\mathfrak{b}$ resp to $\mathfrak{a}$.

Note that $\bar{f}$ is injective iff $\mathfrak{a} = \mathfrak{b}^c$ $\square$

Proposition 4.14. Suppose that $A \xrightarrow{f} B$ integral, and $S \subset A$ is multiplicatively closed. Then $S^{-1}A \xrightarrow{S^{-1}f} S^{-1}B$ is a ring homomorphism that is also integral.

In fact here is a stronger result (we state with subrings to avoid notation). If $A \subseteq B$ and C is the integral closure of A in B , then $S^{-1}C$ is the integral closure of $S^{-1}A$ in $S^{-1}B$.

Proof. We will prove the stronger result.

- i) Every element of $c/s \in S^{-1}C$ is integral over $S^{-1}A$ because $c/s = 1/s \cdot c/1$. and $1/s \in S^{-1}A$ (note that this uses c being integral over A).
- ii) Let b/s be integral over $S^{-1}A$, we need to show that $\exists c/s'$ with $c \in C$ and $s' \in S$ s.t

$$b/s = c/s' \in S^{-1}B$$

Note since $b/1 = b/s \cdot s/1$ is also integral over $S^{-1}A$ it is enough to show taht

$$b/1 = c/s' \quad c \in C, s' \in S$$

wlog we get $s = 1$. We have an equation of the form

$$(b/1)^n + \left(\frac{a_{n-1}}{t_{n-1}}\right)(b/1)^{n-1} + \cdots + \left(\frac{a_1}{t_1}\right)(b/1) + \left(\frac{a_0}{t_0}\right) = 0$$

Clear the denominators by multiplying $t = u^n = (t_0 \cdots t_{n-1})^n$, get $\exists T \in S$ s.t

$$T\left(\left(\frac{ub}{1}\right)^n + (a_{n-1}t_0 \cdots t_{n-2})\left(\frac{ub^{n-1}}{1}\right) + (a_{n-2}t_0^2t_1 \cdots t_{n-2}^1t_{n-1}^2)\left(\frac{ub}{1}\right)^{n-2} + \cdots + a_0\right) = 0$$

call the weird terms $\alpha_{n-1}, \alpha_{n-2}, \dots \in A$ and call $ub = \beta$ therefore

$$T(\beta^n + \alpha\beta^{n-1} + \dots + \alpha_0) = 0$$

Multiply by T^{n-1} to get a relation of integral dependence of $T\beta$ over A . We deduce that $T\beta \in C$. So we write $c = T\beta = Tub$ so $b/1 = \frac{Tub}{Tu} = \frac{c}{Tu} \in S^{-1}C$.

□

Next family of results.

- **"Lying over theorem"**. every $\mathfrak{p} \in \text{Spec } A$ is $\mathfrak{q}^c$ for some $\mathfrak{q} \in \text{Spec } B$ i.e $\text{Spec } B \rightarrow \text{Spec } A$ is surjective.
- **"incompatibility theorem"**. if $\mathfrak{q}_1 \subseteq \mathfrak{q}_2$ primes and $\mathfrak{q}_1^c = \mathfrak{q}_2^c$ then $\mathfrak{q}_1 = \mathfrak{q}_2$.
- **"finiteness of fibres theorem"**. if B is Noetherian, the given $\mathfrak{p} \in \text{Spec } A$, there are only finitely many $\mathfrak{q}$'s with $\mathfrak{q}^c = \mathfrak{p}$.
- **"Going up theorem"**. Say $\mathfrak{p}_1 \subseteq \mathfrak{p}_2$ primes of A and let $\mathfrak{q}_1$ be a prime of B with $\mathfrak{q}_1^c = \mathfrak{p}_1$. Then $\exists \mathfrak{q}_2$ primes of B s.t $\mathfrak{q}_1 \subset \mathfrak{q}_2$ and $\mathfrak{q}_2^c = \mathfrak{p}_2$. (Note that since $\mathfrak{q}_1^c \neq \mathfrak{q}_2^c$, we have that $\mathfrak{q}_1 \subsetneq \mathfrak{q}_2$).

Remark 4.15. Going up for 2 primes $\implies$ going up for any finite family of primes.

$$\begin{array}{ccccccc} \mathfrak{p}_1 & \hookrightarrow & \mathfrak{p}_2 & \hookrightarrow & \mathfrak{p}_3 & \hookrightarrow & \dots \hookrightarrow \mathfrak{p}_n \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \mathfrak{q}_1 & \dashrightarrow & \mathfrak{q}_2 & \dashrightarrow & \mathfrak{q}_3 & \dashrightarrow & \dots \dashrightarrow \mathfrak{q}_n \end{array}$$

Example 4.16. Consider $\mathbb{Z} \subset \mathbb{Z}[i]$, with $\mathfrak{p} = 5\mathbb{Z}$, notice that 2 primes lie over $5\mathbb{Z}$, namely $\langle 2+i \rangle, \langle 2-i \rangle$.

$$\begin{aligned} (5\mathbb{Z})^e &= 5\mathbb{Z}[i] \\ &= \langle 2+i \rangle \langle 2-i \rangle \\ &= \langle 2+i \rangle \cap \langle 2-i \rangle \end{aligned}$$

Note that these are 2 different maximal ideals in $\mathbb{Z}[i]$, since

$$\mathbb{Z}[i]/\mathfrak{q} \cong \mathbb{Z}/5\mathbb{Z}$$

Moreover

$$\langle 2+i, 2-i \rangle \ni (2+i)(2-i) = 5, \quad 2+i+2-i = 4$$

Therefore

$$\langle 2+i, 2-i \rangle \ni 1$$

Similarly,

$$\mathbb{Z} \subset \mathbb{Z}[\sqrt{-5}]$$

$$\mathfrak{p} = 3\mathbb{Z} \rightarrow \mathfrak{p}^e = \langle 3, 1+\sqrt{-5} \rangle \cdot \langle 3, 1-\sqrt{-5} \rangle$$

Picture. Let $A \subseteq B$ be a ring extension, and consider the induced map on spectra

$$\varphi^* : \operatorname{Spec} B \longrightarrow \operatorname{Spec} A, \quad \mathfrak{q} \longmapsto \varphi^{-1}(\mathfrak{q}).$$

From HW we know that the image of φ^* is dense in $\operatorname{Spec} A$, i.e.

$$\overline{\operatorname{Im}(\varphi^*)} = \operatorname{Spec} A.$$

Equivalently, every nonempty open subset of $\operatorname{Spec} A$ intersects $\operatorname{Im}(\varphi^*)$.

$$\operatorname{Spec} B \xrightarrow{\varphi^*} \operatorname{Spec} A$$

$$\overline{\operatorname{Im}(\varphi^*)} = \operatorname{Spec} A$$

Lying over + incomparable $\implies$ finiteness of fibres. Let $\mathfrak{p} \in \operatorname{Spec} A$ we know that $\exists \{q_i \mid i \in I\}$ with $\mathfrak{q}_i^c = \mathfrak{p}$. But no $\mathfrak{q}_i$ contains another. **Claim.** This forces the $\mathfrak{q}_i$'s to be minimal over $\mathfrak{p}^e$.

But B is Noetherian, so $V(\mathfrak{p}^e)$ only has finitely many irreducible components and we only have finitely many primes over $\mathfrak{p}^e$.

Indeed, if $\mathfrak{q}_i \supseteq \hat{\mathfrak{q}} \supseteq \mathfrak{p}^e$, then

$$\mathfrak{p} = \mathfrak{q}_i^c \supseteq \hat{\mathfrak{q}}^c \supseteq \mathfrak{p}$$

so $\mathfrak{q}_i^c = \hat{\mathfrak{q}}^c$ and hence $\mathfrak{q}_i = \hat{\mathfrak{q}}$ (by incomparability). $\square$

Goals. Lying over, incomparability and going-up.

Lemma 4.17. Suppose A, B are domains with $A \subset B$ and B integral over A . Then A is a field $\iff B$ is a field.

Proof. $\implies$ if A is a field, we want to show that B is a field.

way 1. Every element $b \in B$ is algebraic over A , view as $A \subset$ field of fraction L of B . So $A \subseteq A[b]$, since b is algebraic, $A[b] = A(b) \subseteq B$ is the field extension. So if $b \neq 0$, $b^{-1} \in A[b] \in B$ and therefore B is a field.

way 2. Say $b \in B$ and $b \neq 0$, we know that $\exists$ monic equation

$$b^n + a_{n-1}b^{n-1} + \cdots + a_1b + b_0 = 0, \quad a_i \in A$$

Note that we can arrange for $a_0 \neq 0$ since $b \neq 0$. But then

$$-a_0 = b(b^{n-1} + a_{n-1}b^{n-2} + \cdots + a_1) \implies b^{-1} = (-a_0^{-1}) \cdot (b^{n-1} + \cdots + a_1) \in B$$

$\Leftarrow$ Suppose that B is a field. Take $a \neq 0 \in A$, I know that $a^{-1} \in B$ exists, therefore $a^{-1} \in B$ exists, so a^{-1} is integral over A . Therefore

$$a^{-N} + \alpha_{N-1}a^{-N+1} + \cdots + \alpha_0 = 0 \quad \alpha_i \in A$$

multiply by a^{N-1} , we get

$$a^{-1} + \alpha_{N-1} + \cdots + \alpha_0 a^{N-1} = 0 \implies a^{-1} \in A$$

□

Corollary 4.18. *Let $A \rightarrow B$ be any integral homomorphism, let $\mathfrak{q}$ be a prime of B and $\mathfrak{p} = \mathfrak{q}^c \in \text{Spec } A$, then $\mathfrak{q}$ is maximal (in B) $\iff \mathfrak{p}$ is maximal (in A).*

Proof. We have

$$A/\mathfrak{p} \hookrightarrow B/\mathfrak{q}$$

This is now an injective integral extension of domains, so apply the lemma. □

Theorem 4.19 (Lying over). *Let $A \subset B$ with B integral over A . Then the map $\text{Spec } B \rightarrow \text{Spec } A$ is surjective.*

In the more general case we have

$$A \rightarrow A/\ker f \hookrightarrow B$$

then

$$\begin{aligned} f^*(\text{Spec } B) &= \text{Spec}(A/\ker f) \\ &= V(\ker f) \subseteq \text{Spec } A \end{aligned}$$

Remark 4.20. Exercise: this map is in fact closed, in fact $A \xrightarrow{f} B$ is any integral extension then

$$f^*: \text{Spec } B \rightarrow \text{Spec } A$$

satisfies

$$f^*(\text{closed in Spec } B) = \text{closed in Spec } A$$

Proof. Let $\mathfrak{p} \in \text{Spec } A$, localize A and B are at $S = A - \mathfrak{p}$, therefore

$$S^{-1}A \subseteq S^{-1}B$$

and this inclusion is still integral,

$$A_{\mathfrak{p}} \subseteq B_{\mathfrak{p}}$$

Note that $B_{\mathfrak{p}} \neq 0$ (since $A_{\mathfrak{p}} \neq 0$). So let's take a maximal ideal $M \subset B_{\mathfrak{p}}$, then the contraction of M under $(A_{\mathfrak{p}} \rightarrow B_{\mathfrak{p}})$ must be maximal in $A_{\mathfrak{p}}$. But $A_{\mathfrak{p}}$ has only $\mathfrak{p}A_{\mathfrak{p}}$ as maximal ideal, so we have a diagram of ring homom.

$$\begin{array}{ccc} A & \hookrightarrow & B \\ \pi_1 \downarrow & & \downarrow \rho_1 \\ A/\mathfrak{p}_1 & \xrightarrow{\bar{f}} & B/\mathfrak{q}_1 \\ \pi_{21} \downarrow & & \downarrow \rho_{21} \\ A/\mathfrak{p}_2 & \xrightarrow{\bar{f}_2} & B/\mathfrak{q}_2 \end{array}$$

with $\mathfrak{p}A_{\mathfrak{p}}$ contracts to $\mathfrak{p}$. □

Theorem 4.21 (incomparability). *Say $A \subset B$, integral extension. Suppose that $\mathfrak{p}, \mathfrak{q}_1, \mathfrak{q}_2$ are primes of A, B, B respectively with*

$$\mathfrak{q}_1^c = \mathfrak{q}_2^c = \mathfrak{p}, \quad \mathfrak{q}_1 \subseteq \mathfrak{q}_2$$

then

$$\mathfrak{q}_1 = \mathfrak{q}_2$$

Proof. Like in the previous theorem, localize at $\mathfrak{p}$ then

$$\mathfrak{q}_1 B_{\mathfrak{p}} \subseteq \mathfrak{q}_2 B_{\mathfrak{p}}$$

are both primes of $B_{\mathfrak{p}}$ which both contract to the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. So they are both maximal, therefore

$$\mathfrak{q}_1 B_{\mathfrak{p}} = \mathfrak{q}_2 B_{\mathfrak{p}}$$

Hence

$$\mathfrak{q}_1 = \mathfrak{q}_2$$

□

Theorem 4.22 (Going up theorem). *Let $A \subseteq B$ be integral and let $\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{q}$ be primes in A, A, B respectively s.t $\mathfrak{p}_1 \subset \mathfrak{p}_2$ and $\mathfrak{q}_1^c = \mathfrak{p}_1$. Then $\exists \mathfrak{q}_2 \in \text{Spec } B$ with $\mathfrak{q}_1 \subset \mathfrak{q}_2$ and $\mathfrak{q}_2^c = \mathfrak{p}_2$.*

(This can easily adapted to $A \xrightarrow{f} B$)

Proof. We have the integral map

$$A/\mathfrak{p}_1 \hookrightarrow B/\mathfrak{q}_1$$

Then $\mathfrak{p}_2/\mathfrak{p}_1$ is a prime in $A/\mathfrak{p}_1$, therefore by lying over we get a prime $\mathfrak{q}_2/\mathfrak{q}_1$ in $B/\mathfrak{q}_1$ that contracts to $\mathfrak{p}_2/\mathfrak{p}_1$.

$$\begin{array}{ccc} A & \hookrightarrow & B \\ \downarrow & & \downarrow \\ A/\mathfrak{p}_1 & \hookrightarrow & B/\mathfrak{q}_1 \\ \downarrow & & \downarrow \\ A/\mathfrak{p}_2 & \hookrightarrow & B/\mathfrak{q}_2 \end{array}$$

□

Example 4.23. View picture, very interesting stuff. Very good intuition for going down.

Theorem 4.24 (Going down). *Let $A \subset B$ be domains, with*

1. *B integral over A .*
2. *A is integrally closed (in its field of fractions K).*

Let $A \supset \mathfrak{p}_2 \supset \mathfrak{p}_1$, let $B \supset \mathfrak{q}_2$ with $\mathfrak{q}_2^c = \mathfrak{p}_2$. Then $\exists \mathfrak{q}_1 \in \text{Spec } B$ s.t $\mathfrak{q}_2 \supset \mathfrak{q}_1$ and $\mathfrak{q}_1^c = \mathfrak{p}_1$.

Remark 4.25. Let $A \subset K \subset L$ with A integrally closed and K field of fraction and L some big field (say algebraic over K). Let $\beta \in L$ integral over A then therefore the irreducible monic polynomial of β over K has coefficients in A . i.e

$$f(x) = x^n + k_{n-1}x^{n-1} + \cdots + k_0 = (x - \beta)(x - \beta') \cdots (x - \beta^*)$$

Proposition 4.26. Let $A \subset K \subset L \subset \overline{K}$ with A domain, K field of fractions and L an algebraic extension of K (usually we get K as the field of fractions of some B which is integral over A). Let C be the integral closure of A in K , and $\beta \in L$ be integral over A (typically $\beta \in B$). Then the irreducible (monic) polynomial of β over K naturally lies in $C[x]$.

Proof. Write $f(x) = x^n + k_{n-1}x^{n-1} + \cdots + k_1x + k_0 \in K[x]$ to be the minimal polynomial of β over K . We can factor in $\overline{K}$ as

$$f(x) = (x - \gamma_1)(x - \gamma_2) \cdots (x - \gamma_n), \quad \beta = \gamma_1$$

However, we also know that β is a root of a monic polynomial in $A[x]$ so $\exists g(x) \in A[x]$ with

$$g(x) = x^l + a_{l-1}x^{l-1} + \cdots + a_0, \quad g(\beta) = 0$$

note that g factors in $\overline{K}[x]$ as

$$g(x) = (x - \delta_1) \cdots (x - \delta_l)$$

By the existence of g , all the δ_i are integral over A . OTOH since $f(x)$ is irreducible we have

$$f(x) \mid g(x) \in K[x] \implies f \mid g \in \overline{K}[x]$$

Therefore

$$[\gamma_1, \cdots, \gamma_n] \subseteq [\delta_1, \cdots, \delta_l]$$

In particular, all the γ_i 's are integral over A . However the coefficients in $f(x)$ are symmetric polynomials over γ_i 's, i.e

$$k_{n-1} = -(\gamma_1 + \cdots + \gamma_n)$$

$$k_{n-2} = (\gamma_1\gamma_2 + \cdots + \gamma_{n-1}\gamma_n)$$

$$\vdots$$

$$k_0 = (-1)^n \gamma_1\gamma_2 \cdots \gamma_n$$

Conclude that $k_0, \cdots, k_{n-1}$ are also integral over A , but they belong to K therefore

$$k_0, \cdots, k_{n-1} \in C$$

□

4.1 Integrality over an Ideal.

Definition 4.27. Let $\mathfrak{a} \subseteq A \subseteq B$ $\mathfrak{a}$ an ideal and B any ring. We say that $b \in B$ is integral over the $\mathfrak{a}$ iff b is a root of a monic polynomial

$$f = x^n + a_{n-1}x^{n-1} + \cdots + a_0, \quad a_0, \cdots, a_{n-1} \in \mathfrak{a}$$

Proposition 4.28. In the above setting, let C be the integral closure of A in B . Then $b \in B$ is integral over $\mathfrak{a}$ iff $b \in \text{rad}(\mathfrak{a}C)$.

Proof. $\implies$ b integral over $\mathfrak{a} \implies b \in C$. Also we have an equation

$$b^n = -a_{n-1} - \cdots - a_1b - a_0 \quad b \in C \quad \mathfrak{a}_i \in \mathfrak{a}$$

Therefore

$$b^n \in \mathfrak{a}C \implies b \in \text{rad}(\mathfrak{a}C)$$

$\Leftarrow$ Given $b^l \in \mathfrak{a}C$ for some l . Therefore $\exists$ finite expression

$$b^l = \alpha_1 c_k + \cdots + \alpha_k c_k, \quad \alpha_i \in \mathfrak{a}, \quad c_i \in C$$

Let

$$C_0 = A[c_1, \cdots, c_k]$$

Note that C_0 is a finitely generated A -module and A -algebra. Let's take $y_1, \cdots, y_n$ to be A -generators of C_0 . We claim that

$$b^l C_0 \subseteq \mathfrak{a}C_0$$

check the details, use the fact that $y_i y_j$ is an A -linear combination of $y_1, \cdots, y_n$. So we get

$$b^l = \aleph_{11} y_1 + \cdots + \aleph_{1n} y_n$$

$$\vdots$$

$$b^l y_M = \aleph_{N1} y_1 + \cdots + \aleph_{NN} y_N$$

with $\aleph_{ij} \in \mathfrak{a}$, by the usual argument we get

$$\det(b^l I - \aleph_{ij}) = 0$$

but this is a polynomial which is monic in b^l while other factors lie in $\mathfrak{a}$.

$$b^{Nl} + s_{N-1}b^{(N-1)l} + \cdots + s_0 = 0 \quad s_j \in \mathfrak{a}$$

so b is integral over $\mathfrak{a}$.

We get that $b \in B$ is integral over $\mathfrak{a}$ iff $b \in \text{rad}(\mathfrak{a}C)$. □

Corollary 4.29. If b_1, b_2 are integral over $\mathfrak{a}$ then so are $b_1 \pm b_2$ and $b_1 b_2$.

Proposition 4.30. *Suppose we have*

$$\mathfrak{a} \subseteq A \subset K \subset L$$

with A an integrally closed domain, K field of fractions and L is usually the field of fractions of some B with B integral over A .

Then β is integral over $\mathfrak{a}$ iff the minimal polynomial of β in $K[x]$ has all coefficients $\in \text{rad } \mathfrak{a}$ (except the leading one).

Proof. $\Leftarrow$ If β satisfies an equation

$$\beta^n + r_{n-1}\beta^{n-1} + \cdots + r_0 = 0 \quad r_i \in \text{rad } \mathfrak{a}$$

then β is integral over A so $\beta \in \tilde{B}$ (the integral closure of A in L), $\beta^n \in (\text{rad } \mathfrak{a})\beta \subseteq (\text{rad } \mathfrak{a})\tilde{B}$. Therefore $\beta \in \text{rad}(\mathfrak{a}\tilde{B})$ so β is integral over $\mathfrak{a}$.

$\Rightarrow$ Imitate the proof of the less general proposition above. Say the irreducible polynomial of β over K is

$$x^n + k_{n-1}x^{n-1} + \cdots + k_0 = (x - \gamma_1) \cdots (x - \gamma_n)$$

we know that β satisfies an equation

$$g(x) = x^l + \alpha_{l-1}x^{l-1} + \cdots + \alpha_0 \quad \alpha_i \in \mathfrak{a}$$

but

$$g(x) = (x - \delta_1) \cdots (x - \delta_l)$$

therefore as before each γ_i is a δ_j hence each γ_i is integral over A . Then the k_i 's are symmetric polynomials in the γ_i 's, so each k_i is integral over $\mathfrak{a}$ and in K . Applying our last result with $B = K$ so $C = A$; so each $k_i \in \text{rad}(\mathfrak{a}A) = \text{rad}(\mathfrak{a})$. □

Proof of Going Down theorem. Given $A \subseteq B$ domains, A integrally closed and B integral over A . We have primes $\mathfrak{p}_1, \mathfrak{p}_2 \in \text{Spec } A, \mathfrak{q}_2 \in \text{Spec } B$ with

$$\begin{cases} \mathfrak{p}_2 \supset \mathfrak{p}_1 \\ \mathfrak{q}_2^c = \mathfrak{p}_2 \end{cases}$$

Then we want to find $\mathfrak{q}_1$ prime of B s.t

$$1. \mathfrak{q}_1 \subset \mathfrak{q}_2$$

$$2. \mathfrak{q}_1^c = \mathfrak{p}_1$$

Let $S = A - \mathfrak{p}_1$ and $T = B - \mathfrak{q}_2$, note that we want

$$(i) \mathfrak{q}_1 \supseteq \mathfrak{p}_1 B \text{ (so } \mathfrak{q}_1^c \supseteq \mathfrak{p}_1)$$

$$(ii) \mathfrak{q}_1 \cap S = \emptyset \text{ (to ensure } \mathfrak{q}_1^c \subseteq \mathfrak{p}_1)$$

$$(iii) \mathfrak{q}_1 \cap T = \emptyset \text{ (to ensure } \mathfrak{q}_1 \subseteq \mathfrak{q}_2)$$

It's enough to show that

$$ST \cap \mathfrak{p}_1 B = \emptyset$$

(ST is multiplicatively closed, take $\mathfrak{q}_1$ maximal among ideals containing $\mathfrak{p}_1 B$ and not meeting ST).

Suppose that we have $\beta = st \in \mathfrak{p}_1 B$.

Step 1. $t \in T = B - \mathfrak{q}_2 \subseteq B$, therefore t is integral over A , A integrally closed so the monic polynomial has the form

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0, \quad a_i \in A$$

i.e

$$K \subseteq K[t] \subseteq L$$

from $\beta = st$ and

$$t^n + a_{n-1}t^{n-1} + \cdots + a_0 = 0$$

we obtain that (by multiplying by S^n)

$$\beta^n + (a_{n-1}S)\beta^{n-1} + \cdots + a_0S^n = 0$$

So the polynomial

$$x^n + (a_{n-1}S)x^{n-1} + \cdots + a_0S^n \in A[x]$$

has β as a root, but it is also irreducible in $K[x]$ because $x^n + a_{n-1} + \cdots + a_0$ is irreducible in $K[x]$. i.e.

$$x^n + (a_{n-1}s)x^{n-1} + \cdots + a_0s^n$$

is the minimal polynomial of β over K . However, if $\beta \in \mathfrak{p}_1 B$ then $\beta \in \text{rad}(\mathfrak{p}_1 B)$ so apply our earlier result with this A and B ($C = B$ the integral closure of A in B). We get that β is integral over $\mathfrak{p}_1$. Therefore by the result above all elements

$$a_{n-1}s, \cdots, a_0s^n \in \text{rad}(\mathfrak{p}_1) = \mathfrak{p}_1$$

But $S = A - \mathfrak{p}_1$ so all of

$$a_{n-1}, \cdots, a_0 \in \mathfrak{p}_1$$

therefore $t \in B$ is integral over $\mathfrak{p}$, hence

$$t \in \text{rad}(\mathfrak{p}_1 B) \subseteq \text{rad}(\mathfrak{p}_2 B) \subseteq \text{rad}(\mathfrak{q}_2)$$

Therefore $t \in \mathfrak{q}_2$ but $t \in B - \mathfrak{q}_2$ contradiction! □

5 Krull dimension of a ring

We are aiming to show the following results:

1. Noether Normalization theorem

2. Hilbert Nullstellensatz
3. Krull dimension of a commutative ring

The first two apply to "finitely generated k -algebra", i.e. rings of the form

$$A = \frac{k[x_1, \dots, x_n]}{\mathfrak{a}} = k[\alpha_1, \dots, \alpha_n]$$

The main idea is that in this case, $\text{Spec } A$, relates to the set of solutions to the polynomials in $\mathfrak{a}$.
View the picture with $pt \subset curve \subset surface$ which are all irreducible, this corresponds

$$V(\mathfrak{p}_0) \subset V(\mathfrak{p}_1) \subset V(\mathfrak{p}_2)$$

with

$$\mathfrak{p}_2 \subsetneq \mathfrak{p}_1 \subsetneq \mathfrak{p}_0$$

Definition 5.1. The Krull dimension of a commutative ring A , if it exists is the maximum of the length of all proper chains of prime ideals in A

$$\mathfrak{p}_0 \supsetneq \mathfrak{p}_1 \supsetneq \dots \supsetneq \mathfrak{p}_r$$

- Remark 5.2.**
1. There may not exist an upper bound, in that case we say that $\dim A = \infty$.
 2. In the Noetherian case, if we start with some $\mathfrak{p}_0$, then the length descending chains of primes starting at a fixed $\mathfrak{p}_0$ is bounded above, therefore $\dim A_{\mathfrak{p}_0}$ must be finite. We call this the height of $\mathfrak{p}_0$.
(Note that the fact that A is Noetherian doesn't prevent it from having infinite Krull dimension, but A Noetherian and local implies that $\dim A < \infty$).
 3. Even in a Noetherian ring we can have the situation

diagram

without more intermediate primes existing.

Proposition 5.3. If $A \subset B$ is an integral extension, then

$$\dim A = \dim B$$

Proof. Follows from lying over, going up and incomparability. More precisely a chain of primes in A

$$\mathfrak{p}_0 \supset \dots \supset \mathfrak{p}_r$$

Then find a $\mathfrak{q}_r \in \text{Spec } B$ with $\mathfrak{q}_r^c = \mathfrak{p}^r$ (lying over), use going up to find

$$\mathfrak{q}_0 \subset \dots \subset \mathfrak{q}_r, \quad \mathfrak{q}_i^c = \mathfrak{p}_i$$

Note that since the contractions are different, this is clearly a proper chain hence

$$\dim A \leq \dim B$$

Conversely, if I have a chain of primes of length r in B , i.e.

$$\mathfrak{q}_0 \subset \cdots \subset \mathfrak{q}_r$$

Therefore

$$\mathfrak{q}_0^c \subset \cdots \subset \mathfrak{q}_r^c$$

But this is a proper chain in A by incomparability. So this gives

$$\dim B \leq \dim A$$

□

5.1 Noether Normalization Lemma.

This is an amazing technical tool for algebras of finite type over a field.

Theorem 5.4. *Let k be a field, and let $A = k[\alpha_1, \dots, \alpha_n]$ be an algebra of finite type over k . Then $\exists \beta_1, \dots, \beta_r \in A$ (poss $r = 0$), which are "algebraically independent" such that*

$$A' := k[\beta_1, \dots, \beta_r] \subset A$$

and A is integral over A' .

Moreover, if k is infinite (not a finite field), then we can take

$$\begin{aligned} \beta_1 &= c_{11}\alpha_1 + \cdots + c_{1n}\alpha_n \\ &\vdots \\ \beta_r &= c_{r1}\alpha_1 + \cdots + c_{rn}\alpha_n \end{aligned}$$

with $c_{ij} \in K$

Remark 5.5. In the context above, algebraically independent means that the homomorphism

$$k[y_1, \dots, y_r] \rightarrow A$$

with $y_i \mapsto \beta_i$ is injective. i.e. if $f(\beta_1, \dots, \beta_r) = 0$ then $f = 0$.

Remark 5.6. If k is algebraically closed, then k is infinite.

Remark 5.7. A is then a finitely generated A' -module and $\dim A = \dim A' = \dim k[x_1, \dots, x_r]$ (which we expect to be r).

Clearly we have that

$$\dim k[x_1, \dots, x_r] \geq r$$

(think of point, line, plane, ...) the chain is

$$\langle x_1, \dots, x_r \rangle \supsetneq \langle x_1, \dots, x_{r-1} \rangle \supsetneq \langle x_1 \rangle \supsetneq 0$$

Remark 5.8. In the case where $\beta_1, \dots, \beta_r$ are linear combinations of $\alpha_1, \dots, \alpha_n$ we have

$$k[\alpha_1, \dots, \alpha_n] = \frac{k[x_1, \dots, x_n]}{\mathfrak{a}}$$

with $k[y_1, \dots, y_r]$ injects into this. therefore

$$\text{Spec } \frac{k[x_1, \dots, x_r]}{\mathfrak{a}} = V(\mathfrak{a}) \hookrightarrow \text{Spec } k[y_1, \dots, y_r]$$

(view the picture).

Lemma 5.9. For the proof of Noether Normalization, let $0 \neq f(x_1, \dots, x_n) \in k[x_1, \dots, x_n]$ polynomial algebra. Then $\exists y_1, \dots, y_n$ s.t. $k[x_1, \dots, x_n] = k[y_1, \dots, y_n]$ (still algebraically independent) "change of coordinates" s.t. in terms of the y_i 's we get

$$f = cy_n^t + f_{t-1}(y_1 \cdots y_{n-1})y_n^{d-1} + \cdots + f_1(y_1 \cdots y_{n-1})y_n + f_0(y_1 \cdots y_{n-1})$$

with $c \in k^*$ (so f is basically monic), moreover if k is infinite we can take this change of coordinates to be linear, i.e.

$$\begin{aligned} x_1 &= c_{11}y_1 + \cdots + c_{1n}y_n \\ &\vdots \\ x_n &= c_{n1}y_1 + \cdots + c_{nn}y_n \end{aligned}$$

with $(c_{ij}) \in GL_n(k)$.

Proof. First do the case where k is infinite. Make a change of variables:

$$\begin{aligned} x_1 &= y_1 + \lambda_1 y_n \\ x_2 &= y_2 + \lambda_2 y_n \\ &\vdots \\ x_{n-1} &= y_{n-1} + \lambda_{n-1} y_n \\ x_n &= y_n \end{aligned}$$

with $\lambda_1, \dots, \lambda_{n-1} \in K$ to be determined. [write out the change of coordinate matrix, clearly it has $\det = 1$]

Write f in terms of its "homogeneous components" in the x_i and then substitute in terms of the y_i 's. In general

$$f(x_1, \dots, x_n) = h_d(x_1, \dots, x_n) + h_{d-1}(x_1, \dots, x_n) + \cdots + h_0(x_1, \dots, x_n)$$

h_d is a k -linear combination of terms $x_1^{j_1} \cdots x_n^{j_n}$ with $\sum j_i = d$. Now substitute, in terms of the y_i 's we get

$$\begin{aligned} f &= h_d(y_1 + \lambda_1 y_n, \dots, y_{n-1} + \lambda_{n-1} y_n, y_n) + h_{d-1}(\bullet) + \cdots \\ &= (h_d(\lambda_1, \dots, \lambda_{n-1}, 1) y_n^d + \text{lower degree in } y_n) + \text{terms of degree } \leq d-1 \end{aligned}$$

We get that coefficient of y_n^d is

$$h_d(\lambda_1, \dots, \lambda_{n-1}, 1)$$

This is a nonzero polynomial in the λ_i 's (since $h_d(x_1, \dots, x_n)$ is homogeneous of pure degree d in its arguments, view the example). Easy fact (BAI, Ch2): if the λ_i 's are taken from an infinite field, we can find such λ_i 's s.t. $h_d(\lambda_1, \dots, \lambda_{n-1}, 1) \neq 0$ in K . Therefore after the change of variable,

$$f = c y_n^d + \text{lower degree terms in } y_n \text{ involving } y_1, \dots, y_{n-1}$$

In the case where k is a finite field, make a change of coordinates

$$\begin{aligned} x_1 &= y_1 + y_n^N \\ x_2 &= y_2 + y_n^{N^2} \\ x_3 &= y_3 + y_n^{N^3} \\ &\vdots \\ x_{n-1} &= y_{n-1} + y_n^{N^{n-1}} \\ x_n &= y_n \end{aligned}$$

for some big N , therefore if

$$f = \sum c_{i_1, \dots, i_n} x_1^{i_1} \cdots x_n^{i_n}$$

Then the term includes a pure power of y_n , $i_1 N + i_2 N^2 + \cdots + i_{n-1} N^{n-1} + i_n$ with coefficient $c_{i_1, \dots, i_n}$. Take $N > \max_j i_j$, then all the expression as above are distinct: they all give different powers of y_n (base N notation); so the largest one "wins" and f is essentially monic in y_n . $\square$

Proof of the Noether Normalization theorem. We will induct on n , we can do the cases $n = 0$ ($A = k$), and $n = 1$ ($A = k[\alpha]$) by hand ourselves ($A = k[x]$ or $A = k[x]/\langle \text{monic poly} \rangle$).

So suppose we know Noether normalization for $n - 1$ variables. Given $A = k[\alpha_1, \dots, \alpha_n]$, if the α_i 's are already algebraically independent then we don't need to do anything ($\beta_i = \alpha_i$, $A' = A$). Otherwise $\exists f(x_1, \dots, x_n) \neq 0 \in k[x_1, \dots, x_n]$ s.t. $f(\alpha_1, \dots, \alpha_n) = 0$, by the lemma there exists a change of variables s.t. $y_1, \dots, y_n$ replace $x_1, \dots, x_n$ and their images $\gamma_1, \dots, \gamma_n$ (in A), replace $\alpha_1, \dots, \alpha_n$ s.t. f is monic (pseudo-monic) in y_n . Hence, we get that $A = k[\gamma_1, \dots, \gamma_n]$ and we have a relation between the γ_i 's of the form

$$c \gamma_n^d + g(\gamma_1, \dots, \gamma_{n-1}) \gamma_n + \cdots + g_0(\gamma_1, \dots, \gamma_{n-1}) = 0$$

with $c \neq 0$ and $c \in k$ This is the reexpression of the relation $f[\gamma_1, \dots, \gamma_n] = 0$ in terms of the γ_i 's.

At this point, we see that

$$A \supseteq A_0 = k[\gamma_1, \dots, \gamma_{n-1}]$$

and $A = A_0[\gamma_n]$ is integral over A_0 . Now use Noether normalization on A_0 (which has $n - 1$ variables), therefore after a further change of variable we get that $A_0 \supseteq k[\beta_1, \dots, \beta_r]$ integral extension, we take $A' = k[\beta_1, \dots, \beta_r]$ then A is integral over A_0 which is integral over A' and we are done. $\square$

Corollary 5.10. $\dim A = \dim A'$ (in the context of the theorem above), notice that

$$\dim A' \geq r$$

due to

$$0 \subsetneq \langle z_1 \rangle \subseteq \dots \subseteq \langle z_1, \dots, z_r \rangle$$

Corollary 5.11. Suppose that $A = k[\alpha_1, \dots, \alpha_n]$ is a field, then in fact A is a finite algebraic extension of k with $\dim_k A \leq n$ and all elements of A are algebraic over k . In particular we can embed A into the algebraic closure $\bar{k}$ of k .

Proof. Take a Noether normalization, so

$$A \supseteq A' = k[\beta_1, \dots, \beta_r]$$

integral extension, therefore by an earlier result, since A is field then A' is also a field. Therefore if $r \geq 1$ we have that A' cannot be a field, hence $r = 0$ and $A' = k$ so A' . $\square$

Theorem 5.12 (Hilbert's Nullstellensatz (1)). Let $A = k[x_1, \dots, x_n]$ and let $\mathfrak{a} \subseteq A$, TFAE

1. $\mathfrak{a} \neq \langle 1 \rangle$ ($\mathfrak{a}$ is proper).
2. $\exists \underline{a} = (a_1, \dots, a_n) \in \bar{k}^n$ s.t. $\forall f \in \mathfrak{a}$ we have that

$$f(a_1, \dots, a_n) = 0$$

3. $\exists L \supset k$ some field (need not be algebraic) and some $\underline{a} \in L^n$ s.t. $\forall f \in \mathfrak{a}$

$$f(\underline{a}) = 0$$

~~(2) $\Rightarrow$~~ (3) easy.

(3) $\Rightarrow$ (1) If $\mathfrak{a} = \langle 1 \rangle$ then we can not find a common zero over any field.

(1) $\Rightarrow$ (2) Suppose that $\mathfrak{a} \subset k[x_1, \dots, x_n]$, we know that $\exists \mathfrak{m}$ a maximal ideal with $\mathfrak{a} \subset \mathfrak{m} \subsetneq A$ then $A/\mathfrak{m}$ is a field, so by the previous corollary we can embed

$$A/\mathfrak{m} \hookrightarrow \bar{k}$$

Then consider the canonical reduction map

$$\varphi: A \rightarrow A/\mathfrak{m}$$

Call $\varphi(x_i) = a_i \in \bar{k}$, then $f \in \mathfrak{a} \implies f \in \mathfrak{m}$, hence

$$\varphi(f) = 0$$

Hence

$$f(\underline{a}) = 0$$

□

Definition 5.13. If $\mathfrak{a} \subset k[x_1, \dots, x_n]$, define

$$V'(\mathfrak{a}) = \{ \underline{a} \in \bar{k}^n \mid \forall f \in \mathfrak{a} \ f(\underline{a}) = 0 \}$$

If $X \subseteq \bar{k}^n$ then

$$I_k(X) = \{ f \in k[x_1, \dots, x_n] \mid \forall \underline{a}, f(\underline{a}) = 0 \}$$

Its easy to check that $I_k(X)$ is an ideal of $k[x_1, \dots, x_n]$.

Theorem 5.14 (Hilbert's Nullstellensatz (2)). *In the setting above we have*

$$I_k(V'(\mathfrak{a})) = \text{rad } \mathfrak{a}$$

Proof. $\subseteq$ Note that

$$\begin{aligned} f \in \text{rad } \mathfrak{a} &\iff \exists N \ f^N \in \mathfrak{a} \\ &\iff \text{In } (A/\mathfrak{a})_f \text{ is } \frac{1 + \mathfrak{a}}{1} = 0 \end{aligned}$$

So let's study

$$(A/\mathfrak{a})_f = A_f / \mathfrak{a} A_f$$

Note that

$$A_f = A[f^{-1}] \cong A[y] / \langle yf - 1 \rangle$$

So

$$A_f \cong k[x_1, \dots, x_n, y] / \langle yf - 1 \rangle$$

So

$$A_f / \mathfrak{a} A_f \cong k[x_1, \dots, y] / \langle \mathfrak{a}, yf - 1 \rangle$$

If $\frac{1+\mathfrak{a}}{1} \neq 0$ in $(A/\mathfrak{a})_f$, then

$$\langle \mathfrak{a}, yf - 1 \rangle \subsetneq k[x_1, \dots, x_n, y]$$

So it would have a common vanishing point $(a_1, \dots, a_n, b) \in \bar{k}^n$ that satisfy: $\forall h \in \mathfrak{a}$

$$h(a_1, \dots, a_n) = 0, \quad bf(a_1, \dots, a_n) - 1 = 0$$

If $f \in I_k(V'(\mathfrak{a}))$, this isn't possible because this would mean that $(a_1, \dots, a_n) \in V'(\mathfrak{a})$ and $f(a_1, \dots, a_n) \neq 0$. Contradiction.

$\supseteq$ If $f^N \in \mathfrak{a}$ and $\underline{a} \in V'(\mathfrak{a})$ then

$$f^N(\underline{a}) = 0$$

so $f(\underline{a}) = 0$, this holds for all $\underline{a} \in V'(\mathfrak{a})$ so $f \in I_k(V'(\mathfrak{a}))$.

□

Let $\bar{k}$ be an algebraically closed field, if $\mathfrak{b} \subseteq k[x_1, \dots, x_n]$ we have

$$V'(\mathfrak{b}) = \{\underline{a} = (a_1, \dots, a_n) : \forall f \in \mathfrak{b} f(\underline{a}) = 0\}$$

More generally for any $\Sigma \subseteq k[x_1, \dots, x_n]$ we have

$$V'(\Sigma) = \{\underline{a} = (a_1, \dots, a_n) : \forall f \in \Sigma f(\underline{a}) = 0\} = V'(\langle \Sigma \rangle)$$

Similarly if $X \subseteq \bar{k}^n$ we get

$$I(X) = I_{\bar{k}} = \left\{ f \in \bar{k}[x_1, \dots, x_n] : \forall \underline{a} \in X, f(\underline{a}) = 0 \right\}$$

The first version of the Nullstellensatz gives that if $\mathfrak{b} \subseteq \bar{k}[x_1, \dots, x_n] \implies V'(\mathfrak{b}) \neq \emptyset$ and the second version says that $I(V'(\mathfrak{b})) = \text{rad } \mathfrak{b}$.

The Zariski topology on $\bar{k}^n$ says that closed sets are those of the form $V'(\Sigma)$ for some Σ ,

$$\Sigma \subseteq \bar{k}[x_1, \dots, x_n]$$

Note that we can stick to ideals since $V'(\Sigma) = V'(\langle \Sigma \rangle)$, moreover

$$V'\left(\bigcup_i \Sigma_i\right) = \bigcup_i V'(\Sigma_i), \quad V'\left(\sum_i \mathfrak{b}_i\right) = \bigcap_i V'(\mathfrak{b}_i)$$

Note that

$$V'(\{fg : f \in \Sigma_1, g \in \Sigma_2\}) = V'(\Sigma_1) \cup V'(\Sigma_2)$$

and

$$V'(\mathfrak{b}_1 \mathfrak{b}_2) = V'(\mathfrak{b}_1) \cup V'(\mathfrak{b}_2)$$

This defines a topology.

Remark 5.15. In this Zariski topology we have

$$V(I(X)) = \bar{X}$$

Remark 5.16. $\bar{k}^n$ can be identified with

$$\text{MaxSpec } \bar{k}[x_1, \dots, x_n] = \left\{ \mathfrak{m} \subseteq \bar{k}[x_1, \dots, x_n] \right\}$$

via

$$\underline{a} = (a_1, \dots, a_n) \in \bar{k}^n \rightarrow \mathfrak{m}_a = \langle x_1 - a_1, \dots, x_n - a_n \rangle = I(\{a\})$$

All maximal ideals are in this way, because $\mathfrak{m} \subsetneq \bar{k}[x_1, \dots, x_n]$ hence

$$V'(\mathfrak{m}) \neq \emptyset$$

take $\underline{a} \in V'(\mathfrak{m})$, then $\mathfrak{m} \subseteq \mathfrak{m}_a$ so $\mathfrak{m} = \mathfrak{m}_a$. We have

$$\bar{k}^n \cong \text{MaxSpec} \subset \text{Spec}(\bar{k}[x_1, \dots, x_n])$$

and

$$V'(\mathfrak{b}) = V(\mathfrak{b}) \cap \text{MaxSpec}$$

i.e. $\underline{a}$ s.t. $\forall f \in \mathfrak{b}, f(\underline{a}) = 0$ and $\forall f \in \mathfrak{b} f \in \mathfrak{m}_a$ $\mathfrak{b} \subseteq \mathfrak{m}_a$.

We also have that

$$V'(I(V'(\mathfrak{b}))) = V'(\mathfrak{b})$$

and

$$I(V'(I(X))) = I(X)$$

Moreover

$$I(V'(\mathfrak{b})) = \text{rad } \mathfrak{b}, \quad V'(I(x)) = \bar{X}$$

We have that

$$\text{MaxSpec}\left(\frac{\bar{k}[x_1, \dots, x_n]}{\mathfrak{b}}\right) \cong_{\text{homeo}} V'(\mathfrak{b}) \subset \bar{k}^n$$

We have the mutually inverse bijections

$$\text{all Zariski closed subsets } Z \subseteq \bar{k}^n \xleftrightarrow{I} \text{all radical ideals } \mathfrak{b} \subset \bar{k}[x_1, \dots, x_n]$$

Culture, the last version of the Nullstellensatz says that

$$\text{rad } \mathfrak{b} = \bigcap_{\substack{\mathfrak{m} \supseteq \mathfrak{b} \\ \mathfrak{m} \text{ maximal}}} \mathfrak{m}$$

equivalently any prime ideal of $\bar{k}[x_1, \dots, x_n]$ is an intersection of maximal ideals.

5.2 Dimension of Affine rings

These are the rings of the form

$$k[\alpha_1, \dots, \alpha_n] \cong \frac{k[x_1, \dots, x_n]}{\mathfrak{b}}$$

We saw that

$$A \supseteq A' = k[\beta_1, \dots, \beta_r]$$

with β_i 's algebraically independent and A is integral over A' and $\dim A = \dim A' \geq r$ (we want to show $= r$).

We need the notion of transcendence degree and transcendence basis: In general we have $k \subseteq L$ with k a field and L any field extension (not necc algebraic).

Definition 5.17. A set of elements $\{\beta_1, \dots, \beta_n\} \subset L$ is said to be algebraically independent over K iff the homomorphism

$$k[x_1, \dots, x_n] \ni x_i \mapsto \beta_i \in L$$

iff there is no nonzero $f \in [x_1, \dots, x_n]$ s.t. $f(\beta_1, \dots, \beta_n) = 0$.

Otherwise, they are said to be algebraically dependent.

Definition 5.18. $\beta_1, \dots, \beta_n \in L$ algebraically generate L (over K) iff L is an algebraic field extension of $k(\beta_1, \dots, \beta_n)$.

Definition 5.19. $\gamma_1, \dots, \gamma_n \in L$ are a transcendence basis for L (over K) iff they are algebraically independent and they algebraically generate L .

Example 5.20. If $A = k[\alpha_1, \dots, \alpha_n]$ is a domain, and $A' = k[\beta_1, \dots, \beta_r] \subset A$ is a Noether normalization of A ; then we can look at the field of fractions K of A' and L of A . Note that

$$K = (A' - \{0\})^{-1} A', \quad L = (A' - \{0\})^{-1} A$$

Then L is algebraic over $k(\beta_1, \dots, \beta_r)$ and $\beta_1, \dots, \beta_r$ is the transcendence basis of L over k .

By abuse of terminology $\{\beta_1, \dots, \beta_r\}$ is a transcendence basis of A over K .

Theorem 5.21. Any two transcendence bases for L/k have the same cardinality; we call this the transcendence degree of L over k we write

$$\text{tr. deg}_k(L)$$

Note that this implies the following theorem:

Theorem 5.22. Let $A = k[\alpha_1, \dots, \alpha_n]$ be a domain that is a f.g. k -algebra. Then

$$\dim A = \text{tr. deg}_k(A)$$

By the above theorem; this is the cardinality of any transcendence basis.

Corollary 5.23. $\dim k[x_1, \dots, x_n] = n$.

To prove the above we first need

Proposition 5.24. If $A = k[\alpha_1, \dots, \alpha_n]$ is a domain and $\mathfrak{p} \subset A$ is a nonzero prime ideal; then

$$\text{tr. deg}_k(A/\mathfrak{p}) < \text{tr. deg}_k(A)$$

Proof. Given $A = k[\alpha_1, \dots, \alpha_n] \supseteq A' = k[\beta_1, \dots, \beta_r]$ a Noether normalization; r is the transcendence degree of A over k . We are also given $\mathfrak{p} \neq 0 \subset A$ we want to show that

$$\text{tr. deg}_k(A/\mathfrak{p}) \leq r - 1$$

Note that $\mathfrak{p}^c \neq 0 \subset A'$; moreover

$$A'/\mathfrak{p}' \hookrightarrow A/\mathfrak{p}$$

Also $A/\mathfrak{p}$ is still integral over $A'/\mathfrak{p}'$; hence $\text{tr. deg.}_k A/\mathfrak{p} = \text{tr. deg.}_k (A'/\mathfrak{p}')$ we want to show that

$$\text{tr. deg. } A'/\mathfrak{p}' \leq r - 1$$

Note that we also have

$$k \subset k(\gamma_1, \dots, \gamma_l) \subset_{\text{alg}} (A'/\mathfrak{p}') \subset_{\text{alg}} (A/\mathfrak{p})$$

a transcendence basis for $A'/\mathfrak{p}'$ will be a basis for $A/\mathfrak{p}$! hence

$$\text{tr. deg.}(A'/\mathfrak{p}') = \text{tr. deg. } A/\mathfrak{p}$$

By incomparability, $\mathfrak{p}' \neq 0$ so $\exists f(\beta_1, \dots, \beta_r) \in \mathfrak{p}'$ and then $\overline{\beta_1}, \dots, \overline{\beta_r} \in A'/\mathfrak{p}'$ generate $A'/\mathfrak{p}'$ algebraically but they are algebraically dependent since

$$f(\overline{\beta_1}, \dots, \overline{\beta_r}) = 0$$

Therefore some proper subset of $\overline{\beta_1}, \dots, \overline{\beta_r}$ is a transcendence basis of $A'/\mathfrak{p}'$ over k so

$$\text{tr. deg. } A'/\mathfrak{p}' \leq r - 1$$

as claimed. □

Now let's use this to prove the theorem

Proof. Let $A = k[\alpha_1, \dots, \alpha_n]$ is a domain; we want to show that

$$\dim A = \text{tr. deg.}_k(A)$$

Let $r = \text{tr. deg.}_k A$; we know that $A \supset A' = k[\beta_1, \dots, \beta_r]$ some Noether normalization with $\dim A = \dim A' \geq r$ (uses going up theorem). Conversely, let us consider a chain of prime ideals in A

$$A \supset \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \dots \subsetneq \mathfrak{p}_{s-1} \subsetneq A/\mathfrak{p}_0$$

with s the length of the chain; we show that $s \leq r$. By the proposition we have

$$\text{tr. deg. } A > \text{tr. deg.}(A/\mathfrak{p}_{s-1}) > \text{tr. deg.}(A/\mathfrak{p}_{s-2}) > \dots > \text{tr. deg. } 0$$

we go down s steps, losing at least one degree each time; at the end we get $r - s \geq 0 \implies r \geq s$. □

We still need to tackle the theory of transcendence bases;

Lemma 5.25. *Say $k \subset L$ fields and $\delta_1, \dots, \delta_n \in L$ are algebraically dependent; then $\exists r \geq 1$ s.t. δ_r is algebraic over $k(\delta_1, \dots, \delta_{r-1})$.*

Moreover if the δ 's generate L , then $\{\delta_1, \dots, \delta_{r-1}\}$ still generate L .

Proof. Find the first r for which $\delta_1, \dots, \delta_r$ are alg. dependent. so $\delta_1, \dots, \delta_{r-1}$ are alg. independent.

Therefore $\exists f(y_1, \dots, y_r) \neq 0$ with

$$f(\delta_1, \dots, \delta_r) = 0$$

but

$$f = g_N(y_1, \dots, y_r)y_r^N + \dots + g_1(y_1, \dots, y_r) + g_0(y_1, \dots, y_{r-1})$$

But note that

$$g_N(\delta_1, \dots, \delta_{r-1}) \neq 0$$

so evaluating at δ_i 's we get that δ_r is the root of a polynomial in $k(\delta_1, \dots, \delta_{r-1})[x]$ as in the first claim.

We get $k \subset k(\delta_1, \dots, \delta_{n-1}) \subseteq_{\text{alg}} k(\delta_1, \dots, \delta_n) \subseteq L$; hence $\delta_1, \dots, \delta_{n-1}$ algebraically generate L . $\square$

Let $k \subset L$ and $\beta_1, \dots, \beta_n \in L$ algebraically generate L over k iff L is an algebraic extension of $k(\beta_1, \dots, \beta_n)$.

Goal. The number of elements of a transcendence basis of L over k is well determined.

Note that we already showed that if $k \subset L$ and $\delta_1, \dots, \delta_n \in L$ algebraically generate L but are algebraically dependent then $\exists r \geq 1$ s.t. $\delta_1, \dots, \delta_{r-1}$ is algebraically independent but $\delta_1, \dots, \delta_r$ is algebraically dependent. Hence

$$k(\delta_1, \dots, \delta_r) \text{ is algebraically generated by } \delta_1, \dots, \delta_r$$

Corollary 5.26. Any field extension L over K that is algebraically finitely generated has a transcendence basis.

Proof. Start with an algebraically generating set and successively remove redundant elements as in the lemma. $\square$

Remark 5.27. a different approach (which works even for infinite transcendence bases) is to use Zorn's lemma to produce a maximal algebraically independent set.

Proposition 5.28. In order to show that any 2 transcendence bases have the same number of elements it's enough to show that if $\{\alpha_1, \dots, \alpha_l\}$ are algebraically independent and $\{\gamma_1, \dots, \gamma_l\}$ generate L (over k) then $l \leq n$

Proof. We start with the ordered list $\{\gamma_1, \dots, \gamma_n\}$ algebraic generators and "put α_1 at the start" to get the ordered set

$$\{\alpha_1, \gamma_1, \dots, \gamma_n\}$$

This is algebraically dependent because α_1 is algebraic over $k(\gamma_1, \dots, \gamma_n)$. Apply the lemma to find (and remove) some element which is algebraically dependent on the previous elements.

We can't remove α_1 since $\alpha_1, \dots, \alpha_r$ are algebraically independent. Reordering the γ_i 's we can assume that γ_1 is algebraically dependent over $\{\alpha_1\}$. We get $\{\alpha_1, \gamma_2, \dots, \gamma_n\}$

still algebraically generate L over k . Now squeeze in α_2 , the ordered set $\{\alpha_1, \alpha_2, \dots, \gamma_n\}$ is again algebraically dependent. Hence once again we can find and remove some element that is dependent on the previous ones, and this removed element is not one of the α 's. After reordering γ_i 's we can remove γ_2 and get $\{\alpha_1, \alpha_2, \gamma_3, \dots, \gamma_n\}$ is again a algebraic generating set of. Keep repeating this until we get to $\{\alpha_1, \dots, \alpha_l, \gamma_{l+1}, \dots, \gamma_n\}$ we can insert l α 's and for each new α we can remove a γ . Note that we cannot run out of γ 's before getting all of the α 's because $\alpha_1, \dots, \alpha_l$ are algebraically independent. Hence

$$l \leq n$$

□

Applying the result above to any 2 transcendence bases $\{\beta_1, \dots, \beta_m\}$ and $\{\gamma_1, \dots, \gamma_k\}$ thens since the β, γ 's are alg. generating

$$m \leq k, k \leq m \implies k = m$$

The last result for these "affine domains" of the form $k[\alpha_1, \dots, \alpha_n] \cong k[x_1, \dots, x_n]/\mathfrak{p}$: Suppose we have $A = k[\alpha_1, \dots, \alpha_n] \supset \bar{P} \supsetneq \{0\}$ where where $\bar{P} = P/\mathfrak{p}$ with $P \supsetneq \mathfrak{p}$ in $k[x_1, \dots, x_n]$. Then we saw

$$\text{Tr. deg } k[x_1, \dots, x_n]/P < \text{tr. deg } k[x_1, \dots, x_n]/\mathfrak{p}$$

Proposition 5.29. *If the difference in transcendence degree is d , then we can squeeze in primes in between and get*

$$\mathfrak{q}_0 = P \supsetneq \mathfrak{q}_1 \supsetneq \dots \supsetneq \mathfrak{q}_d = \mathfrak{p}$$

Remark 5.30. In an affine domain of dimension n any chain of primes can be extended to a maximal chain of length n , note that this is not true in general rings (not even true in finitely generated k algebras.).

Remark 5.31. Note that any finitely generated k -algebra A is Noetherian, so we have finitely many irreducible components. Moreover

$$\begin{aligned} \dim A &= \max(\dim \text{ irreducible components}) \\ &= \max \text{tr. deg.}(A/\mathfrak{p}) \end{aligned}$$

With $\mathfrak{p}$ a minimal prime of A .

Look up catenary rings; what we are saying is that polynomial k -algebras are catenary rings. The transcendence degrees in these finitely generated k -algebras measures what happens with this dimensionality.

To do this we need a stronger version of Noether Normalization!

Theorem 5.32 (Noether Normalization for ring-ideal). *Let $A = k[\alpha_1, \dots, \alpha_n]$ be a domain and let $\mathfrak{a} \subsetneq A$ be an ideal. Then $\exists$ elements $y_1, \dots, y_r \in A$ algebraically independent (and if k is infinite we can get $y_i = \sum_j c_{ij}x_j$) s.t.*

1. $A' = k[y_1, \dots, y_r] \subset A$ is integral.
2. $\mathfrak{a} \cap A' = \langle y_{d+1}, \dots, y_r \rangle$ for some d , $0 \leq d \leq r$.

In particular if we let $\mathfrak{a}' = \mathfrak{a} \cap A'$ we have

$$k[y_1, \dots, y_d] \cong A'/\mathfrak{a}' \hookrightarrow A/\mathfrak{a}$$

and this injection is also a Noether normalization for $A/\mathfrak{a}$.

Remark 5.33. Note that this implies that $\deg_k A = r$ and $\text{tr. deg}(A/\mathfrak{a}) = d$.

Also note that A' is a UFD, so A' is integrally closed. So going-down holds for $A' \subset A$ with

Note that the theorem implies the result on prime chains in $k[x_1, \dots, x_n] \supset P \supset \mathfrak{p}$ as follows. Let $\mathfrak{a} = P/\mathfrak{p} \subset A = k[x_1, \dots, x_n]/\mathfrak{p} \supset A' = k[y_1, \dots, y_r]$. [view diagram] with

$$r = \text{tr. deg } A/\mathfrak{p}, \quad d = \text{tr. deg } A/P$$

Hence we get

$$P = \mathfrak{q}_0 \supset \mathfrak{q}_1 \supset \dots \supset \mathfrak{q}_l = \mathfrak{p}$$

chain of length $l = r - d$.

Proof of the theorem. $A \supset B = k[x_1, \dots, x_r] \supset B' = k[x_1, \dots, x_{r-1}, f] \supset \mathfrak{b} \supset \mathfrak{b}' = B \cap \mathfrak{a} \supset \mathfrak{b}' = B' \cap \mathfrak{a}$ Note that $A/\mathfrak{a} \supset B/\mathfrak{b}$ and

$$\text{tr. deg } B/\mathfrak{b} = \text{tr. deg } A/\mathfrak{a} = d$$

If $\mathfrak{b} = 0$, we are done! otherwise $\exists f \neq 0 \in \mathfrak{b}$ and after a change of variables we replace $x_1, \dots, x_r$ by new variables s.t.

$$f = cx_r^l + O(x_r^{l-1})$$

Note that this means that B is integral over $k[x_1, \dots, x_{r-1}, f]$ since x_r is integral over this ring. Also note that $\{x_1, \dots, x_{r-1}, f\}$ are algebraically independent over k since

$$\text{tr. deg}_k B' = r$$

Hence we have reduced our situation to

$$B' = k[z_1, \dots, z_r] \supset \mathfrak{b}' \ni z_r$$

and we look for a Noether Normalization with ideal of $(B', \mathfrak{b}')$. We will now do induction on the number of variables, assume this result is known for k -algebras with $r - 1$ generators. We have

$$B' = k[z_1, \dots, z_r] \supset \mathfrak{b}' \ni z_r$$

Note that we have

$$B' \supset C = k[z_1, \dots, z_r] \text{ not integral}$$

But

$$C \supset \mathfrak{c} = C \cap \mathfrak{b}'$$

but C has $r - 1$ variables, therefore by inductive hypothesis $\exists C' \supset C$ with

$$C' = k[y_1, \dots, y_{r-1}], \quad y_i \in C$$

and we get some $\mathfrak{c}' \subset \mathfrak{c}$ with

$$\mathfrak{c}' = \langle y_{d+1}, \dots, y_{r-1} \rangle = \mathfrak{b}' \cap C' = \mathfrak{a} \cap C'$$

Therefore

$$B' \supset_{\text{integral}} C'[z_r] = k[y_1, \dots, y_{r-1}, z_r]$$

but

$$C'[z_r] \supset \mathfrak{b}' \cap C'[z_r] \supset \langle y_{d+1}, \dots, y_{r-1}, z_r \rangle$$

Now our goal is to show that $\mathfrak{b}' \cap C'[z_r] = \langle y_{d+1}, \dots, y_{r-1}, z_r \rangle$. Then we can let

$$A' = C'[z_r], \quad \mathfrak{a}' = \langle y_{d+1}, \dots, y_{r-1}, z_r \rangle$$

Let $h(y_1, \dots, y_{r-1}, z_r) \in C'[z_r] \cap \mathfrak{b}'$ we want to show that

$$h \in \mathfrak{c}'C'[z_r] + z_r C'[z_r]$$

Note that we can always separate the monomials in h that contain z_r from the monomials that only involve $y_1, \dots, y_{r-1}$. Therefore

$$h = z_r g(y_1, \dots, y_{r-1}, z_r) + k(y_1, \dots, y_{r-1})$$

note $h, z_r g \in \mathfrak{b}'$ so

$$k \in \mathfrak{b}' \cap C' = \mathfrak{c}'$$

Hence we get that

$$h = z_r g + k \in \langle z_r \rangle + \mathfrak{c}'C'[z_r]$$

□

5.2.1 Hilbert Basis theorem

Proposition 5.34 (Hilbert Basis Theorem). *If A is Noetherian then so is $A[x_1, \dots, x_n]$: hence every ideal in $k[x_1, \dots, x_n]$ or $\mathbb{Z}[x_1, \dots, x_n]$ is finitely generated.*

Proof. Let $I \subset A[x]$, we show that I is finitely generated. Consider the chain of ideals

$$\mathfrak{q}_0 \subseteq \mathfrak{q}_1 \subseteq \dots$$

of A with

$$\mathfrak{q}_j = \left\{ a \in A : \exists f = ax^j + \cdots + a_0 \in I \right\}$$

Which is the set of all x^j coefficients of degree $\leq j$ elements of I . We easily check that $\mathfrak{q}_j$ is an A -ideal. Moreover $\mathfrak{q}_j \subseteq \mathfrak{q}_{j+1}$ because $f \in I \implies xf \in I$. We know that A is Noetherian, so this chain stabilizes and $\exists N$ s.t.

$$\mathfrak{q}_N = \mathfrak{q}_{N+1} = \cdots$$

Each of $\mathfrak{q}_0, \dots, \mathfrak{q}_N$ is a finitely generated A -ideal. From 2 take generators $a_1 \cdots a_i$ of $\mathfrak{q}_0, b_1, \dots, b_j$ of $\mathfrak{q}_1 \cdots z_1, \dots, z_\ell$ of $\mathfrak{q}_N$. Hence we get

$$\begin{aligned} f_1 &= a_1, \dots, f_i = a_i \in I \\ g_1 &= b_1x + \cdots, \dots, g_j = b_jx + \cdots \in I \\ &\vdots \\ F_1 &= z_1x^N + \cdots, \dots, F_\ell = z_\ell x^N \in I \end{aligned}$$

We claim that

$$I = J := \langle f_1, \dots, g_1, \dots, F_1, \dots \rangle$$

Indeed clearly $J \subseteq I$. OTOH, let

$$G = \alpha_M x^M + \cdots \in I$$

we need to show that $G \in J$ and we do so by induction on the degree of G (M).

- $M = 0$, then $G = \alpha_0 \in I$ so $\alpha_0 \in \mathfrak{q}_0 = \langle f_1, \dots, f_i \rangle$ so $\alpha_0 \in J$.
- Suppose that we know the result for $M - 1$ and

$$G = \alpha_m x^m + \cdots \in I$$

we have two cases:

- **Case** $m \leq n$. Then $\alpha_M \in \mathfrak{q}_M$. for notational convenience we can pretend $M = 2$. so

$$G = \alpha_2 x^2 + \cdots \in I$$

then

$$\alpha_2 \in \mathfrak{q}_2 = \langle c_1, \dots, c_k \rangle_A$$

Hence $\exists a_1, \dots, a_k \in A$ s.t.

$$\alpha_2 = \sum_{i=1}^k c_i a_i$$

Define

$$h_t = c_t x^2 + \cdots \in I \cap J$$

But then consider

$$G - \sum_t a_t h_t = (\alpha_2 x^2 + \dots) - (\sum_t a_t (c_t x^2 + \dots)) = \text{elements of } I \text{ of degree } \leq M - 1$$

Hence $G - \sum_t a_t h_t \in J$ by the inductive hypothesis and hence $\sum_t a_t b_t \in J$ so $G \in J$ as desired.

– **Case** $m > n$. So $\alpha_M = \mathfrak{q}_M = \mathfrak{q}_N = \langle z_1, \dots, z_\ell \rangle$ define

$$F_t = z_t x^N + \dots \quad 1 \leq t \leq \ell$$

But then

$$x^{M-N} F_t = z_t x^M + \dots$$

once again we can find an A -linear combination

$$J \ni \sum_t a_t x^{M-N} F_t = \alpha_M x^M + \dots$$

Then $G - \sum_t a_t x^{n-N} F_t = \text{element of } I \text{ of deg } \leq M - 1 \in J$, therefore $G \in J$.

□

6 Chain conditions.

Definition 6.1. A module M is said to be Artinian iff M satisfies the DCC iff every nonempty collection of modules has a minimal elements.

Proposition 6.2. $0 \rightarrow N \rightarrow M \rightarrow M/N \rightarrow 0$ exact sequence, then M is Noetherian iff N and M/N are Noetherian and M is artinian iff N and M/N Artinian

Lemma 6.3. Let M be a module with N , $q \subseteq P \subseteq M$, then I have an exact sequence

$$0 \rightarrow \frac{P \cap N}{Q \cap N} \rightarrow P/Q \rightarrow \frac{P + N}{Q + N} \cong \frac{\overline{P}}{\overline{Q}} \rightarrow 0$$

Proof. Easy to do by hand, or apply the snake lemma to the diagram

$$0 \rightarrow P \cap N \rightarrow P \rightarrow \overline{P} \rightarrow 0$$

$$\uparrow i \quad \uparrow j \quad \uparrow k$$

$$0 \rightarrow Q \cap N \rightarrow Q \rightarrow \overline{Q} \rightarrow 0$$

Note that i, j, k are injections, and everything is happening inside

$$0 \rightarrow N \rightarrow M \rightarrow M/N \rightarrow 0$$

We get

$$0 \rightarrow \ker i \rightarrow \ker j \rightarrow \ker k \rightarrow \text{coker } i \rightarrow \text{coker } j \rightarrow \text{coker } k \rightarrow 0$$

Using the above

$$0 \rightarrow \frac{P \cap N}{Q \cap N} \rightarrow P/Q \rightarrow \frac{P + N}{Q + N} = 0$$

□

Corollary 6.4. *If*

$$\begin{cases} P \supseteq Q \\ P \cap N = Q \cap N \\ P + N = Q + N \end{cases}$$

Then

$$P = Q$$

Remark 6.5. Note that $P \subseteq Q$ is necessary. Consider three lines in a plane, then $P + N = Q + N$ and $P \cap N = Q \cap N = 0$.

Proof of the Original Proposition. $\implies$ Easy, a chain in N is a chain in M , a chain in M/N corresponds to a chain in M which all contain N .

$\Leftarrow$ Hinged on the lemma, e.g. in the DCC case:

$$P_1 \supset P_2 \supset \cdots \in M$$

We get

$$P_1 \cap N \supset P_2 \cap N \supset \cdots \in N$$

and

$$P_1 + N \supset P_2 + N \supset \cdots \in M/N$$

But note that at some point both chains stabilize, then by the lemma we have that the P chain stabilizes.

□

6.1 Modules of finite length.

Definition 6.6. A simple (or irreducible) A -module M is one with $M \neq 0$ and if $M \supseteq M' \supseteq 0$ then $M' = 0$ or $M' = M$. Note that the simple modules are precisely those of the form

$$A/\mathfrak{m}$$

for $\mathfrak{m}$ a maximal ideal.

Definition 6.7. A composition series of M (if exists) is a chain of submodules

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_r = M$$

with each M_i/M_{i-1} simple, i.e. I cannot squeeze in any more intermediate submodules.

This series has length r , since for each $1 \leq i \leq r$ we have a jump M_i/M_{i-1} .

Definition 6.8. A module M is said to have finite length iff it has a composition series.

Iff it is both Artinian and Noetherian.

Theorem 6.9. If M has finite length, then any two composition series for M have the same length.

Lemma 6.10. Take one composition series $0 = P_0 \supsetneq P_1 \supsetneq \cdots \supsetneq P_r = M$ for M of length r and let $N \subset M$ then in the series (not necc proper)

$$0 = P_0 \cap N \subseteq P_1 \cap N \subseteq \cdots \subseteq P_r \cap N = N$$

and

$$N = P_0 + N \subseteq P_1 + N \subseteq P_1 + N \subseteq \cdots \subseteq P_r + N = M$$

at each stage $P_{i-1} \cap N \subseteq P_i \cap N$ and $P_{i-1} + N \subseteq P_i + N$ exactly one of the inclusions is proper; clearly its quotient is simple P_i/P_{i-1} .

Proof. By the lemma above, indeed we have an exact sequence

$$0 \rightarrow \frac{P_i \cap N}{P_{i-1} \cap N} \rightarrow \frac{P_i}{P_{i-1}} \rightarrow \frac{P_i + N}{P_{i-1} + N} \rightarrow 0$$

□

Corollary 6.11. After removing duplicates, we get a composition series for N of some length $s \leq r$ and another composition series for M/N (i.e. a series from N to M) of length $r - s$. Splicing them together, we get a composition series for M of length r , including N as one of its terms.

Theorem 6.12. if M has a composition series, then any two composition series for M have the same length $\ell(M)$.

Corollary 6.13. If

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$$

is exact, and M has a composition series. Then $\ell(M) = \ell(M') + \ell(M'')$.

Remark 6.14. A special case of the above: $A \supset \mathfrak{m}$ is a maximal ideal and $A/\mathfrak{m} \cong k$, M with $\mathfrak{m}M = 0$ so M can be viewed as a k vector space. Then $\ell(M) = \dim k$.

Proof. Induction on the length of a composition series of M .

- $M = 0$ one composition series of length 0.
- M is simple, one composition series of length 1.

Suppose we know that if M' has a composition series of length $r - 1$, then all its composition series have length $r - 1$. Given $M = M_r \supsetneq \cdots \supsetneq M_0 = 0$ with one composition series of length r . Let $M = P_s \supsetneq \cdots \supsetneq P_1 \supsetneq P_0 = 0$. We can make a new composition series based on the M_r one but also including P_s

$$M = Q_r \supsetneq \cdots \supsetneq Q_1 = P_1 \supsetneq 0$$

But then M/P_1 has two composition series of length $r - 1$ and $s - 1$ resp. therefore $r - 1 = s - 1$. Hence $r = s$ as desired. □

Structure of Modules of finite length. If M is a module over A (keep in mind the example of a finite abelian group) we have a composition series

$$M = M_r \supsetneq \cdots \supsetneq M_0 = 0$$

With

$$M_i/M_{i-1} \cong A/\mathfrak{m}_i$$

Our theorems about $\text{Supp } M$ say

$$\text{Supp } M = \{\mathfrak{m}_1, \dots, \mathfrak{m}_r\} \subseteq \text{Spec } A$$

In such a situation we have

$$M \cong_{\text{canonical}} M_{\mathfrak{m}'_1} \oplus \cdots M_{\mathfrak{m}'_j}$$

Where the $\mathfrak{m}'_j$'s are the set of $\mathfrak{m}_i$'s after removing duplicates. Each $M_{\mathfrak{m}'_k}$ is a finite length module over $A/\mathfrak{m}'_k$. Moreover $\exists N$ s.t.

$$(\mathfrak{m}'_k)^N M_{\mathfrak{m}'_k} = 0$$

So $M_{\mathfrak{m}'_k}$ is a module over $A/(\mathfrak{m}'_k)^N$. With $\ell(M_{\mathfrak{m}'_k})$ equals the number of elements with quotient $A/\mathfrak{m}'_k$.

6.2 Krull Principal Ideal theorem

Theorem 6.15. A is an Artinian ring iff A is Noetherian and $\dim A = 0$ (i.e. every prime ideal of A is maximal).

Remark 6.16. Note that it follows that A has only finitely many such maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_r$ distinct with

$$A \cong A_{\mathfrak{m}_1} \times \cdots \times A_{\mathfrak{m}_r}$$

Example 6.17. $\mathbb{Z}/24\mathbb{Z} \cong \mathbb{Z}/8\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ where $\mathbb{Z}/8\mathbb{Z}$ is the localization at $\langle 2 \rangle$ and $\mathbb{Z}/3\mathbb{Z}$ is the localization at $\langle 3 \rangle$. with

$$\text{Spec}(\mathbb{Z}/24\mathbb{Z}) = \{\langle 2 \rangle, \langle 3 \rangle\}$$

$$k[x, y]/a \text{ where } \mathfrak{m} = \langle x - a, y - b \rangle \text{ and}$$

$$\mathfrak{m} \supset \mathfrak{a} \supset \mathfrak{m}^N$$

for some N . Spec here is just $\overline{\mathfrak{m}}$

Proof. $\Leftarrow$ A Noetherian and every prime ideal of A is maximal. We show that A has finite length as an A -module. Indeed, all primes ideals of A are also minimal, and there are finitely many of them

$$\mathfrak{m}_1, \dots, \mathfrak{m}_r$$

Therefore

$$r(A) = \mathfrak{m}_1 \cap \cdots \mathfrak{m}_r = \mathfrak{m}_1 \cdots \mathfrak{m}_r$$

since by CRT we know that if $\mathfrak{m}_i + \mathfrak{m}_j = A$. Moreover $r(A)$ is finitely generated (Since A is Noetherian). But we showed that if $r(\mathfrak{a}) = \mathfrak{b}$ is finitely generated then $\exists N$ s.t.

$$\mathfrak{b}^N \subseteq \mathfrak{a} \subseteq r(\mathfrak{a})$$

In our setting, this gives $(\mathfrak{m}_1 \cdots \mathfrak{m}_r)^N = 0$, i.e. in A the zero ideal is a product of maximal ideals. Let's rewrite this as

$$A \supset \mathfrak{m}_1 \supset \mathfrak{m}_1^2 \supset \cdots \supset \mathfrak{m}_1^N \mathfrak{m}_2 \supsetneq \cdots$$

so each quotient is isomorphic to some $c/\mathfrak{m}_i c$ for some $\mathfrak{m}_i$. But then each quotient is viewable as a vector space over the field $k_i = A/\mathfrak{m}_i$ for some i . But also c is finitely generated so $c/\mathfrak{m}_i c$ is finite dimensional k_i vector space. So each quotient is of finite length. So we have finitely many quotients of finite length so A has finite length.

$\Leftarrow$ We first show that in an Artinian ring, all prime ideals are maximal. let $\mathfrak{p} \subset A$ prime and $A/\mathfrak{p} = B$ is an Artinian domain We show that it is a field. If $x \in B$ and $x \neq 0$ we have the descending chain of ideals

$$\langle x \rangle \supseteq \langle x^2 \rangle \supseteq \langle x^3 \rangle \supseteq \cdots$$

At every point we get

$$\langle x^k \rangle = \langle x^{k+1} \rangle$$

In particular

$$x^k = yx^{k+1} \implies yx = 1$$

Therefore B is a field.

Note, if we show that A is Noetherian then we would be done. We first show that there are finitely many such maximal ideals. Suppose we have infinitely many distinct maximal ideals $\mathfrak{m}_1, \mathfrak{m}_2, \dots$ and consider

$$\mathfrak{m}_1 \supseteq \mathfrak{m}_1 \mathfrak{m}_2 \supseteq \cdots$$

Eventually

$$\mathfrak{m}_1 \mathfrak{m}_2 \cdots \mathfrak{m}_k = \mathfrak{m}_1 \mathfrak{m}_2 \cdots \mathfrak{m}_{k+1} \subset \mathfrak{m}_{k+1}$$

So $\mathfrak{m}_{k+1} \in V(\mathfrak{m}_1 \cdots \mathfrak{m}_j) = V(\mathfrak{m}_1) \cup \cdots \cup V(\mathfrak{m}_k) = \{\mathfrak{m}_1\} \cup \cdots \cup \mathfrak{m}_k$, i.e.

$$\mathfrak{m}_{k+1} = \mathfrak{m}_i$$

for some $i \in \{1, \dots, k\}$ contradiction. Hence

$$\text{Spec } A = \{\mathfrak{m}_1, \dots, \mathfrak{m}_r\}$$

and

$$r(A) = \mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_r = \mathfrak{m}_1 \cdots \mathfrak{m}_r =: \mathfrak{a}$$

We would like to show that $a^N = 0$ for some N . Hence consider the chain

$$\mathfrak{a} \supseteq \mathfrak{a}^2 \supseteq \dots$$

Eventually $\exists k$ s.t.

$$\mathfrak{a}^k = \mathfrak{a}^{k+1} = \dots$$

if $\mathfrak{a}^k = 0$ use the argument above to get submodules of A with each quotient c/cm_i and each quotient is an Artinian vspace over k hence finite dimensional and hence Noetherian. Hence A is also Noetherian and we are done.

Could we have $\mathfrak{a} = r(A)$ and $\mathfrak{a}^k = \mathfrak{a}^{k+1} = \dots = \mathfrak{b} = 0$? Consider the set of all ideals I with

$$\begin{cases} I \subseteq \mathfrak{a} \\ I\mathfrak{b} \neq 0 \end{cases}$$

Note that $\mathfrak{a}$ is one of them. Take a minimal such I then $\exists f \in I$ s.t.

$$f\mathfrak{b} \neq 0$$

But then

$$\langle f \rangle \subseteq I \text{ and } \langle f \rangle \mathfrak{b} \neq 0$$

so

$$I = \langle f \rangle$$

but then

$$(fa)\mathfrak{b} = (fa)\mathfrak{a}^{k+1} = f\mathfrak{b} \neq 0$$

Therefore

$$f\mathfrak{a} \subset \langle f \rangle = I$$

Hence $f\mathfrak{a} = \langle f \rangle$ (due to the minimality), so $\exists g \in \mathfrak{a} = r(A)$ s.t.

$$fg = f$$

So

$$f(1 - g) = 0$$

but $1 - g$ is a unit, therefore $f = 0$ which contradicts the fact that $f\mathfrak{b} \neq 0$. Therefore $\mathfrak{b} = \mathfrak{a}^k = 0$.

□

Geometric intuition. The dimension of the vanishing set of a function is one less than the dimension of the whole space. Unless f vanishes on an irreducible component of maximal dimension. Intuitively the height of $\mathfrak{p}$ is the codomain of $V(\mathfrak{p})$ inside the largest component of Spec containing $V(\mathfrak{p})$.

Definition 6.18. Let $\mathfrak{p} \in \text{Spec } A$, then the height of $\mathfrak{p}$ is the supremum of length of all chains of

primes

$$\mathfrak{p} = \mathfrak{p}_r \supsetneq \mathfrak{p}_{r-1} \supsetneq \cdots \supsetneq \mathfrak{p}_0$$

Remark 6.19. $h(\mathfrak{p}) = \dim A_{\mathfrak{p}}$ and

$$\dim A = \sup_{\mathfrak{p}} h(\mathfrak{p})$$

we can limit ourselves to maximal ideals.

$h(\mathfrak{p}) = 0 \iff \mathfrak{p}$ is minimal $\iff \dim A_{\mathfrak{p}} = 0$, when A is Noetherian this gives us that $A_{\mathfrak{p}}$ is artinian. Hence its length is finite. Without assuming Noetherian $\dim A_{\mathfrak{p}} = 0 \iff A_{\mathfrak{p}}$ has only one prime $\mathfrak{p}A_{\mathfrak{p}}$ which is maximal.

Theorem 6.20 (Krull Principal Ideal Theorem). *Let A be Noetherian, let $f \in A$ and $\mathfrak{q}$ a prime that is minimal over $\langle f \rangle$. Then*

$$h(\mathfrak{q}) \leq 1$$

Proof. From now on, all rings are Noetherian unless indicated otherwise. Let $\mathfrak{q} \supsetneq \mathfrak{p}$ minimal over f , and replacing A by $A_{\mathfrak{q}}$ and f by $fA_{\mathfrak{q}}$ and $\mathfrak{p}$ by $\mathfrak{p}A_{\mathfrak{q}}$, we can assume that A is local and $\mathfrak{q}$ is its maximal ideal. We want to that $h(\mathfrak{p}) = 0$, we want to show that $A_{\mathfrak{p}}$ is artinian.

Strategy: $A/\langle f \rangle$ is Artinian ($\mathfrak{q}$ is the only prime hence $\dim = 0$), therefore chains of ideals in A that contain $\langle f \rangle$ satisfy the DCC, recall the n th symbolic power $\mathfrak{p}^{(n)}$ of $\mathfrak{p}$ is (via $A \rightarrow A_{\mathfrak{p}}$) $(\mathfrak{p}^n)^{ec}$ so

$$\mathfrak{p}^{(n)} = \{x \in A \mid \exists s \in A - \mathfrak{p}, sx \in \mathfrak{p}^n\}$$

Therefore

$$\mathfrak{p}^{(1)} + \langle f \rangle \supsetneq \mathfrak{p}^{(2)} + \langle f \rangle \supsetneq \cdots$$

Stabilizes, so for some n we have

$$\mathfrak{p}^{(n)} + \langle f \rangle = \mathfrak{p}^{(n+1)} + \langle f \rangle$$

Therefore

$$\mathfrak{p}^{(n)} \subseteq \mathfrak{p}^{(n+1)} + fA$$

so every $x_n \in \mathfrak{p}^{(n)}$ can be written as

$$x_n = y_{n+1} + fz$$

Where $y_{n+1} \in \mathfrak{p}^{(n+1)}$, $z \in A$ we get

$$fz = x_n - y_{n+1} \in \mathfrak{p}^{(n)}$$

However $\mathfrak{q}$ is minimal over f , but $f \notin \mathfrak{p}$ (or else we could squeeze $\mathfrak{q} \supsetneq \mathfrak{p} \supset \langle f \rangle$) Therefore $z \in \mathfrak{p}^{(n)}$ (since $\mathfrak{p}^{(n)}$ is $\mathfrak{p}$ -primary). Hence

$$\mathfrak{p}^{(n+1)} \subseteq \mathfrak{p}^{(n+1)} + f\mathfrak{p}^{(n)}$$

Also clearly

$$\mathfrak{p}^{(n+1)} + f\mathfrak{p}^{(n)} \subseteq \mathfrak{p}^{(n)}$$

we get

$$\mathfrak{p}^{(n)} = \mathfrak{p}^{(n+1)} + f\mathfrak{p}^{(n)}$$

Now let $M = \frac{\mathfrak{p}^{(n)}}{\mathfrak{p}^{(n+1)}}$, we have obtained

$$M = fM \subseteq \mathfrak{q}M \subseteq M$$

and $f \in \mathfrak{q}$ with $\mathfrak{q}$ maximal ideal of A . Hence by Nakayama's lemma $M = 0$ so

$$\mathfrak{p}^{(n)} = \mathfrak{p}^{(n+1)}$$

So $\mathfrak{p}^{(n)} = \mathfrak{p}^{(n+1)}$, hence in $A_{\mathfrak{p}}$ we have

$$\mathfrak{p}^n A_{\mathfrak{p}} = \mathfrak{p}^{(n+1)} A_{\mathfrak{p}} = \mathfrak{p}(\mathfrak{p}^n A_{\mathfrak{p}})$$

By Nakayama's lemma in $A_{\mathfrak{p}}$, we deduce

$$\mathfrak{p}^n A_{\mathfrak{p}} = 0$$

Therefore

$$r(A_{\mathfrak{p}}) = \mathfrak{p} A_{\mathfrak{p}}$$

And $\dim A_{\mathfrak{p}} = 0$. Hence $\mathfrak{p}$ is minimal in $A (= A_{\mathfrak{q}})$, so when such a $\mathfrak{p}$ exists, $h(\mathfrak{q}) = 1$. $\square$

Corollary 6.21. *If f is not a zero divisor, then $f \notin$ any minimal prime of A . In this case $\mathfrak{q}$ which is minimal above f cannot be minimal in A hence $h(\mathfrak{q}) = 1$.*

Generalization of the Principal Ideal theorem. Let A Noetherian, and $I = \langle f_1, \dots, f_r \rangle$ an ideal generated by r elements. Let $\mathfrak{q} \subset A$ a prime that is minimal over I . Then

$$h(\mathfrak{q}) \leq r$$

Consequence, in a Noetherian ring, every prime has finite height ($\mathfrak{q}$ is minimal over itself and $\mathfrak{q}$ is finitely generated).

Proof. A clever induction on r .

- $r = 0$ is trivial, $I = \emptyset$ and $\mathfrak{q}$ is minimal means $\mathfrak{q}$ has height 0.
- $r = 1$ is the PIT.
- Suppose $r \geq 2$, and we know the result for primes $\mathfrak{p}$ which are minimal over an ideal J with $r - 1$ generators. Wlog, after replacing A by $A_{\mathfrak{q}}$, A is local Noetherian with $\mathfrak{q}$ its maximal ideal. So we have $I = \langle f_1, \dots, f_r \rangle \subseteq \mathfrak{q}$. We want to show that $h(\mathfrak{q}) \leq r$, we take a chain

$$\mathfrak{q} \supsetneq \mathfrak{p} \supsetneq \dots$$

we show that the length is $\leq r$. By Noetherian, we can assume that $\mathfrak{q}$ is minimal over $\mathfrak{p}$ (this is because we can squeeze in some maximal $\tilde{\mathfrak{p}}$ and use it instead of $\mathfrak{p}$). The goal is to show that $\mathfrak{p}$ is minimal over some ideal J generated by $r - 1$ elements.

Note that since $\mathfrak{q}$ is minimal over I and $\mathfrak{q} \supsetneq \mathfrak{p}$, we deduce that $\mathfrak{p} \not\supseteq \mathfrak{I}$, so wlog $f_1 \notin \mathfrak{p}$. Then $f_1 \in I \subset \mathfrak{q}$, then I deduce that $\mathfrak{q}$ is minimal over $\mathfrak{p} + \langle f_1 \rangle$. So $\mathfrak{q} \supseteq \mathfrak{p} + \langle f_1 \rangle \supsetneq \mathfrak{p}$ with no other primes "in between". This is saying that

$$\dim A/\mathfrak{p} + \langle f_1 \rangle = 0$$

If we let $A' = A/\mathfrak{p}$, we get that $\mathfrak{q}A'$ is nilpotent in A' , so $\exists N$ s.t. $f_2^N = p_2 + a_2 f_1$, $f_3^N = p_3 + a_3 f_1, \dots$. Notice that $f_2, \dots, f_r \rightarrow \mathfrak{q}A'$ which is nilpotent. Therefore some power N s.t.

$$f_i^N = p_i + a_i f_1, \quad p_i \in \mathfrak{p}$$

We claim that $\mathfrak{q}$ (which is minimal over $\langle f_1, \dots, f_r \rangle$) is in fact minimal over $\langle f_1, p_2, \dots, p_r \rangle$. This is because

$$\mathfrak{q} \supseteq \mathfrak{q}' \supseteq \langle f_1, \dots, p_r \rangle \ni f_i^N$$

so $f_i \in \mathfrak{q}'$ and $\mathfrak{q}' \ni f_1$.

Finally map to $\overline{A} \supset \overline{\mathfrak{q}} \supset \overline{\mathfrak{p}}$ with

$$\overline{A} = A/\langle p_2, \dots, p_r \rangle$$

Notice that $\overline{J} = \langle \overline{f_1} \rangle \subseteq \overline{\mathfrak{q}}$ (with $J = f_1, \dots, p_r$) so by PIT we have that $h(\overline{\mathfrak{p}}) = 0$, so $\mathfrak{p}$ is minimal over $\langle p_2, \dots, p_r \rangle$ (with $r - 1$ elements) so $h(\mathfrak{p}) \leq r - 1$ for all $\mathfrak{p}' \subsetneq \mathfrak{q}$. Therefore

$$h(\mathfrak{q}) \leq r - 1$$

□

Corollary 6.22. *In an Noetherian ring, every prime $\mathfrak{p}$ has finite height. Since $\mathfrak{p}$ is finite over itself and is finitely generated.*

Theorem 6.23. *If A is Noetherian, and $\mathfrak{p} \subset A$ has height r , then $\exists$ an ideal $\mathfrak{a}$ generated by r elements s.t. $\mathfrak{p}$ is minimal over $\mathfrak{a}$.*

Proof. Induction on $r = h(\mathfrak{p})$.

- $r = 0$. In this case $\mathfrak{p}$ is minimal, i.e. a minimal prime over $\langle 0 \rangle$. But 0 is generated by the emptyset.
- $r = 1$. Here $\mathfrak{p}$ is not minimal, but if $\mathfrak{p} \supsetneq \mathfrak{q}$ then $\mathfrak{q}$ is minimal. We want to find $f \in \mathfrak{p}$ s.t. $\mathfrak{p}$ is minimal over $\langle f \rangle$.

Notice that there are only finitely many minimal primes $\mathfrak{q}_1, \dots, \mathfrak{q}_l \in \text{Spec } A$. $V(\mathfrak{p})$ cuts out a smaller domain (subset of some $\mathfrak{q}_i$); find f that doesn't vanish on any $V(\mathfrak{q}_j)$.

So we have $\mathfrak{p}$ of height 1, so $\mathfrak{p} \not\subseteq$ any of $\mathfrak{q}_1, \dots, \mathfrak{q}_l$ so

$$\mathfrak{p} \not\subseteq \mathfrak{q}_1 \cup \dots \cup \mathfrak{q}_l$$

(prime avoidance). Hence $\exists f \in \mathfrak{p}$ s.t. f is not in $\mathfrak{q}_1 \cup \dots \mathfrak{q}_l$. In particular $\mathfrak{p}$ has to be minimal over f , if not we would have some minimal containing f . So we got our ideal $\langle f \rangle$ with $\mathfrak{p}$ minimal over it.

- **Inductive step.** Say $h(\mathfrak{p}) = r \geq 2$. Once again, take $f \in \mathfrak{p}$ with $f \notin$ any minimal prime of A consider $\bar{A} = A/\langle f \rangle$. Then $\bar{\mathfrak{p}} \subset \bar{A}$ is also prime and let $r' = h(\bar{\mathfrak{p}})$. Claim $r' < r$, indeed if we have a chain

$$\bar{\mathfrak{p}} \supsetneq \overline{\mathfrak{p}_{r'-1}} \supsetneq \cdots \supsetneq \bar{\mathfrak{p}}_0$$

This corresponds to a chain

$$\mathfrak{p} \supsetneq \mathfrak{p}_{r'-1} \supsetneq \cdots \supsetneq \mathfrak{p}_0 \supsetneq \langle f \rangle$$

so $\mathfrak{p}_0$ has height 1 by the previous case. So we conclude that we have a chain of length $r' - 1$ that starts at $\mathfrak{p}$. i.e.

$$r \geq r' + 1 \iff r > r'$$

But now, in $\bar{A} = A/\langle f \rangle$, $\bar{\mathfrak{p}}$ is minimal over an ideal $\bar{\mathfrak{a}}$ which is generated by r' elements

$$\bar{\mathfrak{a}} = \langle \bar{g}_1, \dots, \bar{g}_{r'} \rangle$$

So

$$\bar{A} \supset \bar{\mathfrak{p}} \supset \bar{\mathfrak{a}} \supset 0 \implies A \supset \mathfrak{p} \supset \mathfrak{a} = \langle g_1, \dots, g_{r'} \rangle \supset \langle f \rangle$$

and $\mathfrak{p}$ is minimal over an ideal that can be generated by $r' + 1$ elements, therefore

$$r = h(\mathfrak{p}) \leq r' + 1$$

Hence

$$r = r' + 1$$

hence $\mathfrak{p}$ is minimal over $\mathfrak{a}$ which can be generated by r elements. □

Putting everything together. Let $(A, \mathfrak{m})$ be a Noetherian local ring, then

$$\dim A = h(\mathfrak{m}) = r < \infty$$

where $r = \min \# \{\text{generators of an } \mathfrak{m}\text{-primary ideal } \mathfrak{a}\}$. Because in a local ring, $\mathfrak{m}$ minimal over $\mathfrak{a} \iff r(\mathfrak{a}) = \mathfrak{m} \iff \mathfrak{a}$ is $\mathfrak{m}$ -primary. ($A/\mathfrak{a}$ has finite length).

Remark 6.24. By Nakayama's Lemma, the minimum number of generators of $\mathfrak{m}$ is

$$\dim_{A/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 \geq \dim A$$

Fact. A regular local ring is a domain, in fact it is a UFD. In fact the localization of $A_{\mathfrak{p}}$ of a regular local ring is again regular.

7 Discrete Valuation rings & Dedekind domains

Definition 7.1. A discrete valuation ring (DVR) is a local ring $(A, \mathfrak{m})$ which is a PID and not a field.

Example 7.2. start with any PID $(\mathbb{Z})$, and localize at any nonzero prime p so

$$\mathbb{Z}_p = \left\{ \frac{a}{s} \mid p \nmid s \right\}$$

thus any ideals of $\mathbb{Z}_p$ are $\langle 1 \rangle, \langle p \rangle, \langle p^2 \rangle, \dots$ and 0. And not that $\mathbb{Q}$ is still the field of fractions of $\mathbb{Z}_p$.

Remark 7.3. In $\mathbb{Z}_p$, every nonzero element has the form $p^i U$, $i \geq 0$ and U a unit;

$$p^i \frac{r}{s}, \quad p \nmid r, p \nmid s$$

Notice that the maximal ideal of this local Noetherian ring is principal generated by p .

Theorem 7.4. Let $(A, \mathfrak{m})$ be a local Noetherian local ring, with $\mathfrak{m} = \langle \pi \rangle$ principal. Then either

1. π is nilpotent, A is Artinian, so $\dim A = 0$.
2. π is not nilpotent, A is a domain, A is a PID, $\dim A = 1$ (A is a DVR).

Proof. If π is nilpotent $\pi^N = 0$, then $\mathfrak{m}^N = 0$. So the 0 ideal is a product of maximal ideals. We saw in that case that A since Noetherian must be artinian so 1 is easy.

Now if π is not nilpotent. We first show that every nonzero element $x \in A$ can be written as

$$x = \pi^i u, \quad u \in A^*, i \geq 0$$

Start with x , if x is a unit then we are done. Otherwise, $x \in \mathfrak{m} - \langle \pi \rangle$, so $x = \pi y$. Note, we obtain

$$\langle x \rangle \subset \langle y \rangle$$

and in fact

$$\langle x \rangle = \langle \pi y \rangle = \mathfrak{m} \langle y \rangle$$

and $y \neq 0$. So by Nakayama's Lemma we must have

$$\langle x \rangle \subsetneq \langle y \rangle$$

if y is a unit, we are done. Otherwise keep going. Since A is Noetherian this process must terminate (we get a smaller ideal every time). Therefore we get

$$x = \pi^i u$$

We now show that A is a domain. Indeed if $x, y \neq 0$ we have $x = \pi^i u$, $y = \pi^j v$ so

$$xy = \pi^{i+j} uv$$

but π is not nilpotent. so $xy \neq 0$.

Every ideal is principal, this is clearly true for $\langle 0 \rangle$. if $\mathfrak{a}$ is a nonzero ideal, then it must be

finitely generated by a set of nonzero elements

$$\mathfrak{a} = \langle \pi^i u, \pi^j v, \dots, \pi^m z \rangle, \quad i, j, \dots, m \geq 0$$

Then clearly

$$\mathfrak{a} = \langle \pi^n \rangle, \quad n = \min \{i, j, \dots, m\}$$

So A is a DVR and $\dim A = 1$. □

Remark 7.5. A good example of a DVR is

$$A = k[[z]]$$

with $\mathfrak{m} = \langle z \rangle$, the units here are those with $c_0 \neq 0$ since we can invert these by "infinite division".

$$x = c_7 z^7 + \dots = z^7 (c_7 + \dots)$$

Remark 7.6. In any Noetherian ring, either

$$\mathfrak{m} \supsetneq \mathfrak{m}^2 \supsetneq \dots$$

or by Nakayama

$$\mathfrak{m}^N = \mathfrak{m}^{N+1} = \dots$$

and A is Artinian

Fact. We will show later that

$$\bigcap_i \mathfrak{m}^i = 0$$

7.1 Discrete Valuations

Let A be a DVR as above, its field of fraction

$$K = A_\pi = \left\{ a/\pi^k \right\}$$

So $K = \{0\} \cup \{\pi^n u : U \in A^*, n \in \mathbb{Z}\}$. Also note that $\pi^n u = \pi^m v$, then $n = m$.

Definition 7.7. The valuation on K is the function

$$v: K^* \rightarrow \mathbb{Z}, \quad v(\pi^n u) = n$$

we can extend this to

$$v: K \rightarrow \mathbb{Z}$$

Remark 7.8. $V(\alpha\beta) = V(\alpha) + v(\beta)$, and

$$v(\alpha + \beta) \geq \min \{v(\alpha), v(\beta)\}$$

Note that $\exists \pi$ s.t.

$$v(\pi) = 1$$